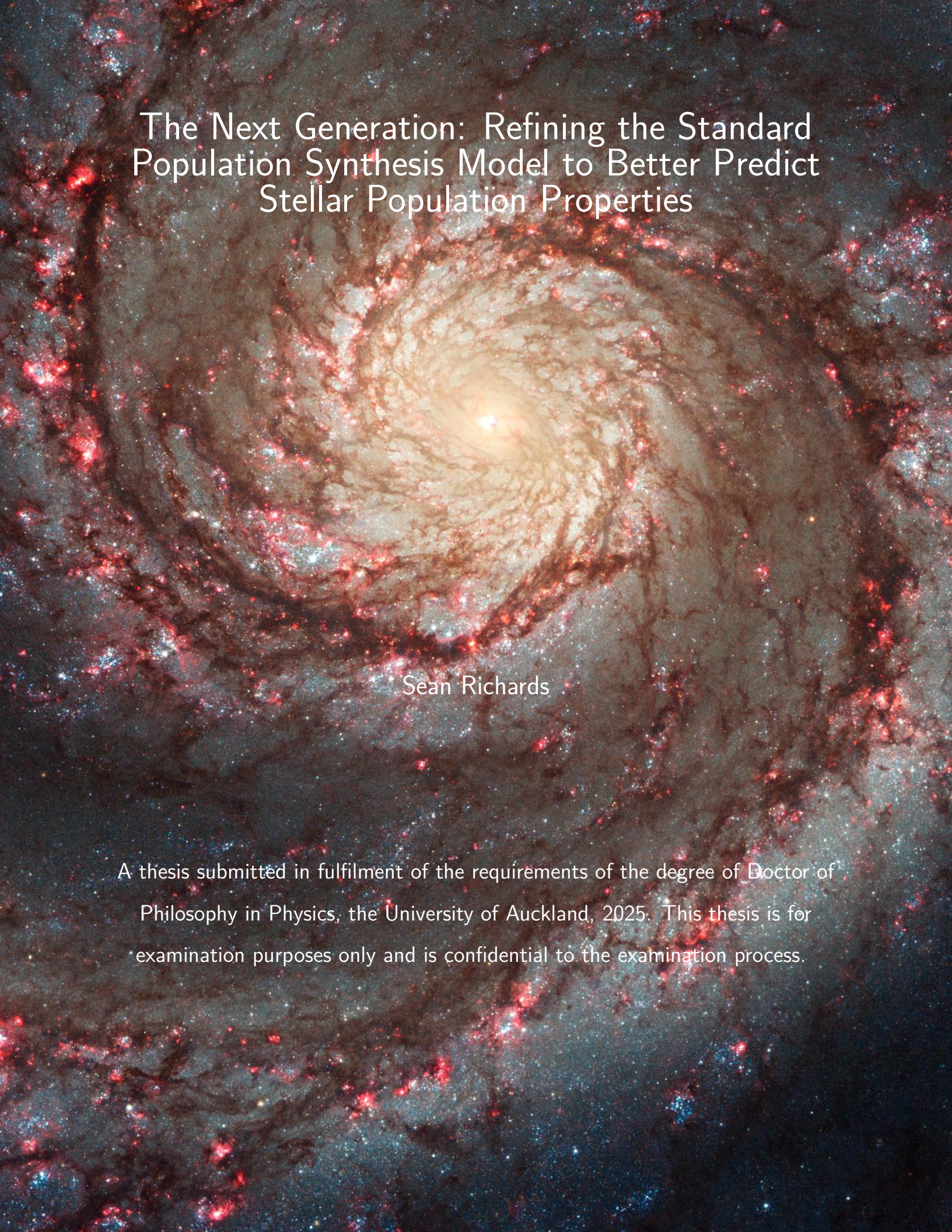




The Next Generation: Refining the Standard Population Synthesis Model to Better Predict Stellar Population Properties

Sean Richards

A thesis submitted in fulfilment of the requirements of the degree of Doctor of
Philosophy in Physics, the University of Auckland, 2025. This thesis is for
examination purposes only and is confidential to the examination process.

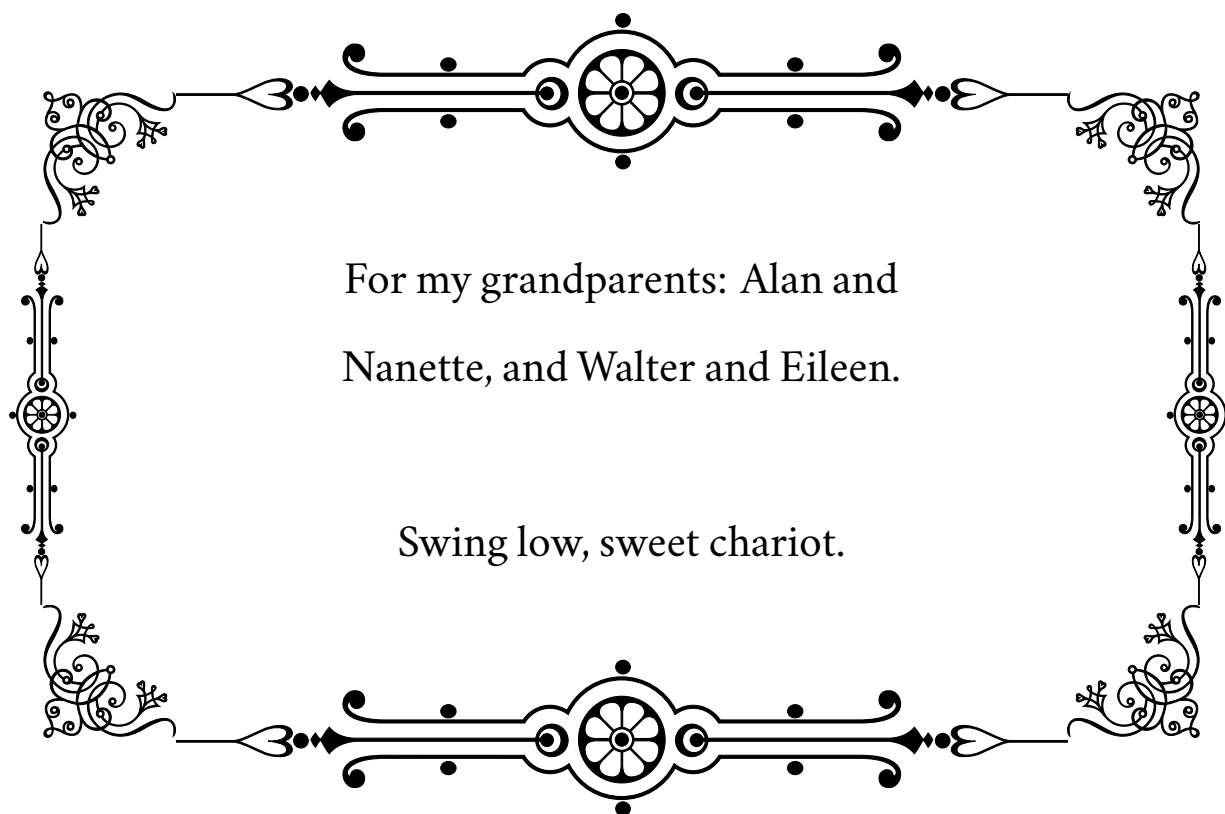
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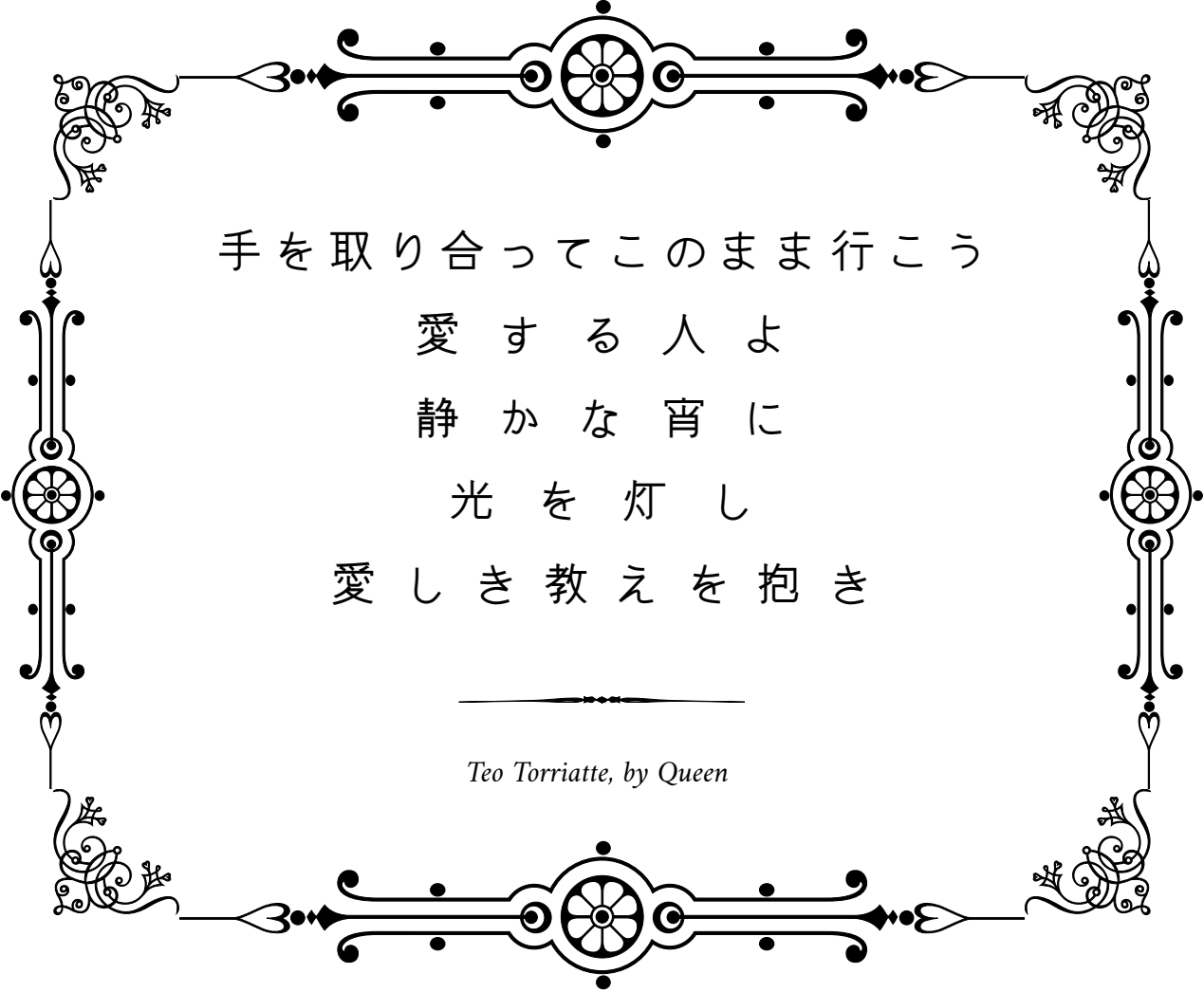


A short description of the work.

Cover Image: The Whirlpool Galaxy (M51). Image is by NASA and released into the public domain:

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手を取り合ってこのまま行こう

愛する人よ

静かな宵に

光を灯し

愛しき教えを抱き

Teo Torriatte, by Queen

Ho there, wanderer! Stay thy course a moment and in-
dulse an old man.

– ELMINSTER AUMAR

Acknowledgements

Nei rā te kaupapa o te reo māori e / He taonga tuku iho nō nga tipuna
he parekarakawa, he mea hirahira e / Toku reo, toku ohoohe mapihi maurea whakakaimarihi

Nei rā te kaupapa, BY TE OKEROA HURIWAKA AND TAWHIRO MAXWELL

Waipapa Taumata Rau (The University of Auckland) sits within the rohe of several Māori iwi/hapū, including Ngāti Whātua, Ngāti Whātua o Ōrākei, Te Kawerau a Maki, Ngāti Tamaoho, Te Ākitai Waiohū, Ngāti Maru, Te Patukirikiri, Ngāti Pāoa, Ngāti Tamatera, Ngāi Tai ki Tāmaki, Ngāti Te Ata, and Ngāti Whanaunga, and Waikato-Tainui. Ipipiri (the Bay of Islands) sits within the rohe of Ngāpuhi and Ngāti Rēhia. I acknowledge these iwi and hapū as the tāngata whenua of Tāmaki Makaurau, Ipipiri, and Aotearoa, I acknowledge that sovereignty was never ceded under Te Tiriti o Waitangi, and I pay my respects to kuia and kaumatua of these iwi and hapū, past, present, and future. Tūturu whakamaui, kia tina! Tihei mauri ora!

And I came from the Dreaming / From the dusty red-soil plain / I am the ancient heart / The keeper of the flame.
I stood upon that rocky shore / I watched those tall ships come / when for eighty thousand years I've been
The first Australian

I am Australian, AS PERFORMED BY THE BUSHWACKERS

Throughout my travels for this PhD, I was privileged to visit several lands I never dreamed I could as a child. In Australia, I worked on lands traditionally occupied by peoples of the Kulin Nations (predominantly the Wurundjeri people), and the Jagera, Yuggera and Ugarapul peoples. I acknowledge their sovereignty and traditional ownership and custodianship of the lands and waterways on which Swinburne University of Technology and the University of Southern Queensland (Springfield) are located. I acknowledge their Elders, past, present, future and emerging, and I acknowledge that sovereignty was never ceded. I thank the kind hospitality of the Polish, Belgian, Singaporean, German, Dutch, and Australian peoples, all of whose lands I visited throughout this thesis. I have learned so much, sitting on the shores of the Eternal Sea that is the Milky Way.

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This work made use of v2.2 of the Binary Population and Spectral Synthesis (BPASS) models as described in Eldridge et al. (2017) and Stanway & Eldridge (2018).

Raise a glass to all of us
Tomorrow there’ll be more of us
Telling the story of tonight.

FOUND/TONIGHT (BEN PLATT AND LIN-MANUEL MIRANDA)

I am the proud grandchild of immigrants, and by my count, the first person who identifies as Sāmoan to graduate with a PhD from the Department of Physics at the University of Auckland in (at least) 24 years. I’m also one of the few gender non-conforming students in my Department. To all my fellow BIPOC students, all my queer whānau, and all my other underrepresented students in the sciences, I see you, I recognise you, I feel you. Kia kaha, kia māia, kia manawanui.

Who am I? / I am a girl who loves my island / And the girl who loves the sea, it calls me / I am the daughter of the village chief / We are descended from voyagers / Who found their way across the world / They call me

I Am Moana FROM MOANA

It would be impossible for me to list my entire whakapapa. It is known as far back as nine generations in parts, tracing back to around the mid-1700s – including to the first Lord Mayor of the City of Sheffield, a Prime Minister of Samoa, several Ministers in Samoa – including a former Minister of Finance – and Chair of Samoa’s National Bank, and some of the major clans in Samoa. We suspect there may be an American President in there too. At last count we know of about 500 relatives – my Great-Great-Aunt’s father-in-law had thirteen children in the mid-1800s. But I want to especially mihi to my mother Sue, my late aunt Leslie, my late grandfather Alan, my grandmother Nanette and her late mother Mary “Sugalu” Scott, my aunt Alison and uncle Pete, my great-uncle Ben and great-aunt Lulu, my cousins Jeremy, Jaime, Charlotte, Candice, Ngareta, Wairaka, Marihi, and Tama. To the rest of my ‘aiga: *e lautele le vateatea, ae le o le avanoa iinei: Ou te alofa ia te outou uma.*

Life before death. Strength before weakness. Journey before destination.

FIRST IDEAL OF THE KNIGHTS RADIANT

For Jaffe, Bailey, Riegel, Ray, Mercer, Johnson, Willingham, and O’Brien: Vox Machina, the Mighty Nein, Bells Hells, Gilmore, Allura, Delilah, Pumat, Xandis, and Ryn. For Iyengar, Carazo, Ray, Riegel, Willingham, Wilson, and Mulligan: the Ring of Brass, Vespine, Maya, and Kir. For Lewis: Peter, Edmund, Susan, Lucy, Aslan and Jadis. For Sanderson: Kaladin, Syl, Shallan, Jasnah, Vin, Elend, Kelsier, Rysn, and Hoid. For Vincke, Wilde, English, Newbon, Downie, Solomon, Béart, Jones, Wiles, Mercer, Gregory, Simmons, Isaacs, and Robertson: Lae’zel, Shadowheart, Astarion, Gale, Wyll, Karlach, Halsin, Jaheira, Minsc, Minthara, Ketheric, Gortash, and Orin the Red. For those who imagine the strange and beautiful and complex and weird and mystical

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I argue that we choose to be good because of our bonds with other people and our innate desire to treat them with dignity. Simply put, we are not in this alone.

CHIDI ANAGONYE

It occurs to me that there are precious few moments in one's life to acknowledge and thank the people who contributed to where one is now. I hold the belief that a person is defined by the sum of their actions and interactions, so I would like to acknowledge a few of the people that have made me the person I am today.

I'd like at this point to acknowledge friends: Connor, Frazer, Girish, Hunter, Pierce, Ryan, Scott, and Sean, with whom I shared some of my first physics and mathematics courses. Emma, the self-appointed creative director of this thesis. Josh, Tessa, Hamish, Jasmine, Ryan: my first experience with D&D was with Eldrin. Andrew, George, Hamish, Connor, Callum, and Mark: Katerina was my longest player experience; playing her taught me so much about myself (and Andrew). Josh, Liam, Arthur, Connor, Andrew, Oliver: Ayo and the rest of the Blue Originators (you never learned their party's name) owe you a debt of gratitude. Even good old Necromancer Bob says hi.

Thank you to my office mates, Max, Bushra, Jonathan, Petra, Hamish, Erin, and Laura. On the topic of Big Dog and Big L: I can't say what I got told they'd do if I didn't acknowledge them. They know what they said. But thank you for long walks on beaches, fish and chips, drinks at My Bar (we need to get a better bar), and generally being a welcome distraction whilst writing this thesis. Also, get therapy, the both of you.

Once a king or queen in Narnia, always a king or queen.

ASLAN

I've lost touch with some of the people I did high school with but the formative experiences there still sit inscribed in indelible ink in my mind. Caitlin, who has been one of my best friends for the better part of a quarter-century (and who would *not* let me live it down were I to forget her). Callum, Susannah, Kyle, Grayson, Tomek "if you're going to try to pronounce my last name, put some **EMPHASIS** on it" Pietkiewicz and the entire business and commerce crew. Sam, Louise, Crooks, the IT Ninjas. Isabella, Chyna, Evan, Parth, Jay, Matt, all the people at NZSDC 2015, even if I had to argue in favour of alternative medicine and exclusive schools for LGBTQIA youth, those moments will be with me forever. Finn and Susannah: I know they said they never ranked us in maths classes, but they ranked us in maths classes, and I know we always vied for the top three places. Finn is an architect now and Susannah is admitted to the High Court – does that mean I finally win the ranking?

I'd also like to thank Di Eastwood, Maddy Haydock, Melissa Johansson, and Lauren Mulcahy (our debating coach, and my Year 13, 12, and 11 English teachers). Not all of you are teaching anymore, but I still remember

vivid parts of your lessons. Miss Mulcahy commending my essay structure (for not opening an essay with “The question I am answering is...” but actually motivating why I enjoyed *To Kill a Mockingbird* and *Romeo+Juliet*.). Miss Johansson, for showing faith in the process of my creative writing portfolio (“Sean, I want to send you to counselling for this story. Change nothing, this is an E.”). Mrs Haydock, firstly for Mufasa but also for believing in my weird and experimentalist poetry (I was going through a bit of a fourth-wall breaking phase with poetry), for letting me rant for 30 minutes on why *The Matrix* is actually a Marxist allegory and the (concerningly frequent) themes from Freudian psychoanalysis we can pull from it. Moreover, I remember in English class, you congratulating me for using “moreover” correctly in a sentence. Mrs Eastwood for always challenging me to do better in debating. I’ve said it before and I will say it again: I credit a mix of you four for my erudition and eloquence.

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Right, I’m almost done here, I swear.

Physics is a collaborative endeavor, and I would be remiss to not acknowledge my research compatriots. Thank you to the BPASS team, both in Auckland and abroad: John, Heloise, Max, Sohan, Wouter, Petra, Erin, Adib, Conor, Gareth, Ashley, and Ankur. We’ve also been joined by a few students who have since left the team (because they graduated!): Gleb, Emma, Louis, Ethan, Siris, Celina, and Amanda. Thank you to our PIs, Dr Elizabeth Stanway and Prof. Jan Eldridge.

Speaking of Jan! Thank you to Prof. Jan Eldridge and Assoc. Prof. Nick Rattenbury. I genuinely cannot image a more perfect supervision team. Both carry a large amount of mana, and I have learned an immense amount from each of them. You’ll notice that all of my contractions of people’s titles either omit (in the case of “Dr”) or contain periods (in the case of basically every other title). Nick was very pedantic about this in my Masters thesis and now I subconsciously check for this when reading other people’s prose. Thank you by extension to Prof. Richard Easter. Jan, Nick, and Richard were my lecturers for PHYSICS 107/G in Semester Two of 2016. I needed extra papers to fill out my first year, as there were only three required papers for my (at that time) Computer Science degree. I took PHYSICS 120 and 150 because I was good at it in High School (see Mr Lumkong, Mr Dyson, and Mr Whitaker above), and PHYSICS 107/G because it sounded fun. I left that course a physics double major, and no joke, restructured my entire degree around taking PHYSICS 356. Jan, Nick, and Richard are a large part of why I am a scientist today. I cannot thank them enough for that. From the bottom of my heart, thank you, ngā mihi, vinaka vakalevu, fa’afetai lava.

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¹This is not a joke or a quip, my Year 13 physics teacher played as a number eight in the 1999 Rugby World Cup for the USA.

You'll be in my heart.
No matter what they say.
You'll be here in my heart.
Always.

You'll be in my heart BY PHIL COLLINS

When I started this PhD, I had two alive grandparents. I now have only one, my grandmother Nanette. For Alan, Walter, and Eileen, I carry your memories with me always. In the words of Liam O'Brien: *I was richer for having known you, friend.*

I could not be more keen, for some plain old boring precedented times.

CR. RICHARD HILLS

I've noticed that the more theses I write, the more people I thank. My PGDipSci dissertation thanked five people. My Masters thesis thanked some fifty-five, a Government, and several coffee companies. Did I beat it this time?

Run, you clever [girl], and be a Doctor.

CLARA OSWIN OSWALD, *Doctor Who*

No, thank YOU. One thing about trains... it
doesn't matter where they're going. What matters
is deciding to get on.

THE CONDUCTOR, *The Polar Express*

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Glossary



Notation	Description
3D MHD	3D Magnetohydrodynamics.
AGB	Asymptotic Giant Branch.
BH	Black Hole.
BNS	Binary Neutron Star.
BPASS	Binary Population and Spectral Synthesis.
CCSN	core-collapse supernovae.
CDF	Cumulative Distribution Function.
CEE	Common Envelope Evolution.
CME	coronal mass ejection.
CO	Compact Object.
CO core	Carbon-Oxygen core.
CPTA	Chinese Pulsar Timing Array.
DTD	Delay-Time Distribution.
EOS	Equation of State.
EPTA	European Pulsar Timing Array.
GR	General Relativity.

Notation	Description
IMF	Initial Mass Function.
ISCO	innermost stable circular orbit.
ISM	Interstellar Medium.
JWST	James Webb Space Telescope.
K-S	Kolmogorov-Smirnov.
LIGO	Laser Interferometer Gravitational Wave Observatory.
LISA	Laser Interferometer Space Antenna.
LMC	Large Magellanic Cloud.
LVK Collaboration	LIGO-Virgo-KAGRA Collaboration.
MLT	Mixing Length Theory.
MS	Main Sequence.
MWEG	Milky-Way Equivalent Galaxy.
NANOGrav	North American Nanohertz Observatory for Gravitational Waves.
NS	Neutron Star.
NS-BH	Neutron Star - Black Hole Binary.
O3	Operating Run 3.
O4	Operating Run 4.
ONe core	Oxygen-Neon core.
PDF	Probability Density Function.
PPTA	Parkes Pulsar Timing Array.
RGB	Red Giant Branch.
RLOF	Roche Lobe overflow.
SFH	Stellar Formation History.

Notation	Description
SFRD	Star Formation Rate Density.
SMC	Small Magellanic Cloud.
SPH	Smoothed-particle hydrodynamics.
TAMS	Terminal-Age Main Sequence.
USSN	ultra-stripped supernova.
WD	White Dwarf.
WR	Wolf Rayet.
ZAMS	Zero-Age Main Sequence.

Thesis To-do List



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Co-Authorship Form

Email:

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Please indicate the chapter/section/pages of this thesis that are extracted from a co-authored work and give the title and publication details or details of submission of the co-authored work.

Parts of Chapter 4 are based on "The variable evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material", submitted to MNRAS and available on the NASA ADS here:
<https://ui.adsabs.harvard.edu/abs/2024arXiv241103000R/abstract>

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Extent of contribution by PhD candidate (%)	70	

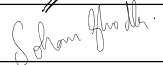
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Jan Eldridge	Wrote part of text, criticism, commentary, and review
Max Briel	Criticism, commentary, and review
Sohan Ghodla	Criticism, commentary, and review

Certification by Co-Authors

The undersigned hereby certify that:

- ❖ the above statement correctly reflects the nature and extent of the PhD candidate's contribution to this work, and the nature of the contribution of each of the co-authors; and
- ❖ that the candidate wrote all or the majority of the text.

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Co-Authorship Form

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Nature of contribution by PhD candidate	Wrote majority of the paper, developed most parts of the testing pipeline, later data generation, performed the statistical tests.
Extent of contribution by PhD candidate (%)	90

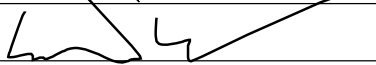
CO-AUTHORS

Name	Nature of Contribution
Jan Eldridge	Original data generation, guidance on project, wrote clarification on the difference between BPASS primary and secondary models, criticism on manuscript
Max Briel	Developing ideas, code commentary, criticism on manuscript
Héloïse Stevance	Developing the test for Ultra-Stripped Supernovae, guidance on project, criticism on manuscript
Reinhold Willcox	Providing single-star pulsar dataset from Willcox+ 2021, criticism on manuscript

Certification by Co-Authors

The undersigned hereby certify that:

- ❖ the above statement correctly reflects the nature and extent of the PhD candidate's contribution to this work, and the nature of the contribution of each of the co-authors; and
- ❖ that the candidate wrote all or the majority of the text.

Name	Signature	Date
Jan Eldridge		11/11/2024
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The only way to learn is by playing. The only way to win
is by learning. And the only way to begin is by beginning.

– SAM REICH

Chapter

1

Introduction

ARGUABLY THE GREATEST ADVANCEMENT in modern astronomy was the direct detection of gravitational waves – the rippling of space-time as a result of the motion of massive celestial bodies – on September 14, 2015. The Laser Interferometer Gravitational Wave Observatory (LIGO) collaboration detected the chirp of merging binary black holes, with masses $m_1 = 36^{+5}_{-4} M_\odot$ and $m_2 = 29^{+4}_{-4} M_\odot$ (Abbott et al., 2016). This detection sparked the so-called “gravitational wave era”, with detections in Operating Run 4 (O4) reading a cadence of hundreds per year.¹ These detections afford researchers probes of various physical phenomena. For instance, these merger events are some of the most energetic events in the Universe – GW150914 peaked at a gravitational-wave luminosity of $3.6 \times 10^{56} \text{ erg s}^{-1}$ (Abbott et al., 2016), compared with the typical energy release of a core-collapse supernovae (CCSN) of $\sim 10^{53} \text{ erg}$ (Hartmann, 1999) – and the merger of binary neutron stars are the predominant formation channels for most of the elements heavier than iron (see, among others, Eichler et al., 1989; Arcavi et al., 2017; Pian et al., 2017; Smartt et al., 2017).

In addition, the stochastic background of gravitational waves is a direct probe on early-Universe cosmology (Christensen, 2018), though the frequency band that these gravitational waves span is between 10^{-9} and 10^{14} Hz (Christensen, 1992, 2018), making detection of the stochastic background difficult. Indeed, it took eight years after the detection of gravitational waves for the stochastic background to be detected by the collaborative effort of North American Nanohertz Observatory for Gravitational Waves (NANOGrav) (Agazie et al., 2023), the European Pulsar Timing Array (EPTA) (Antoniadis et al., 2023), the Parkes Pulsar Timing Array (PPTA) (Reardon et al., 2023), and the Chinese Pulsar Timing Array (CPTA) (Xu et al., 2023).

Beyond the stochastic background, the vast majority of gravitational wave sources involve stars at some point in their lives – whether that be the merger of Compact Object (CO), the detonation of CCSN, or the rotation of an isolated massive source such as a Neutron Star (NS) or Black Hole (BH). In these cases stars are either the immediate sources of gravitational waves, or the progenitors of the objects that are the sources. Only in extreme cases, such as the merger of galaxies or the motion of supermassive black holes, are stars not directly involved in the generation of gravitational waves (and even then, this is debatable).

Ergo, any attempt at understanding gravitational waves must be founded in a strong theoretical understanding

¹From the start of O4 in May 2023 to the end of December 2024, approximately 167 unretracted significant (false alarm rate of $< 1 \text{ month}^{-1}$) detections, averaging to a cadence of $\sim 107 \text{ yr}^{-1}$.

of stellar structure and evolution. Without theory, any observation is no more than a curiosity; it is with theoretical underpinning and models that insight can be gained. As most astrophysical sources of gravitational waves (i.e., not the stochastic background) are formed by the merger of two “compact remnants” – black holes, white dwarfs, and neutron stars – one must understand their formation and dynamics, and thus the evolution of progenitor stars into these remnants, as well.

Furthermore, a sound understanding of the evolution of stars has impacts beyond gravitational wave astronomy. As the interiors of stars are complex nuclear furnaces, they are responsible for producing most of the elements in the Universe heavier than Hydrogen (Burbidge et al., 1957). In fact, of the elements lighter than Uranium, the only elements not substantially produced in stellar processes (such as through kilonovae or in supernovae) are technetium, promethium (which both lack stable isotopes),² boron, beryllium (which both are produced via cosmic ray fission), and some formation channels of lithium.

From the perspective of Universal building blocks, stars are also the focal point of planetary systems, and provide the theoretical right conditions for the formation of life. Without a solid theory of stellar evolution, one is unable to describe the likelihood of extraterrestrial life within the Universe.

1.1 | Observational astronomy

If the stars should appear one night in a thousand years, how would
men believe and adore; and preserve for many generations the
remembrance of the city of God which had been shown!

Nature, BY RALPH WALDO EMERSON

There are approximately 9000 stars visible in the night sky to the naked eye (with magnitudes brighter than 6.5), across both hemispheres (Hoffleit, 1964), which are displayed in figure 1.1. The dark-adapted eye can see approximately as far as η Carinae,³ which is only approximately 2300 pc away.

2300 pc is an order of magnitude less than the size of Galaxy, which is approximately 26.8 kpc in diameter (Goodwin et al., 1998), which in turn pales in comparison to the Local Group at ~ 3 Mpc (Redd, 2018), and the observable Universe at 28.5 Gpc (Bars & Terning, 2010). There are, in addition, $\sim 1.7 \times 10^{22}$ stars in the observable Universe, assuming 1.7×10^{11} galaxies, with approximately 10^{11} stars in each (Marov, 2015). Thus, the birth, evolution, and death of stars is of fundamental importance to understanding the local structure and evolution of astrophysical objects on any scale.

Given the number of stars visible in the vast expanse of night, and their relatively stable positions in the sky, it is no surprise that they are a fixture in many religions and cultures around the world. Ancient Greece saw a difference in movement of some celestial objects, and called these $\pi\lambda\alpha\nu\eta\tau\eta\varsigma$, which morphed into *planētes* and eventually the English word “planets”. Babylonia knew of Halley’s comet and tracked observations of it as late as 87 BCE (Stephenson et al., 1985). Records of eclipses in Chinese culture exist as far back as 1600 BCE (Fang,

²There are technetium stars, and promethium has been observed in some atmospheres (Cowley et al., 2004), however these processes do not generally pollute their environment.

³This is a debated topic; we use the definition that a “star” includes binaries and that the star is in the catalogue of Hoffleit (1964), which rules out V762 Cas, the other commonly held “farthest star”.

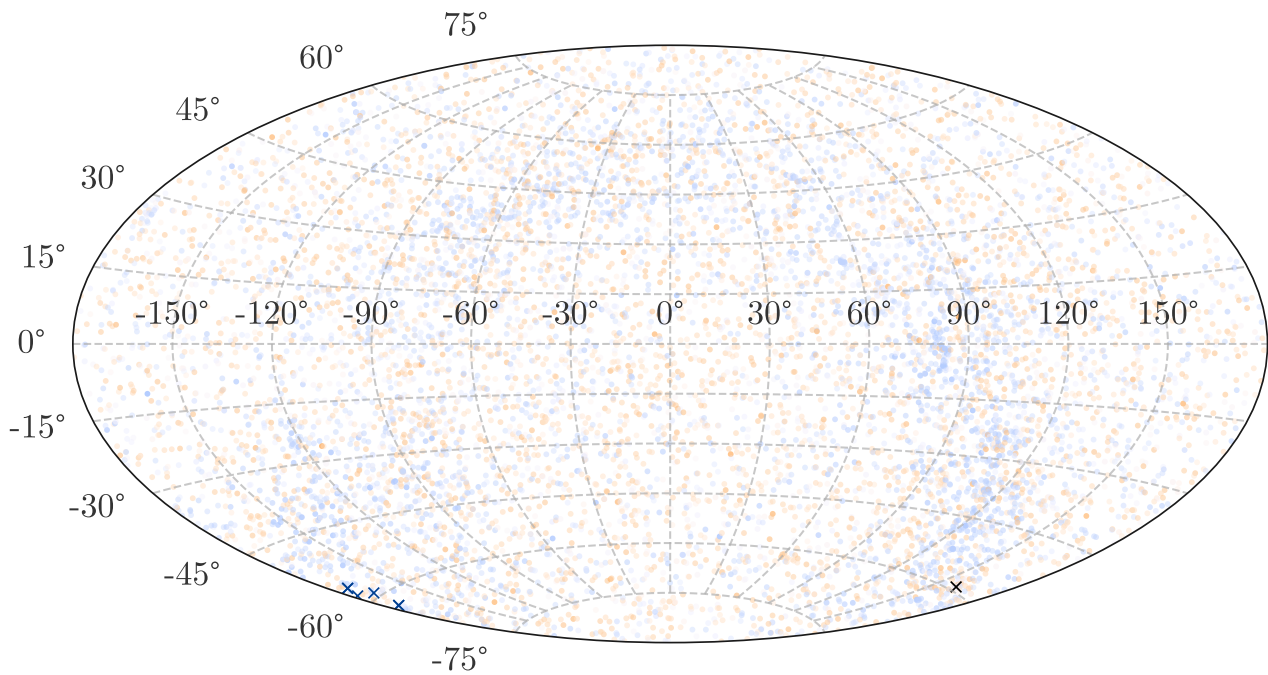


Figure 1.1: The locations of the ~ 9000 stars in the night sky visible to the naked eye, as reported in Hoffleit (1964). The colors are computed by converting the B-V color indices to temperatures using the fits in Ballesteros (2012), and then converted to RGB via the algorithm in this StackOverflow post: <https://stackoverflow.com/a/45497817>. The Southern Cross (Crucis) is shown fully opaque in blue towards the bottom left of the plot as a point of reference, and η Carinae is shown fully opaque in black in the bottom right.

2015). Knowledge of the movements of stars guided Polynesian wayfarers as they crossed the South Pacific (Buck, 1938; Hamacher, 2022), were embedded in the cultural mythos of First Nations peoples in Australia (Noon & De Napoli, 2022), allowed them to forecast weather patterns (Hamacher et al., 2019), and were plausibly recorded into oral histories (Fuller & Hamacher, 2017). Even in the arts, the knowledge of the lunar perigee was written into *Othello* by Shakespeare:

It is the very error of the moon;
She comes more near the earth than she was wont,
And makes men mad.

– Shakespeare (ca. 1603)

Knowledge of oceanic tides existed in the time of Ancient Greece, and was described by Ptolemy, who wrote in *Tetrabiblos* that

ἀμπώτεις καὶ αἱ παλίρροιαι, καὶ αἱ τῶν πνευμάτων
 δὲ τροπαὶ μάλιστα περὶ τὰς τοιαύτας τῶν φωτῶν
 κεντρώσεις ἀποτελοῦνται πρὸς οὓς ἂν τῶν ἀνέμων
 ἐπὶ τὰ αὐτὰ τὸ πλάτος τῆς σελήνης τὰς
 προσνεύσεις ποιοῦμενον καταλαμβάνηται.

Ptolemy & Proclus (ca. 168)

For the hour by hour intensifications and
 relaxations of the weather vary in response to
 such positions of the stars as these, in the same
 way that the ebb and flow of the tide respond to
 the phases of the moon, and the changes in the
 air-currents are brought about especially at such
 appearances of the luminaries at the angles, in
 the direction of those winds towards which the
 latitude of the moon is found to be inclining.

Robbins (1940)

This knowledge flowed through to Dante, who wrote in the third part of the Divine Comedy, *Paradiso*:

E come 'l volger del ciel de la luna
 cuopre e discuopre i liti sanza posa,
 così fa di Fiorenza la Fortuna:

Alighieri (ca. 1317)

And, as the turning of the lunar sphere covers
 and endlessly uncovers the edges of the shore,
 thus does fortune deal with Florence.

Corbett & Webb (2016)

In religious scripture, celestial navigation has oft been a fixture, such as in Chapter 2 of the Gospel according to Matthew:⁴

2:7 – Then Herod, when he had priuily called the Wise men, enquired of them diligently what
 time the Starre appeared:

2:8 – And he sent them to Bethlehem, and said, Goe, and search diligently for the yong child,
 and when ye haue found him, bring me word againe, that I may come and worship him also.

2:9 – When they had heard the King, they departed, and loe, the Starre which they saw in the
 East, went before them, till it came and stood ouer where the young childe was.

– Various Authors (1611)

Despite this deep cultural connection to the stars, what we would now call Astronomy is a somewhat recent invention. Arguably conventional (and in some sense, “correct”) astronomy arose in the 16th century CE with Copernicus proposing the heliocentric model of the solar system, which was confirmed by Galileo in the 17th century CE (Carroll & Ostlie, 2017) – although Galileo’s claims did result in his being convicted of heresy for 300 years.

⁴This text is as reproduced from the King James Version of the Bible, as written in Early Modern English as faithful as possible to the script.

The early 20th century saw a veritable renaissance in both the observational and theoretical spheres. On the one hand, developments such as the theories of Special Relativity in 1905 (Einstein, 1905), General Relativity (GR) in 1917 (Einstein, 1917), and the predictions of gravitational waves (Poincaré, 1906; Einstein, 1916) led to theoretical revolutions. Around this time, Arthur Eddington began working on determining the internal structure and evolution of stars (Eddington, 1916, 1920), and Gamow (1928) put forth the idea of quantum tunneling allowing elements to overcome the Coulomb barrier. On the other hand, the innovations of Payne (1925) in showing that stars are composed of mostly Hydrogen and Helium, Baade & Zwicky (1934) in distinguishing “super-novae” from “common novae”, Chandrasekhar (1935) in establishing the idea of the “Chandrasekhar mass” (the limiting mass of a white dwarf), Hoyle (1946) in developing the foundation of the nuclear reaction of elements heavier than Hydrogen in stars, Salpeter (1955) in developing the Initial Mass Function (IMF), Burbidge et al. (1957) in truly developing the theory of stellar nucleosynthesis, Henyey et al. (1955) and Hayashi (1961) in exploring the pre-main sequence evolution of stars were key to kickstarting entire branches of astronomy such as spectroscopy and computational astrophysics.

The pace of discovery of new and exotic astronomical phenomena has not slowed into the 21st century. Thorne-Żytkow objects – giant stars with Neutron star cores (Thorne & Żytkow, 1977) – once only existed as theoretical curiosities, yet now have credible observational candidates in Levesque et al. (2014). The aforementioned gravitational waves were detected in 2015 (LIGO Scientific Collaboration and Virgo Collaboration, 2016), a century after they were demonstrated by Einstein (1916) to be a central prediction of GR. More recently, Lau et al. (2022) directly imaged 17 carbon-rich dust rings around WR140 with James Webb Space Telescope (JWST), enriching the Interstellar Medium (ISM) with carbon-rich compounds. In the coming years, the Vera C. Rubin Observatory (Ivezić et al., 2019) and Laser Interferometer Space Antenna (LISA) are expected to vastly increase the number of short-period eclipsing double white dwarfs (Burdge et al., 2020), with the latter able to detect gravitational waves with frequencies between 0.1 mHz and 1 Hz (Amaro-Seoane et al., 2012), compared to the $\sim 10 \text{ Hz} - 1000 \text{ Hz}$ that LIGO-Virgo-KAGRA Collaboration (LVK Collaboration) can detect (Schmitz, 2021). The stochastic gravitational-wave background was detected by Agazie et al. (2023); Antoniadis et al. (2023); Reardon et al. (2023); Xu et al. (2023), via a search of pulsar-timing arrays.

Astronomy as a field has blossomed since the naked-eye era – entering the era of the telescope, the era of the observatory, and now the gravitational wave era. The next era of astronomy is the big data era. With observations reaching sub-day cadences, a large array of theoretical models, and perhaps the development of “on-demand” detailed models is required.

1.2 | Stellar Modelling

Modelling the evolution of a star is a nontrivial endeavour. Stars reach internal temperatures, pressures, and densities unsustainable on Earth. For instance, the core of a star is capable of nuclear fusion, which has not been sustainably recreated on Earth. Moreover, the density of Sirius B – a White Dwarf (WD) in the constellation of Canis Major – is $2.14 \times 10^{10} \text{ kg m}^{-3}$ (Barstow et al., 2005) and the mean density of the Sun is $\sim 1410 \text{ kg m}^{-3}$ (Eddy & Ise, 1979). On Earth, the highest density measured in a laboratory is $\sim 27\,000 \text{ kg m}^{-3}$, requiring pressures on the order of 2.3 TPa (Fratanduono et al., 2020), and observed for only nanoseconds.

It is nigh impossible to achieve and sustain these conditions to experimentally test any physical phenomena. That said, missions such as the Parker Solar Probe have made flyby missions into the atmosphere of the Sun, though broadening the sample size to two stars – using Proxima Centauri, the nearest star to Earth aside from the Sun – would require travelling at approximately $0.1c$ for 40 years to reach. If instead we were to send the Parker Solar Probe travelling at a constant velocity of its top velocity of $\sim 176\,471\text{ ms}^{-1}$, it would take approximately 7200 years to reach Proxima Centauri. Needless to say, physical experimentation with a star is difficult.

1.2.1 | Population Synthesis and Stellar Modelling

Because of these factors, one often turns to computer simulations to model the interiors of stars. The question turns from physical experimentation to mathematical ones, and the idea of a *stellar model* becomes useful. A stellar model is a calculation performed such that the internal structure (the present state of physical quantities) and evolution (how these quantities change over time) are determined. Computing these parameters is nontrivial. As will be developed in chapter 2, the simplest model of a star is a series of four coupled differential equations describing the internal pressure, temperature, luminosity, and mass distributions. However, including “extra” physics greatly increases the complexity of the problem; Henyey et al. (1964) developed methods of solving the equations of structure and evolution that turn the problem into matrix inversion, at the cost of the central “Henyey matrix” being thousands of elements in size.

The diversity and complexity of stellar profiles, interiors, and characteristics discussed earlier demands a careful and varied approach. One cannot abstract away much of the important physics. Although it is common to use approximations such as spherical symmetry to reduce the dimensionality of the problem, and to only evaluate the equations of stellar structure at a fixed number of locations in this one remaining dimension, chosen sufficiently intelligently, a complex computational effort must be undertaken.

This problem is then magnified when attempting to extrapolate out to describe an entire population of stars, such as from a burst of stellar formation activity, or in a shared environment such as a globular cluster. It is inefficient to, for example, model all approximately 10^{11} stars in the Milky Way (Marov, 2015) as independent entities. One must develop a way to take a library of models and determine the large-scale behaviour this way.

Broadly speaking, there are three approaches generating a population of stars, all of which have both merits and drawbacks. These three methods are:

1. Rapid Population Synthesis,
2. Detailed Population Synthesis, and
3. 3D Magnetohydrodynamics (3D MHD) Simulations.

Each of these approaches informs the other two, and all play important roles in stellar modelling. Although we focus on detailed methods, we will briefly outline all three approaches here.

Population Synthesis with Detailed Binary Models

A detailed population synthesis code uses detailed model sets, where the interior state of the stars is computed, and coupled to the orbital evolution equations. In this regime, observables such as mass transfer are explicitly

calculated and solved for from the exact equations of state, structure, and evolution of the binary. By doing this, one follows the binary evolution in a reasonable amount of detail, obfuscating fine detail and (generally) asymmetry in the star in favour of recovering the larger-scale structure.

Rapid Binary Population Synthesis

Rapid binary population synthesis models are instead either compilations of single-star evolutionary tracks that have been solved for using a code such as MESA (Paxton et al., 2011, 2013, 2015, 2018, 2019; Jermyn et al., 2023), fit to using approximations such as those of Hurley et al. (2000) or Hurley et al. (2002), or interpolated between. These models are then coupled to the orbital evolution *a posteriori* using a binary evolution framework, effectively approximating the computationally expensive part of binary evolution models to varying degrees of detail. This allows them to generate binary models on-the-fly, and optimise for this interaction, at the potential cost of missing more precise facets of binary evolution.

3D MHD Simulations

3D Magnetohydrodynamics (3D MHD) and Smoothed-particle hydrodynamics (SPH) simulations are the most detailed method of stellar modelling. The base principle is that the star is evaluated as a network of particles in a continuous media, and the evolution of the star is followed in theoretically atomic detail. For example, PHANTOM (Price et al., 2018) is a SPH code which essentially models the star as a fluid, where the Lagrangian hydrodynamical equations are solved for in two-dimensional cells (termed “particles”). Although these present the most detailed models, they are also the most time-intensive to solve. For a SPH code, the famous Navier-Stokes equation is invoked, which has no analytical solution but is computed in excruciating detail. As a result, most SPH models take weeks to run and simulate a star for kiloseconds of its lifetime, rather than for the millions (if not billions) of years detailed models are able to compute.

Though it may seem that they serve the same niche and are therefore competing to be the “better” method of stellar modelling, detailed 1D models and 3D MHD/SPH models exist harmoniously with each other. The advancements in 3D MHD/SPH modelling inform the evolution of 1D models in a process known as “3-to-1”, and the computational flexibility of 1D models offers insights into where 3D models wish to target to evolve, in a lock-step advancement method.

1.3 | This work

We now describe the placement of this work within the field, and outline the key aims this thesis sets out to answer. This thesis can be summarised as an endeavour to model the interior physics of interacting binary stars over certain near-extreme phenomena. We also aim to examine the post-supernova kinematics of binary systems. Thus, the area of interest of this thesis is effectively the entire lifetime of a binary system post-birth, with the critical exceptions of the moment of supernovae, and the interior physics of compact objects.

1.3.1 | Research themes

There are essentially three major themes guiding this research.

- (I) Can we methodologically overcome the typical cases in which stellar models fail:
 - (a) RLOF,
 - (b) Common Envelope Evolution (CEE), and
 - (c) the helium flash
- (II) Can we develop a testing regime for the natal supernova kick, which relies on more observations than just isolated pulsar velocities?
- (III) What modifications are required to be made to our stellar modelling code in preparation for the next major BPASS release?

1.3.2 | Thesis outline

Six impossible things. Count them, Alice.

ALICE IN WONDERLAND

This thesis is, broadly, a tour of stellar evolution from Zero-Age Main Sequence (ZAMS) to the eventual merger of two compact objects, with a few omissions along the way (such as the moment of supernova). The thesis is structured as follows:

1. In chapter 2, we provide the necessary theoretical background for future chapters, critically examine the pieces of software and numerical methods employed in this work, and discuss the logistics of running this single-thread grid-based code as a multiprocessed code on a distributed cluster.
2. In chapter 3, we present the results of our stellar evolution codes in the interesting case when both stars almost (or in some instances, do) touch. (Theme I)
3. In chapter 4, we explore the regime in which mass transfer is assumed to become helium-rich, and the effects on the future evolution of these binaries. (Themes I, III)
4. In chapter 5, we explore the population properties of binaries which underwent helium-rich mass transfer (Themes I, III).
5. In chapter 6, we examine the kinematics of the post-supernova inspiral and the mechanics of the natal supernova kick. (Theme II)
6. In chapter 7, we review the thesis and conclude. (Themes I, II, III)
7. In appendix A, we provide the KAITIAKI code necessary to generate the models for certain stellar evolution plots throughout the thesis. (Themes I, III)

8. In appendix B, we list the codes employed and their versions.
9. In appendix C, we present period-eccentricity distributions for the LVK Collaboration O3 contour (Theme II)
10. In appendix D, we list the default parameter choices we make for our stellar evolution models (Themes I, III).

We now move into the more detailed background, and outlay the theory that underpins all of our models.

Se'i lua'i lou le 'ulu taumamao.

Pick the breadfruits on the far-off branches first

– SĀMOAN PROVERB

Chapter

2

Theoretical Background

IN THIS CHAPTER, WE PROVIDE A brief account of the evolution of a star from a theoretical perspective, from its formation to post-supernova. We will discuss both the case of a single isolated star, and an isolated binary star system. It is pertinent to note that our binaries will be assumed to have formed together at birth – thus having the same age – rather than by, for example, the dynamical capture of a runaway star in a globular cluster, which introduces extra effects and considerations.

2.1 | Stellar Formation

Nature and nature's laws lay hid in night: God said, "Let Newton be!" and all was light.

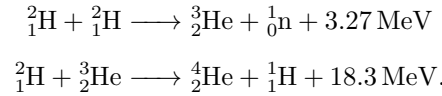
ALEXANDER POPE

From the point of view of stellar evolution, which starts in earnest at the ZAMS, the formation of stars is interesting but ultimately has little to no impact on the nuclear evolution of the star besides setting the initial mass. We provide a short recap of stellar formation here as scene-setting, but our stellar evolution models assume that a gas cloud has already collapsed and we have a star at its ZAMS.

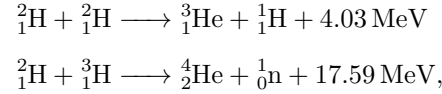
Stars are born in clouds of gas and dust. Due to inhomogeneities in the dust formation, some regions become more dense than others. When a region of this gas with adiabatic constant γ and mean molecular weight μ reaches the so-called Jeans Mass,

$$M_J = \left(\frac{10k_B T}{3(\gamma - 1)\mu m_H G} \right)^{\frac{3}{2}} \left(\frac{3}{4\pi\rho} \right)^{\frac{1}{2}} \sim \left(\frac{10k_B T}{G} R \right)^{\frac{3}{2}} \left(\frac{1}{M} \right)^{\frac{1}{2}}, \quad (2.1)$$

where the similarity is taking $\gamma = \frac{5}{3}$, $\mu = \frac{1}{2}$ for an adiabatic gas composed of H(II), then gravity becomes the dominant force in the clumps of gas. This causes a protostar to form. Generally speaking, this protostar slowly gains more mass from its surroundings (McKee & Ostriker, 2007) and, as it does so, increases in temperature. When the core temperature reaches 10^7 K, the hydrogen in the core begins to fuse into helium via the reaction chain



This is aided by two further hydrogen to helium reactions:



or, via a master equation,



with intermediate products which remain stable and are thus omitted for simplicity. The burning of ${}^2_1\text{H}$ – deuterium – occurs for a short fraction of the lifetime of the star, before core hydrogen burning begins in earnest. This energy release will sustain the star throughout its lifetime, providing outwards pressure resisting gravity pulling the star inwards, in a balance termed *hydrostatic equilibrium*. Once hydrogen fusion has begun, we say that the star has reached ZAMS, and is now on the *main sequence*, where it will spend approximately 90% of its lifetime.

2.2 | Stellar Evolution

It did not last: the Devil howling “Ho! Let Einstein be!” restored the
status quo.

Answer to Pope’s Epitaph for Sir Isaac Newton, BY J. C. SQUIRE

Now that our protostar has evolved onto the main sequence, it has become a star in its own right. Over the next few million years, it will continue to burn hydrogen into helium, forming a helium core. It is at this point that the evolution encounters a number of bifurcation points, depending on the mass and chemical composition of the star. In general, while the star lives, the four equations of stellar structure hold:

$$\left. \begin{aligned}
\frac{dm}{dr} &= 4\pi r^2 \rho && \text{(mass continuity),} \\
\frac{dp}{dr} &= -\frac{GM\rho}{r^2} && \text{(hydrostatic equilibrium),} \\
\frac{dT}{dr} &= -\frac{3\kappa\rho L}{64\pi r^2\sigma T^3} && \text{(energy transport), and} \\
\frac{dL}{dr} &= 4\pi r^2 \rho \epsilon && \text{(thermal equilibrium).}
\end{aligned} \right\} \quad (2.3)$$

where ϵ is the rate of energy energation per unit mass, κ is the opacity of the material, σ is the Stefan-Boltzmann constant, m is the mass internal to radius r , p is pressure, ρ is density, M is total mass of the star, and L is the luminosity.

In addition to equation (2.3), an Equation of State (EOS) demanding $\rho = \rho(P, T)$ is desired. We take our EOS from Pols et al. (1995), which built upon that in Eggleton et al. (1973). Critically, Pols' EOS includes the dissociation of molecular hydrogen, Coulomb interactions, and pressure ionization, which Eggleton et al. (1973) lacked. The EOS is an approximation to Fermi-Dirac statistics, and originates from the free energy

$$F(V, T, N_e, N_e) = -\frac{1}{3}aT^4V + kT \sum_i N_i \left(\ln \frac{N_i h^3}{\omega_i V (2\pi m_i kT)^{\frac{3}{2}}} - 1 + \frac{\chi_i}{kT} \right) + F_e(N_e, V, T) + \Delta F_e(N_e, V, T) \quad (2.4)$$

These equations represent conservation of mass, hydrostatic equilibrium, energy transport, and thermal equilibrium, respectively. Knowing constants ϵ, κ , we have four equations with four unknowns. It is usually convenient to use the *Rosseland mean* opacity,

$$\frac{1}{\kappa} = \frac{\pi}{acT^3} \int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B}{\partial T} d\nu \quad (2.5)$$

where a is the radiation density constant ($a = 7.57 \times 10^{-15} \text{ erg cm}^{-3} \text{ K}^{-4}$) and B is the black-body intensity at frequency ν and temperature T . The function κ_ν is obtainable from atom-electron-photon interaction (Eldridge & Tout, 2004a): bound-bound, bound-free, free-free (bremsstrahlung) transitions, and electron scattering. κ_ν is thus related to the scattering cross-section $\sigma_\nu = \int_0^{2\pi} \int_0^\pi \frac{d\sigma}{d\Omega} \sin \theta d\theta d\phi$. The Rosseland mean becomes

$$\kappa = \frac{T^5 ac^3 k_B}{\pi h^2 \int_0^\infty \frac{\nu^4}{\left(\cosh\left(\frac{h\nu}{T k_B}\right) - 1 \right) \kappa_\nu} d\nu}, \quad (2.6)$$

though the integral cannot be analytically solved further. On the other hand, ϵ is obtainable from knowing the nuclear reactions occurring in the star's core:

$$\epsilon = \sum_i^N \text{energy output of reaction } i. \quad (2.7)$$

[JJE: the energy generation rates come from nuclear physics! normally from calculations and direct measurement - e.g. the NACRE reaction rates]

All told, in the simplest sense, the internal *structure* of the star can be modelled by hand by solving equation (2.3) from $m = 0$ to $m = M$.

2.2.1 | The chemical composition of stars

There's a potion that can make you shrink.

FIRST OF THE SIX IMPOSSIBLE THINGS

A star contains at its core a complex nuclear furnace. Since stars are born in hydrogen-rich regions of space

and obtain high enough pressures in the core to ignite fusion, the star will be sustained by the burning of predominantly hydrogen and helium. We thus group the elements into hydrogen, helium, and metals (which we define as anything with atomic number ≥ 3 , i.e., lithium and heavier elements). It is convenient to speak of the mass fraction of the element, i.e., the fraction by mass of a star that is a certain element. For our purposes, this is expressible as

$$X + Y + \sum_{i=0}^{115} \mathcal{Z}_i = 1, \quad (2.8)$$

where X is the mass fraction of hydrogen, Y of helium, and $\mathcal{Z}_i$ as the metal¹ with atomic number $i + 3$.

[JJE: What about isotopes?] When speaking about the chemical composition of a star it is also convenient to speak of the *metallicity*, the fraction at ZAMS of the star that is metal. For this, we define the metallicity as

$$Z \equiv \sum_{i=0}^{115} \mathcal{Z}_i, \quad (2.9)$$

such that equation (2.8) becomes

$$X + Y + Z = 1. \quad (2.10)$$

2.2.2 | Evolutionary Pathways

In this section, we examine the three major regimes of stellar evolution for single-stars: low-mass (?? 3), intermediate-mass (?? 3), and massive stars (?? 3). We do not consider stars about $M_{\text{ZAMS}} \approx 80 M_{\odot}$ in this thesis **[TODO: why?]**

Low-Mass Stars: $0.08 M_{\odot} < M < 2 M_{\odot}$

Stars below $\approx 0.5 M_{\odot}$ do not ignite helium, and become helium WDs. Above this, they ignite helium degenerately in their cores, leading to thermal runaway. The star can peak at a luminosity close to $10^{11} L_{\odot}$ (approximately 100 times the peak luminosity of SN 1987A, per Figure 5 of Arnett et al. (1989)). However, this peak in luminosity is attributable to nuclear fusion in the core, and the energy does not reach the surface, let alone escape into the ISM – rather, it is absorbed into the outer layers of the star. During this extremely brief thermal runaway, the luminosity begins to settle and the star is able to continue evolving. Models of low mass stars fail at this stage and care is required to evolve them. This phenomena is called the *helium flash* and is discussed further in ??.

After the ignition of helium, these stars cross the so-called Hertzsprung Gap – the region where very few stars are observed empirically. This region is bounded (on a colour-magnitude diagram, which is a transform of the HR diagram) by the points $(M_v, B - V) = \{(2, 0.5), (-7, 0.1), (-6, 1.6)\}$ (Hoyle, 1960), where M_v is the V-band magnitude (magnitude in the visual band), and $B - V$ is the colour index of the star. The phenomenon of a gap existing is attributable to the fact that stars cross this gap exceedingly quickly: whereas it spends $\sim 90\%$ of its life on the main sequence, it spends $\sim 2\%$ of its life crossing the Hertzsprung gap (Hoyle, 1960).

¹It is custom in astronomy to refer to anything with an atomic number greater than 2 as a metal, rather than the conventional definition of a material conductive at absolute zero, including most of the d-block, s-block, f-block, and about half of the p-block on the periodic table.

Intermediate-Mass Stars: $2 M_{\odot} < M < 8 M_{\odot}$

Intermediate mass stars exist above the threshold for the helium flash to occur. Helium ignition is not degenerate and the development of a helium core is not impeded. As the star ascends the Asymptotic Giant Branch (AGB), the core becomes helium-exhausted and it becomes a red giant, with a radius many hundreds or even thousands of times greater than the Sun. Its mass, sustained prior to the AGB via nuclear fusion, begins to decrease. An example of intermediate-mass evolution is shown in figure 2.1.

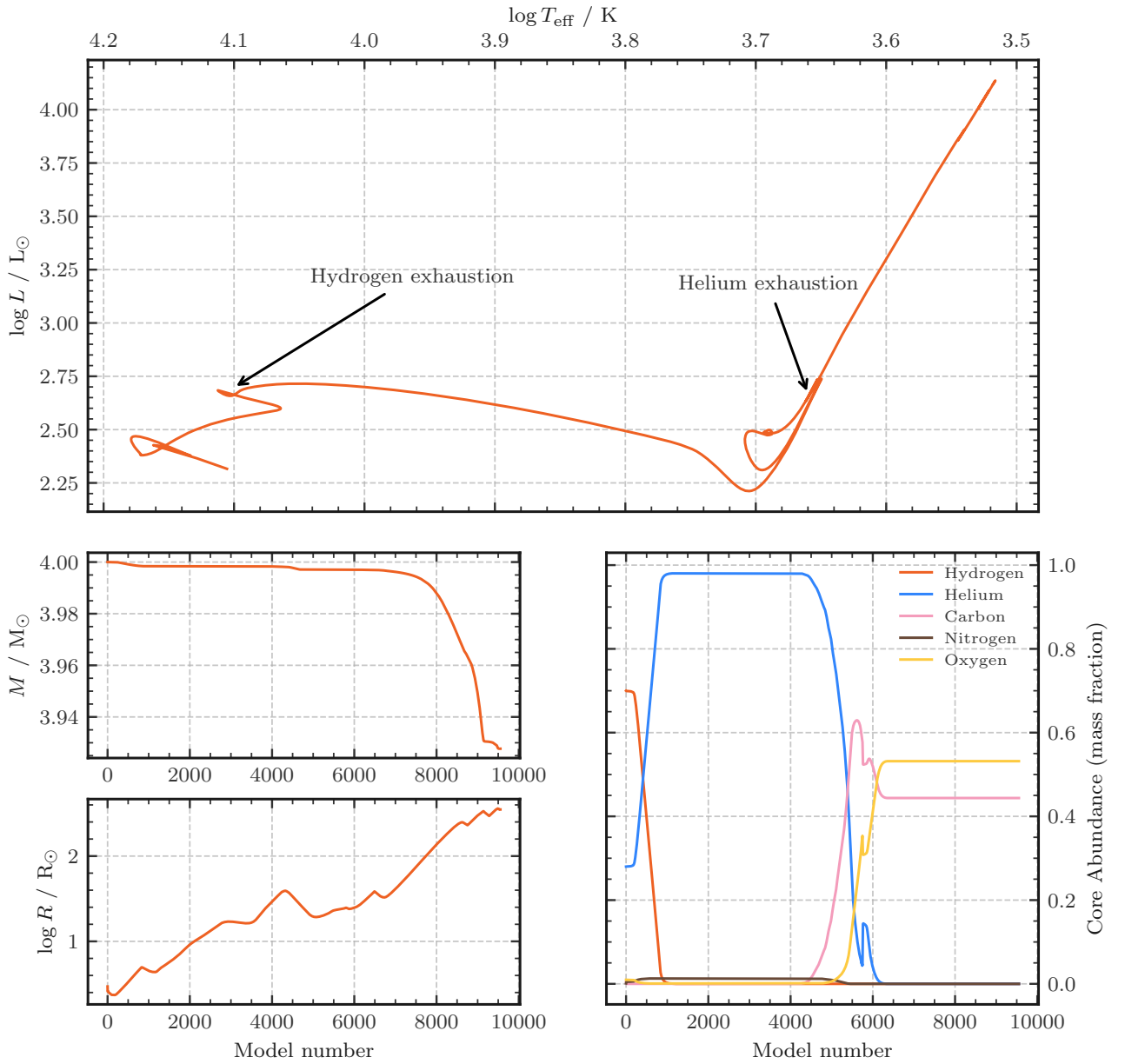


Figure 2.1: The evolution of an $4 M_{\odot}$ star from ZAMS until CO core development. The final core abundances are $Z_{\text{C}} \approx 0.44$ and $Z_{\text{O}} \approx 0.53$.

Massive Stars: $8 M_{\odot} < M < 80 M_{\odot}$

Massive stars are the key focal objects of this thesis, and in many ways offer some of the most exciting phenomena we can probe. In figure 2.2 we show the variety of evolution that this regime has, alongside the ZAMS track from $8 M_{\odot}$ to $90 M_{\odot}$. The key features are as follows. When a massive star exhausts its supply of hydrogen in its core and evolves off the main sequence to begin burning helium, it generally bypasses the Hertzsprung gap and evolves up the Red Giant Branch (RGB). From here, stars below $\approx 15 M_{\odot}$ generally engage in a “blue loop”, where it remains at a roughly constant radius whilst increasing in temperature, before cooling at a slightly higher luminosity. The question of what physical process powers a blue loop is unanswered (Zhao et al., 2023), [JJE: we know, read Walmswell & Eldridge (201x?)] though this thesis is unconcerned with their generation. We note that our models include blue loops, as shown in figure 2.2 and figure 2.3.

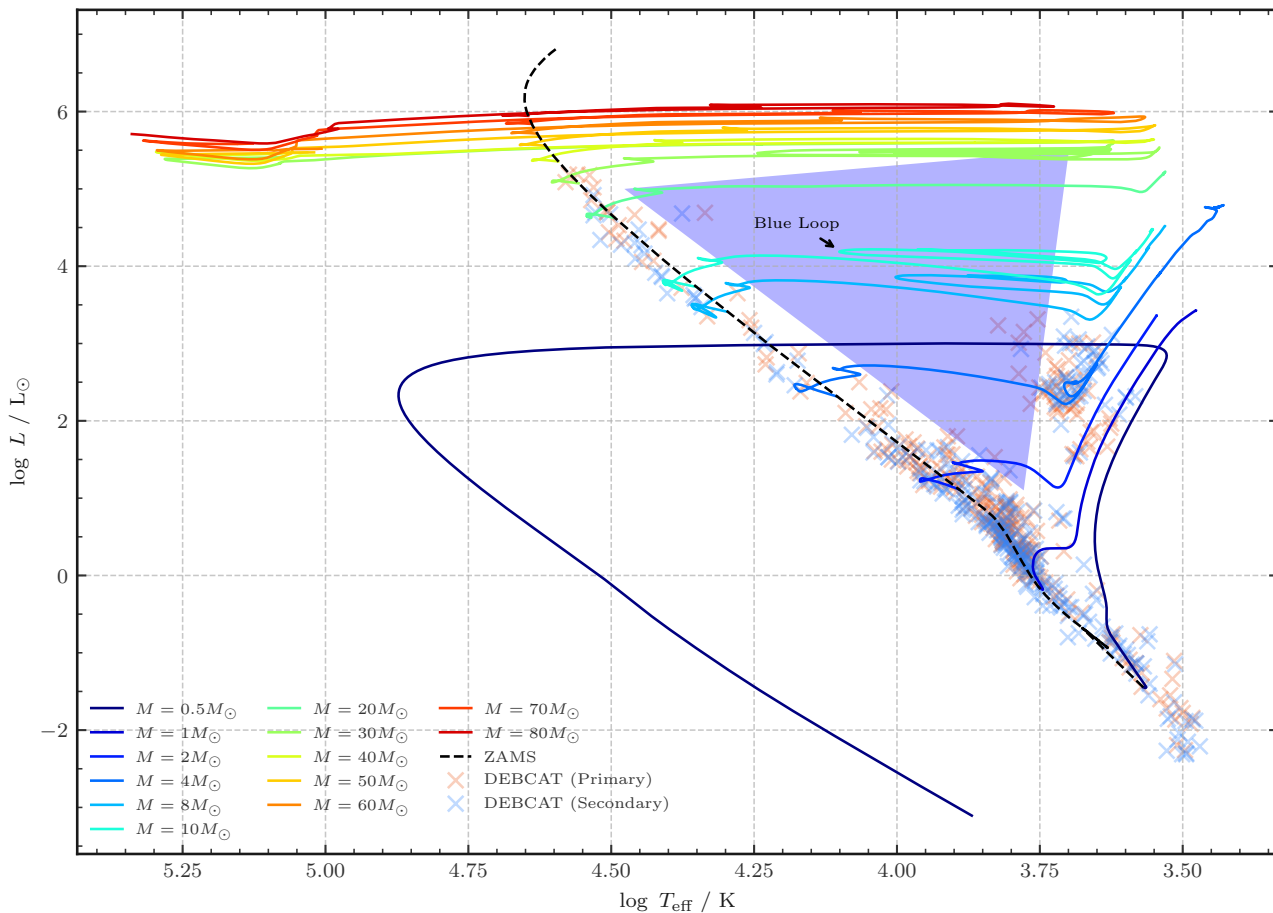


Figure 2.2: The evolution of stars in the range $0.5 - 80 M_{\odot}$. The dashed black line represents the Zero-Age Main Sequence (ZAMS), which is the origin point of all of our models, and (in this plot) extends from $0.5 M_{\odot}$ to $300 M_{\odot}$. The blue triangle is the (highly approximate) bounds of the Hertzsprung gap. Crosses indicate binaries in the Double Eclipsing Binaries catalogue (DEBCAT; Southworth, 2015), and illustrate the apparent “zone of avoidance” that defines the Hertzsprung gap.

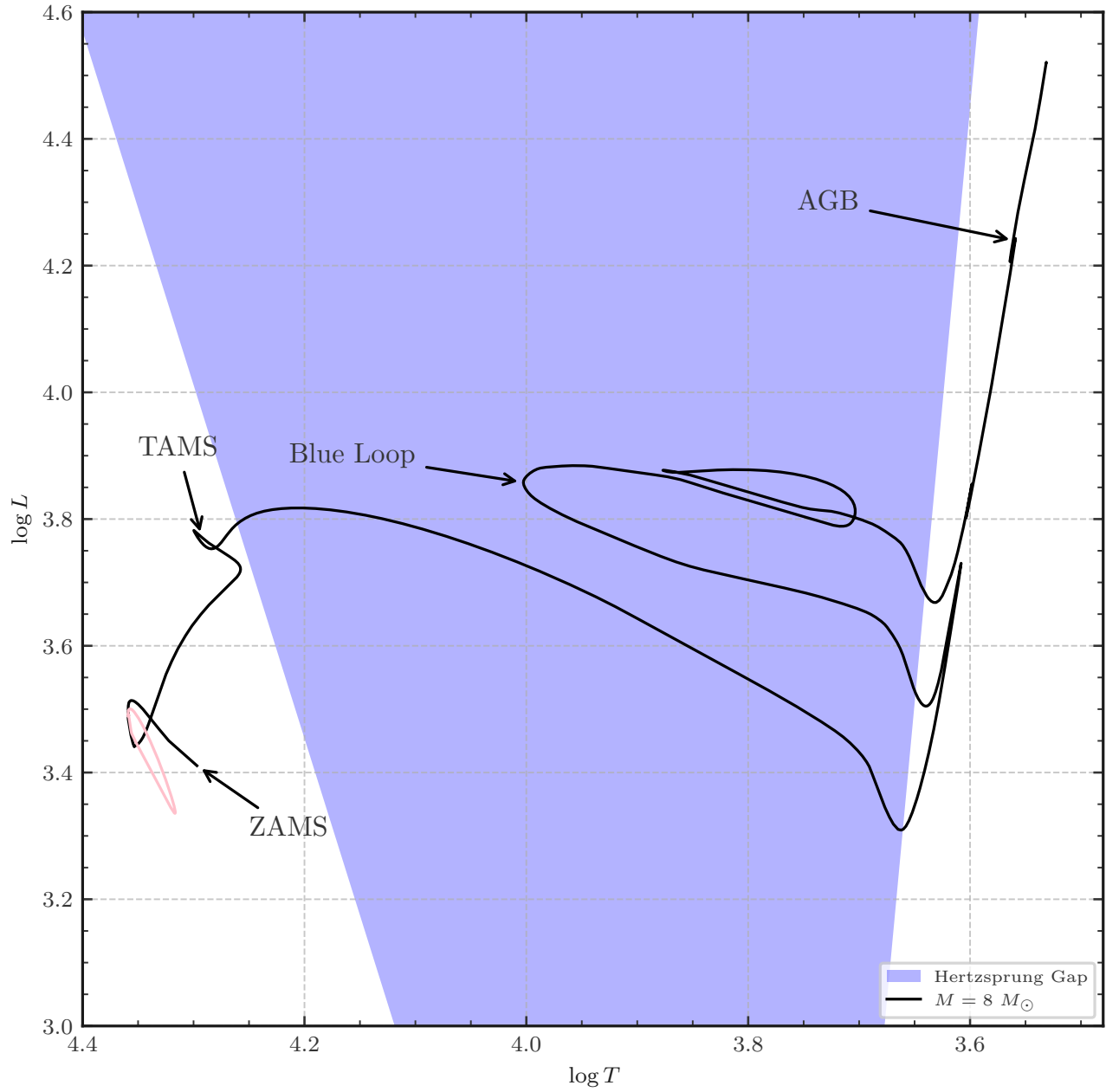


Figure 2.3: The evolution of an $8 M_{\odot}$ star. Highlighted is the star's Zero-Age Main Sequence (ZAMS), Terminal-Age Main Sequence (TAMS), the blue loop, and the Asymptotic Giant Branch (AGB). The Hertzsprung gap is also highlighted. In this particular model, numerical instabilities in the leftmost (hottest) region of the plot have been highlighted in pink. This is normally removed in postprocessing though has been left in as an illustration.

2.2.3 | Relativistic corrections

There's a trick you can use in mathematics called "not worrying about it"

MATT PARKER

This analysis has presumed Newtonian mechanics. However, the mass of stars puts them formally into the general relativistic regime, and thus strictly speaking we should treat this as a problem in full GR. In this approach, we would solve the Einstein Field Equations for the interior of the star:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \quad (2.11)$$

with cosmological constant Λ , stress-energy $T_{\mu\nu}$, and the Einstein tensor $G_{\mu\nu}$ given by

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \quad (2.12)$$

$$= R^\lambda_{\mu\lambda\nu} - \frac{1}{2} g_{\mu\nu} R^\sigma_\sigma \quad (2.13)$$

$$= R^\lambda_{\mu\lambda\nu} - \frac{1}{2} g_{\mu\nu} g^{\sigma\gamma} R_{\sigma\gamma} \quad (2.14)$$

$$= \sum_\lambda R^\lambda_{\mu\lambda\nu} - \frac{1}{2} g_{\mu\nu} \sum_\sigma \sum_\gamma \sum_\alpha g^{\sigma\gamma} R^\alpha_{\sigma\alpha\gamma} \quad (2.15)$$

$$= \sum_\lambda \left(\Gamma^\lambda_{\mu\nu,\lambda} - \Gamma^\lambda_{\mu\lambda,\nu} + \sum_\xi \Gamma^\lambda_{\xi\lambda} \Gamma^\xi_{\mu\nu} - \sum_\xi \Gamma^\lambda_{\xi\nu} \Gamma^\xi_{\mu\lambda} \right) - \frac{1}{2} g_{\mu\nu} \sum_\sigma \sum_\gamma \sum_\alpha g^{\sigma\gamma} \left(\Gamma^\alpha_{\sigma\gamma,\alpha} - \Gamma^\alpha_{\sigma\alpha,\gamma} + \sum_\xi \Gamma^\alpha_{\xi\alpha} \Gamma^\xi_{\sigma\gamma} - \sum_\xi \Gamma^\alpha_{\xi\gamma} \Gamma^\xi_{\sigma\alpha} \right), \quad (2.16)$$

where there is still tensorial shorthand:

$$\Gamma^\gamma_{\beta\mu} \equiv \frac{1}{2} \sum_\alpha g^{\alpha\gamma} (g_{\alpha\beta,\mu} + g_{\alpha\mu,\beta} - g_{\beta\mu,\alpha}), \quad (2.17)$$

and the comma-derivative notation is

$$\phi_{,\mu} \equiv \frac{\partial \phi}{\partial x^\mu}, \quad (2.18)$$

such that, for example,

$$g_{\alpha\beta,\mu} = \frac{\partial g_{\alpha\beta}}{\partial x^\mu}.$$

In equation (2.15), we explicitly reinsert the summation symbols to illustrate the number of simultaneous summations that are taking place implicitly. As one can see, this reasonably simple expression balloons quickly into something intractable by hand and time-consuming to evaluate computationally. Although the metric ten-

space is generally four-dimensional,² such as the Friedmann-Lemaître-Robertson-Walker metric (Friedmann, 1922; Lemaître, 1931; Robertson, 1935; Walker, 1937) describing the expansion of the Universe,

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a(t)^2 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix},$$

it often lacks a closed form, especially on large-scale chaotic behaviour that cannot be locally “washed out” (such as how on Universal scales, individual stars can be ignored to describe the homogeneity and isotropy of the Universe), and is obtained via numerical methods.

This is also not considering complications of writing the stress-energy tensor $T_{\mu\nu}$, which may have its own complex form (again, assuming a closed-form solution *exists*) determined via fluid-mechanical processes. All told, there are 64 equations to be solved instead of 4. In addition, the 64 solutions are “worse” to solve: harder to integrate by hand, and requiring greater time complexity. Numerical relativity codes typically simulate one star for the order of kiloseconds of its lifetime (see, for an example, ud-Doula et al., 2013), and even the most efficient codes take longer than our codes to run. For instance, the PHANTOM 3D MHD code (Price et al., 2018) can take several days to run high-resolution models. Consequently, for the number of stars we wish to simulate, we choose to adopt a Newtonian framework.

2.2.4 | Stellar Atmospheres

As a final note in the evolution of a single star, we must briefly address what occurs about the photosphere of the star. A star is not a planet – there is no firm boundary demarcating the inner of a star from the outer. The common definition of the “surface” of a star is the surface of last scattering (photosphere), which is the point radially in which the mean free path ℓ of a photon emitted from the core of the star becomes meaningfully long (internally, $\ell_\gamma \sim 2$ cm, externally, $\ell_\gamma \sim 4.4$ pc).³ However, outflows from the star as well as the potential interaction of a partner in a common-envelope event mean that this surface is not necessarily firm: there is no sharp turn from stellar material to empty space. Thus, the simulation of stellar atmospheres is of interest to the stellar astrophysicist. However, this introduces another rather large layer of complexity. The synthesis of a stellar spectra, for example, is not a trivial task, as well as simulating the matter content. This thesis is concerned with detailed modelling of the stellar *interior*, and thus the atmosphere is an interesting sidenote, but one we shall not pursue.

²And thus the summation indices run from 1 to 4

³The mean free path is a function of the particle under consideration: for a neutrino, internally $\ell_\nu \sim 3.68$ pc and externally $\ell_\nu \sim 81$ Rpc $\sim 3 \times 10^{18}$ times the diameter of the observable Universe.

2.3 | Supernovae

And on the pedestal these words appear:
 “My name is Ozymandias, king of kings:
 Look on my works, ye Mighty, and despair!”
 Nothing beside remains.

PERCY BYSSHE SHELLEY, *Ozymandias*

Pallida mors æquo pulsat pede pauperum
 tabernas regumque turres.
 Pale death knocks with the same tempo upon the
 huts of the poor and the towers of kings.

HORACE

A star’s lifetime is not infinite. The key assumption underpinning stellar evolution is that of hydrostatic equilibrium: the counterplay of gravity against outwards pressure. Whilst this equilibrium holds, the star is stable and resists explosion. It is when this equilibrium is broken – with gravity the dominant force in this scenario – that the star implodes in a CCSN. From a thermodynamic perspective, as the star evolves, it shrinks as energy is bled off into the interstellar medium. This shrinkage compresses and heats the core, meaning the star’s specific heat capacity is negative. This system is thermodynamically unstable, and thus every star either collapses into a remnant (cf. black holes, neutron stars) or is sustained in some form by quantum mechanical phenomena (cf. White Dwarfs, Thorne-Żytkow objects, and other exotica).

In any case, the breaking of hydrostatic equilibrium results generally results in⁴ a supernova, which can be classified into subtypes, as shown in figure 2.4. In this event, gravity pulls the stellar core inwards, before “bouncing” and ejecting the envelope into the interstellar medium. The physics of this approximately second-long event is highly uncertain, and is in general unable to be modelled by detailed stellar evolution codes. All models we generate – assuming they have not been terminated earlier – will terminate at the moment of supernova.

Importantly for the nucleosynthesis of heavier-than-Iron elements (which cannot be formed via fusion,⁵ as shown in figure 2.5), the supernova shockwave propagating through the core causes a new phase of nucleosynthesis, albeit briefly.

2.4 | Two to tango: Binary Evolution

Introducing a second star into the mix complicates the evolution. Whereas before a star losing mass (through, for example, a coronal mass ejection (CME)) was assumed to eject it into the ISM, it may now accrete that material directly onto its companion. In addition, either star may in principle increase in radius enough to overlap with its companion, or even touch surfaces – introducing new dynamics into the mix.

⁴The extent of the veracity of this statement is an area of active research: Reynolds et al. (2015) presents a short summary.

⁵Strictly speaking, fusion can occur for elements heavier than Iron-56, but the reaction is endothermic; it requires more energy than is released, meaning it is unsustainable.

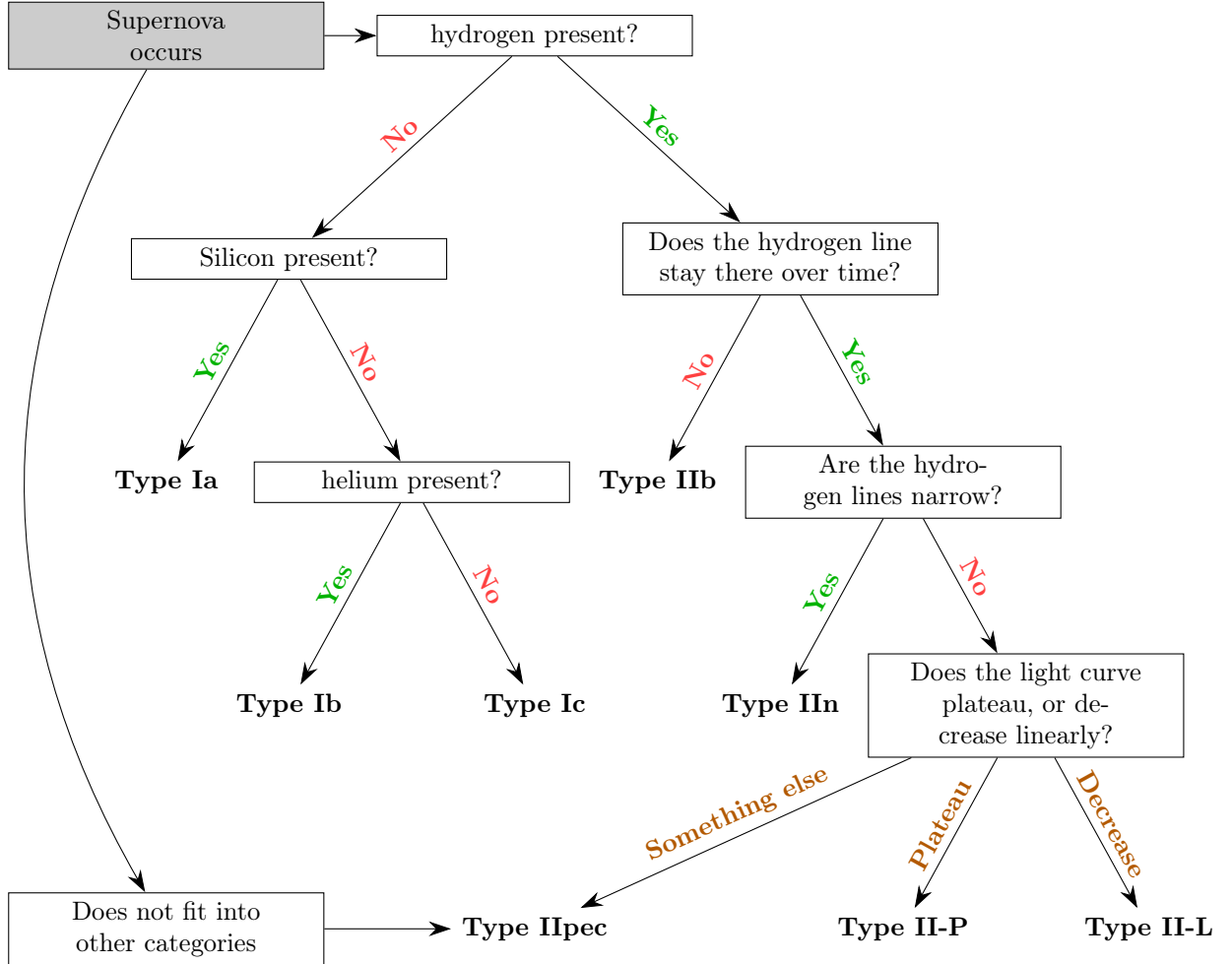


Figure 2.4: A rough outline of supernova classification. One can see that the presence of hydrogen lines distinguishes a Type I from Type II supernova. Type II supernova subtypes are determined by the behaviour of their light curves – how the apparent magnitude changes against time – and thus are determined by the luminosity of the supernova, something detailed models are not able to model. On the other hand, the presence of hydrogen/silicon/helium in the supernova is something that detailed codes *are* able to model.

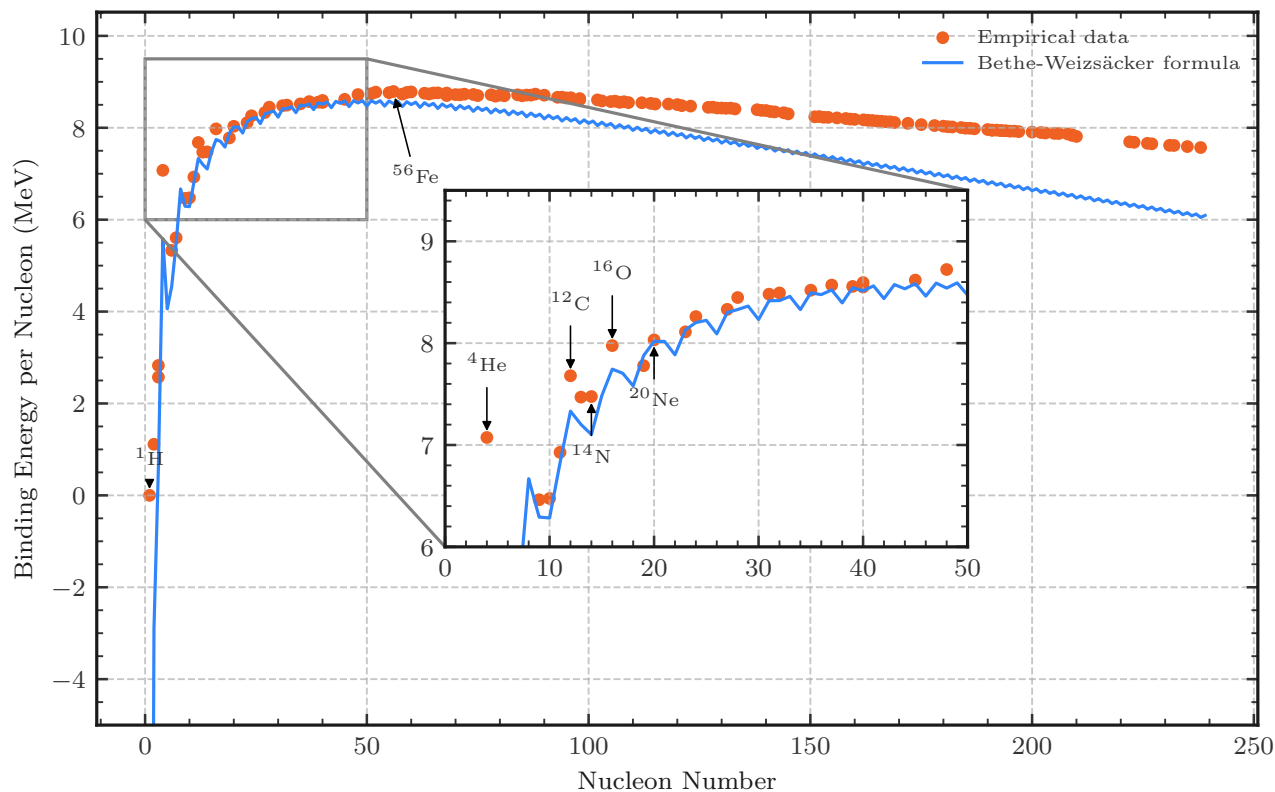


Figure 2.5: The binding energy per nucleon for elements up to Uranium-238. The most stable element of fusion here is Iron-56. Elements lighter than Iron-56 will fuse towards it, and elements heavier will undergo fission to reach it. The Bethe-Weizsäcker formula is the semi-empirical mass formula devised by Bethe and Weizsäcker to describe the liquid drop model of Gamow (1930). The marked atoms on the plot represent the six elements we generally consider in our stellar modelling code – ^1H , ^4He , ^{12}C , ^{14}N , ^{16}O , ^{20}Ne – with the balance assumed to be insufficiently energetic to alter the structure or evolution of a star (Stancliffe et al., 2004). Graph adapted from Fastfission on Wikimedia Commons (who released it into the public domain).

2.4.1 | Binary star kinematics

There are several formation channels for a binary star system. Binaries may be formed from birth (as discussed in section 2.1, with the caveat that in this scenario two clumps of gas form near each other), via fragmentation of a protostar’s disc, dynamical capture of one star by another (comparable to one of the suggested origins of Phobos and Deimos, per Burns (1978)), or via a higher multiplicity system ejecting one or more of its constituent stars, perhaps via a natal supernova kick.

In any case, once formed and the orbit is stabilised, binary systems obey Kepler’s three laws. In the generalised binary regime, however, the assumption that $M_1 \gg M_2$ (which underpins most statements of Kepler’s laws) is not necessarily true – for instance, the system LSS 3074 has two stellar components with ZAMS masses of approximately $30 M_\odot$ and $10 M_\odot$ (Raucq et al., 2017). Moreover, the fact that there are two stars in orbit rather than one object orbiting another means that Kepler’s laws must be stated more precisely. Formally, they can be expressed as (see the appendix of Richards (2021) for derivations):

1. All orbits are ellipses, with both stars orbiting a common barycentre at the focus of the ellipse,
2. A line segment joining the two stars sweeps out equal areas in equal times,
3. The square of the period of the system is proportional to the cube of the semimajor axis of the ellipse (the binary separation), with the proportionality term dependent on the sum of the masses.

The third law can be stated more precisely as

$$T^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3, \quad (2.19)$$

where T is the period of the binary, M_1 and M_2 the constituent masses, $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the universal gravitational constant, and a is the semimajor axis of the binary. A schematic of the orbit of a binary system is given in figure 2.6.

2.4.2 | Roche Lobes

A binary system is complicated by the presence of its companions in several different ways. As alluded to in the previous section, the presence of a companion warps space-time in such a way that material can be quite distorted from a sphere or oblate spheroid but still remain gravitationally bound to the system. In the extremal case, a star can grow to fill or even exceed its Roche Lobe, which is the tear-drop shaped equipotential surface within which material remains gravitationally bound to its star. In figure 2.7 we display these equipotential surfaces for a range of different mass ratios. It is pertinent to note that the shape of all of the Roche Lobes change rapidly at low mass ratios, however at higher mass ratios the shapes change less rapidly. In an equal-mass ratio binary, the binary configuration has rotational symmetry about the $\mathcal{L}_1$ point, and reflectional symmetry about the x -axis and the line $y = \mathcal{L}_1$.

There are two primary Roche Lobe points we consider: the $\mathcal{L}_1$ point and the $\mathcal{L}_2$ point. The $\mathcal{L}_1$ point (also known as the inner Lagrangian point) is the nexus between the two stars, the $\mathcal{L}_2$ point (the outer Lagrangian

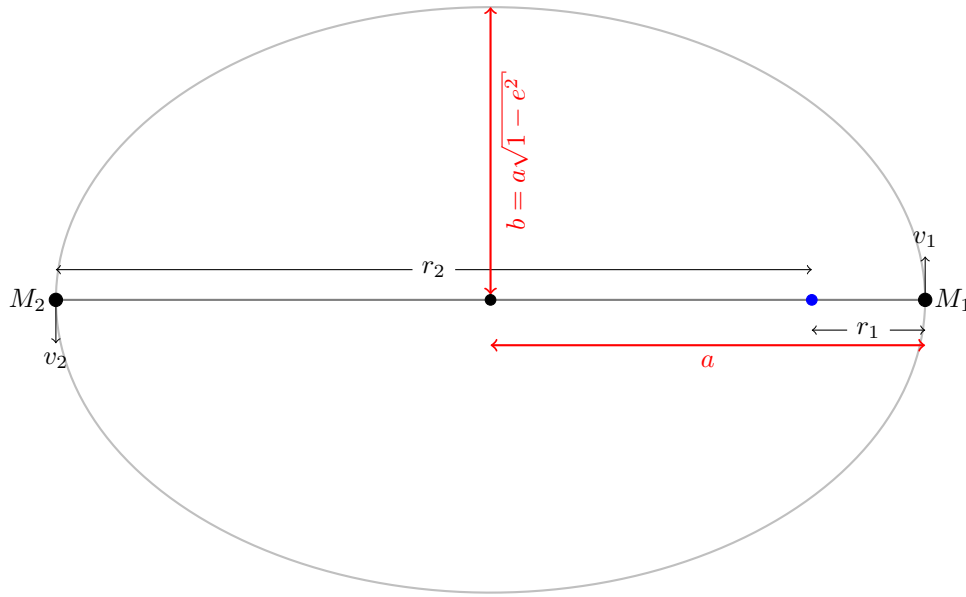


Figure 2.6: A schematic of a simple binary star system with eccentricity e . The centre of mass (barycentre) is shown as a blue dot. Here, M_1 is the heavier star, which orbits at a smaller distance r_1 to the barycentre by Kepler's first and Newton's second laws. The elliptical orbital trajectory is shown in grey. Figure adapted from Richards (2021).

point) is located on the other side of the secondary to the inner point. The dimensionless x -ordinate of the $\mathcal{L}_1$ point, ξ , measured in terms of the separation a , is given as the solution to

$$0 = \xi - \frac{\xi_2}{(\xi - \xi_1)^2} - \frac{\xi_1}{(\xi - \xi_2)^2}, \quad (2.20)$$

where $\xi_1 = \frac{M_2}{M_1 + M_2}$, and $\xi_2 = \frac{M_1}{M_1 + M_2}$.

This is a quintic expression in ξ , and solving this results in the real ξ -ordinates of five extrema. The extrema are labelled $\mathcal{L}_1$ through $\mathcal{L}_5$. $\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3$ are saddle points of the potential, and $\mathcal{L}_4$ and $\mathcal{L}_5$ are maxima. No analytical function for ξ exists unless either or both of ξ_1, ξ_2 are zero (which would only be the case for a single star, and thus the Roche lobe would not exist in the first instance), thus we must make approximations.

The $\mathcal{L}_1$ point

The inner Lagrangian point has the approximation from Eggleton (1983), which is

$$\frac{R_L}{a} \approx \frac{0.49q^{\frac{2}{3}}}{0.6q^{\frac{2}{3}} + \ln\left(1 + q^{\frac{1}{3}}\right)}. \quad (2.21)$$

We show the circle with this radius R_L in figure 2.8 plotted against the exact solution for $\eta = \mathcal{L}_1$. Eggleton's formula is a volume-equivalent one; although his radius $R_L < \xi$, the volume of an Eggleton-Roche sphere $\frac{4}{3}\pi R_L^3$ is approximately equal (to within 1 per cent) of the numerically integrated volume, as is shown in figure 2.9.

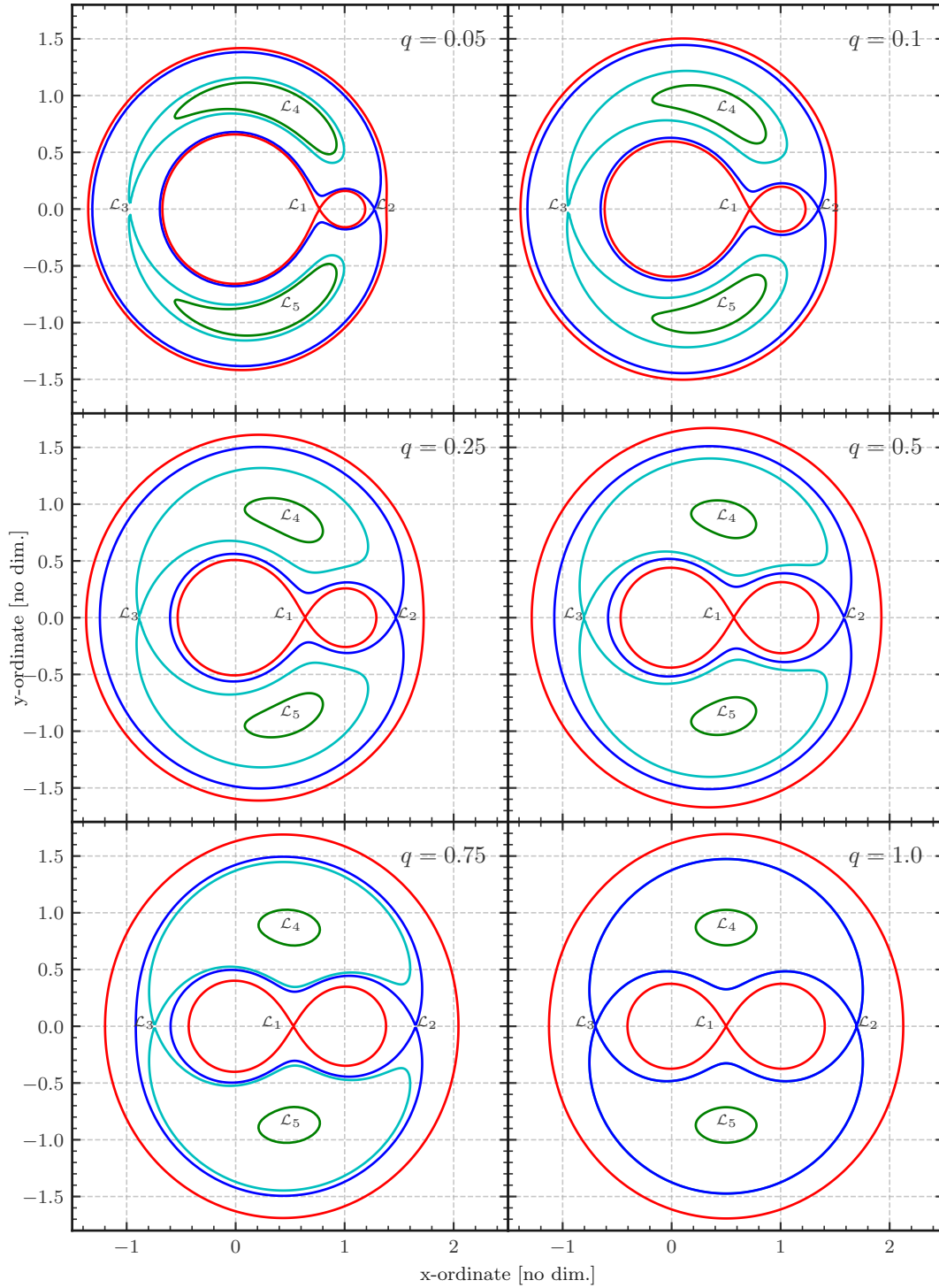


Figure 2.7: The dimensionless Roche potential as a function of mass ratio q . Note that the shape of the Roche potential changes markedly at low mass ratio, and changes more slowly as the mass ratio approaches unity. Distances have been normalised such that the primary star is at $(x, y, z) = (0, 0, 0)$ and the secondary star is at $(x, y, z) = (1, 0, 0)$, i.e., we measure distances in units of separation and that the coordinate system is centered on the primary star.

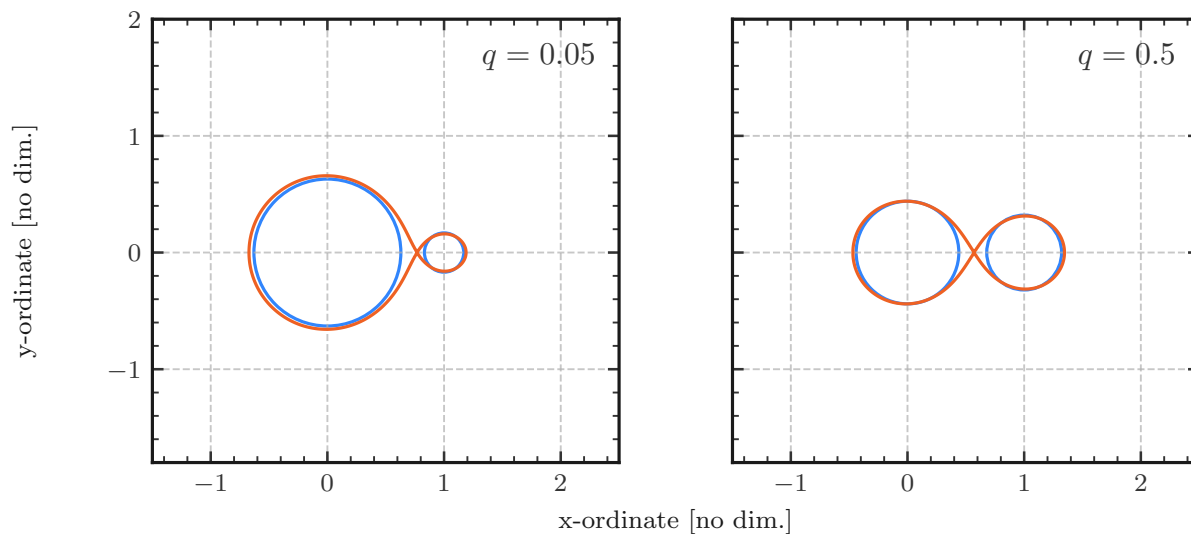


Figure 2.8: A schematic of the Roche lobe formula equation (2.21). Here, the exact solution to the Roche lobe formula ξ in equation (2.20) (orange) vs the fit provided by Eggleton (1983) (blue). Although precise details of the Roche lobe, such as the apex of the teardrop at the point ξ , are lost, the volume is preserved to within 1 per cent accuracy over all mass ratios.

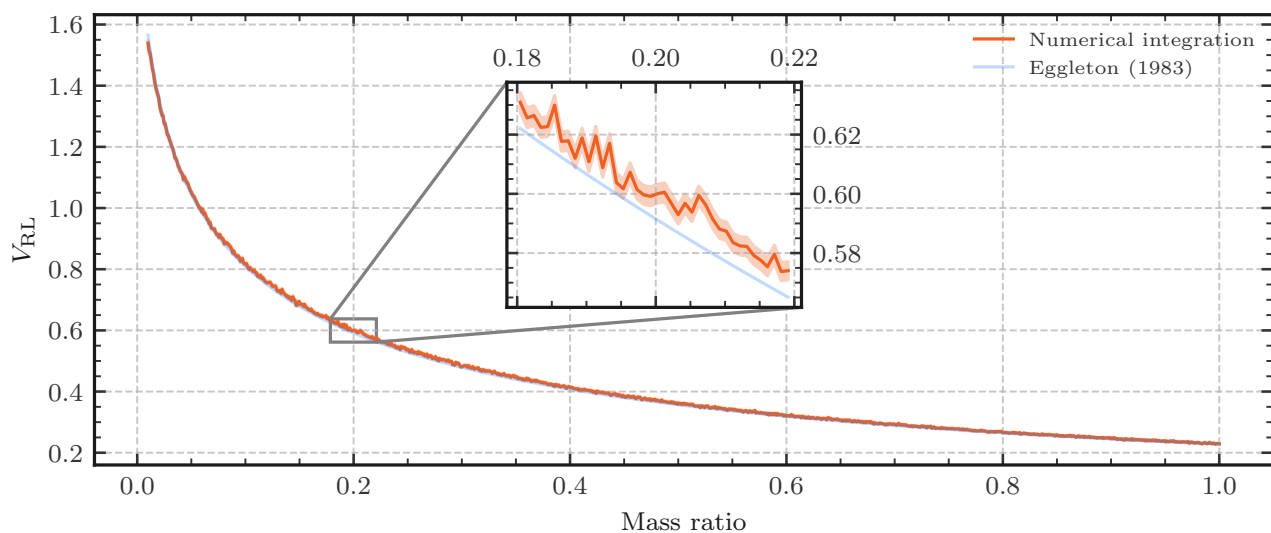


Figure 2.9: The volume of the Roche Lobe computed numerically through Monte-Carlo integration (orange) vs the volume assuming equation (2.21) (blue). The Eggleton approximation fits well to the volume, with around 1 per cent error across the entire mass ratio range (truncated here to $q \leq 1$).

The $\mathcal{L}_2$ point

The outer Lagrangian point, $\mathcal{L}_2$, is given by a different zero of the potential function. The approximation we use is that of Marchant et al. (2016), where

$$R_{L2} = R_L(q) \left(1 + \frac{0.299 \arctan(1.84q^{-0.397})}{q^{0.520}} \right), \quad (2.22)$$

and $R_L(q)$ is given by equation (2.21).

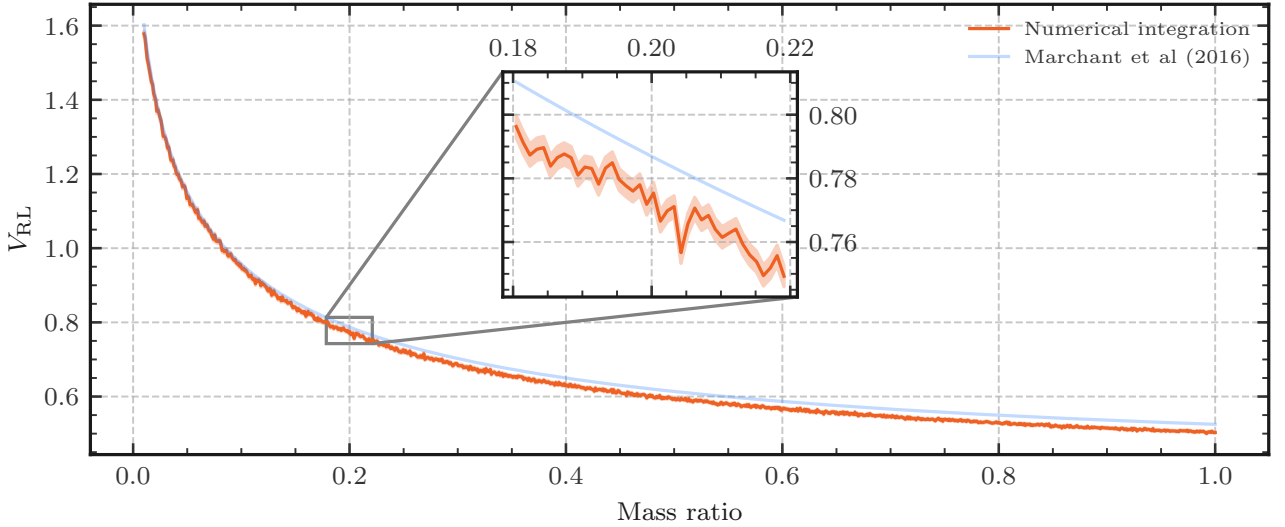


Figure 2.10: As for figure 2.9, but for the $\mathcal{L}_2$ point with formula given by equation (2.22) (Marchant et al., 2016).

We note that Marchant et al. (2016) uses an inverse mass ratio definition compared to Eggleton (1983), that is, Marchant et al. (2016) takes $q_{\text{Marchant}} = \frac{M_1}{M_2}$ whereas Eggleton (1983) takes $q_{\text{Eggleton}} = \frac{M_2}{M_1} = \frac{1}{q_{\text{Marchant}}}$. This is accounted for in equation (2.22) such that the mass ratios are consistent (and we take $q = q_{\text{Eggleton}}$), but other texts do not necessarily differentiate or describe the difference.

2.4.3 | Mass Transfer

And a cake that can make you grow.

SECOND OF THE SIX IMPOSSIBLE THINGS

When two stars are sufficiently close in proximity, events such as CMEs can cause two stars to exchange mass (whereas in the single-star case, this would be matter lost to the ISM). The nature of mass transfer is a complicated affair; most binary star population synthesis codes typically inherit their mass loss prescriptions from papers such as Bondi & Hoyle (1944); Hurley et al. (2002), and this approach is what we detail in this section.

Mass transfer can broadly be sectioned into four main categories:

1. $\dot{M}_{\text{wind}}$, where matter is transferred through stellar winds,
2. $\dot{M}_{\text{RLOF}}$, where matter is transferred through one (or more) of a binary's Lagrange points (cf. section 2.4.2),

3. $\dot{M}_{\text{acc}}$, where matter is transferred through accretion (or interaction with a disk), and
4. $\dot{M}_{\text{Bondi}}$, where matter proceeds through a Bondi & Hoyle (1944) type mechanism.

Our mass loss rate for each star is constructed from these four terms as

$$\dot{M} = \dot{M}_{\text{wind}} - \dot{M}_{\text{Bondi}} + \dot{M}_{\text{RLOF}} - \dot{M}_{\text{acc}}. \quad (2.23)$$

Wind mass-loss

Wind mass loss computations are (generally) performed using the prescription of de Jager et al. (1988) and Vink et al. (2001). The precise scheme is detailed in Eldridge et al. (2017).⁶ There are two critical modifications we make to their scheme.

The first modification surrounds the transition into the Wolf Rayet (WR) regime. Traditionally this has been a sharp transition when the surface hydrogen mass fraction becomes less than 0.4 and the surface temperature is above 10^4 K. This is a reasonable approximation, though if a star reaches this phase and is *not* yet a WR star, the code will fail. To account for this we gradually turn on these enhanced WR rates when this condition approaches, smoothing the transition.

The second modification we have made is that if a star has a hydrogen exhausted core and begins approaching the WR temperature range, we assume that WR mass loss must begin and thus turn on the slowly-turning on WR regime.

Mathematically, the computation for $\dot{M}_{\text{wind}}$ is more involved, though follows the same essential structure as Vink et al. (2001). We construct it as follows. The total pre-WR mass loss is given by de Jager et al. (1988), namely

$$\log \dot{M}_{\text{de Jager}} = - \sum_{n=0}^5 \sum_{\substack{i=0 \\ j=n-i}}^n a_{ij} \cos(i \arccos x_1) \cos(j \arccos x_2), \quad (2.24)$$

$$\dot{M}_{\text{pre-WR}} = \sqrt{\frac{Z}{Z_{\odot}}} 10^{\log \dot{M}_{\text{de Jager}}}, \quad (2.25)$$

where $x_1 = \frac{\log T - 4.05}{0.75}$, $x_2 = \frac{\log L - 4.6}{2.1}$, and the coefficients a_{ij} are given in Table VI of de Jager et al. (1988). We interpolate the bi-stability jump around 25 kK (see, for example, Pauldrach & Puls, 1990; Vink et al., 1999) with the following regime. Define the constants

⁶In other codes such as MESA, this is referred to as the ‘‘Dutch’’ mass loss prescription owing to the nationality of the authors of de Jager et al. (1988) and Vink et al. (2001).

$$R_1 = \left(\frac{Z}{Z_\odot} \right)^{0.13}, \quad (2.26)$$

$$R_2 = -c_1 + c_2 \log \left(\frac{L}{c_{16}} \right) - c_3 \log \left(\frac{M}{c_{15}} \right) - c_4 \log (1.3R_1) + c_5 \log \left(\frac{T}{c_{14}} \right) - c_6 \left(\log \left[\frac{T}{c_{14}} \right] \right)^2 + c_7 \log \left(\frac{Z}{Z_\odot} \right), \quad (2.27)$$

and

$$R_3 = -c_8 + c_9 \log \left(\frac{L}{c_{16}} \right) - c_{10} \log \left(\frac{M}{c_{15}} \right) - c_{11} \log (0.65R_1) + c_{12} \log \left(\frac{T}{2 \times 10^4} \right) + c_{13} \log \left(\frac{Z}{Z_\odot} \right), \quad (2.28)$$

where the constants c_i are given in table 2.1.

Constant	Value	Constant	Value	Constant	Value	Constant	Value
c_1	6.697	c_5	0.933	c_9	2.210	c_{13}	0.85
c_2	2.194	c_6	10.92	c_{10}	1.339	c_{14}	4×10^4 K
c_3	1.313	c_7	0.85	c_{11}	1.601	c_{15}	30 $M_\odot$
c_4	1.226	c_8	6.688	c_{12}	1.07	c_{16}	$10^5 L_\odot$

Table 2.1: The mass loss rate constants for interpolation around the bistability jump (equations (2.26)–(2.28)). This interpolation is shown pictorially in figure 2.11.

Our mass loss rate becomes:

$$\dot{M}_{\text{BS}} = \begin{cases} \frac{(T-T_1)10^{R_3} + (T_2-T)\dot{M}_{\text{pre-WR}}}{2.5 \times 10^3} & T_1 < T < T_2 \\ 10^{R_2} & T_2 \leq T \leq T_3 \\ \frac{(T-T_3)10^{R_2} + (T_4-T)10^{R_3}}{5 \times 10^3} & T_3 < T < T_4 \\ \frac{(6T_1-T)10^{R_2} + (T-5T_1)\dot{M}_{\text{pre-WR}}}{10^4} & 5T_1 < T < 6T_1 \\ \dot{M}_{\text{pre-WR}} & \text{otherwise} \end{cases} \quad (2.29)$$

where $T_1 = 10$ kK, $T_2 = 12.5$ kK, $T_3 = 22.5$ kK, $T_4 = 27.5$ kK. This interpolation is shown in figure 2.11. We define

$$\mathcal{M} = \sqrt{\frac{Z}{Z_\odot}} \times 10^{-13.6 + 1.63 \log \left(\frac{L}{L_\odot} \right) + 2.22 \log X}, \quad (2.30)$$

where X is the surface hydrogen abundance of the primary, and then define

$$\dot{M}_{\text{WNL}} = \begin{cases} 5 \left(\mathcal{M} - \dot{M}_{\text{BS}} \right) (0.42 - X) + \dot{M}_{\text{BS}} & 0.42 > X > 0.4 \\ \mathcal{M} & \text{otherwise} \end{cases}. \quad (2.31)$$

Our total mass loss (provided the star has not depleted its surface helium, which will be discussed below) is now

$$\dot{M}_A = \begin{cases} 10\dot{\mu} (\log T - 3.9) + \dot{M}_{BS} & X < 0.42 \text{ and } \log T > 3.9 \\ \dot{M}_{WNL} & X > 0.42 \text{ and } \log T > 4.0, \\ \dot{M}_{BS} & \text{otherwise} \end{cases} \quad (2.32)$$

where $\dot{\mu} = \dot{M}_{WNL} - \dot{M}_{BS}$. Finally, if the surface helium depletes we modify the mass loss rate further, defining

$$\dot{M}_{WC} = \sqrt{\frac{Z}{Z_\odot}} \times 10^{-8.3+0.84 \log\left(\frac{T}{T_\odot}\right)+2.04 \log X_{\text{He}}+1.04 \log(1-X_{\text{He}})} \quad (2.33)$$

such that our total mass loss rate becomes (provided $X_H < 10^{-3}$ and $\log T > 4$)

$$\dot{M}_B = \begin{cases} 100 (\dot{M}_{WC} - \dot{M}_{WNL}) (\mathcal{X} - 0.02) + \dot{M}_{WNL} & \mathcal{X} > 0.02 \\ \dot{M}_{WNL} & \mathcal{X} < 0.02 \end{cases}, \quad (2.34)$$

where $\mathcal{X} = \frac{\frac{1}{3}X_C + \frac{1}{4}X_O}{X_{\text{He}}}$. The total Vink et al. (2001) mass-loss rate is then

$$\dot{M}_{\text{wind}} = \begin{cases} \dot{M}_B & \text{surface helium depleted} \\ \dot{M}_A & \text{otherwise} \end{cases}. \quad (2.35)$$

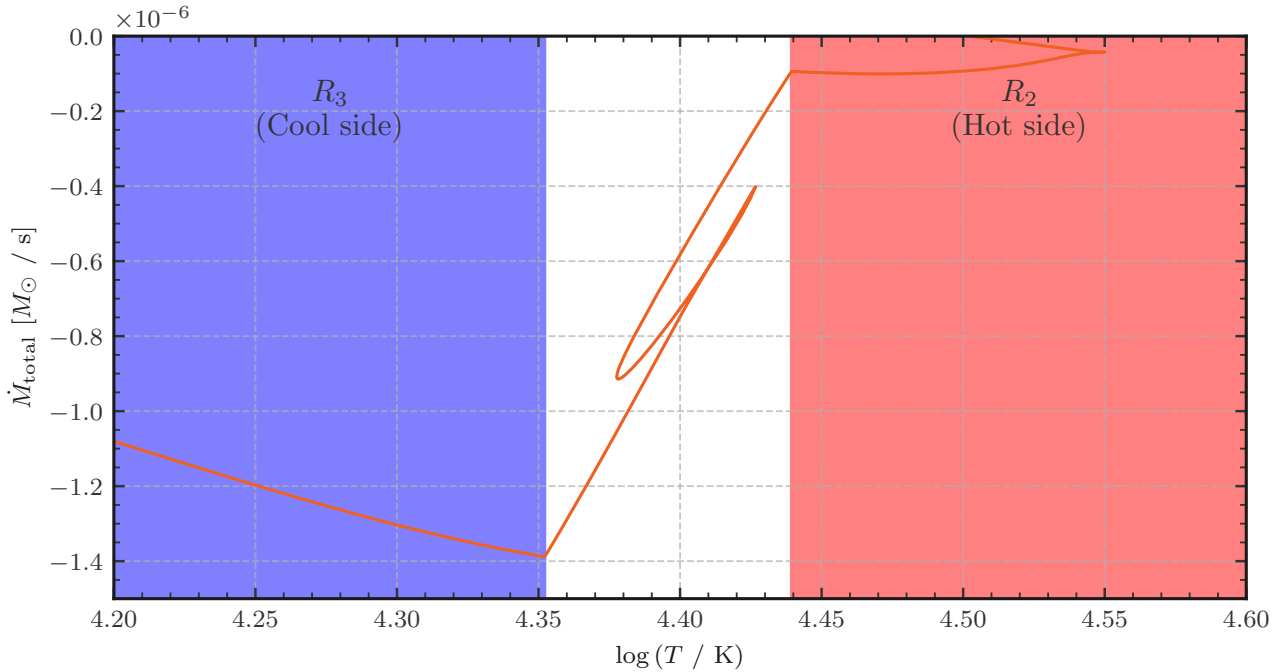


Figure 2.11: An illustration of the bi-stability jump at $\log T \approx 4.4$. This illustration was computed for a $21 M_\odot$ star. Shown in blue is the cool side of the bistability jump, where R_3 (equation (2.28)) is the interpolation, and shown in red is the hot side, where R_2 (equation (2.27)) is the interpolation.

Roche Lobe overflow (RLOF)

Matter extruded from the Lagrangian points is assumed to transfer with a mass loss rate similar to Hurley et al. (2002). We take

$$\dot{M}_{\text{RLOF}} = \min \left(k \mathcal{R} \left(\frac{R}{R_{\text{RL}}} \right)^3, 10^{-2} \right) M_{\odot} \text{ yr}^{-1}. \quad (2.36)$$

Here, k (referred to in our code as RMT) parameterises effectively how much of the star we permit to exist outside of its Roche lobe before it is accreted – i.e., the voracity of RLOF. The choice of this can be somewhat arbitrary. Hurley et al. (2002) for example took $k = 3 \times 10^{-6} \min \{M_1, 5\}^2$, where the coefficient was “chosen by experiment to ensure that the mass transfer is steady”, and warned that for degenerate stars a more appropriate choice would be to increase this by a factor of 10^3 . Instead, we choose to adopt $k = 10^3$ outright with no dependence on the mass, finding instead that this replicated other work – allowing efficient RLOF – best.

Matter exceeding $\mathcal{L}_2$ (the outer Lagrange point) does not currently receive special treatment. The implications of $\mathcal{L}_2$ overflow are significant. This ejection will lead to unstable mass transfer, shrinking the orbit and likely inducing a CEE event (Fragos et al., 2023). There is debate about the appropriate treatment here. Linial & Sari (2017) indexes the $\mathcal{L}_2$ rate to the $\mathcal{L}_1$ rate via an efficiency parameter. Li & Li (2013) considers the case analogous to $\mathcal{L}_1$ outflow, where material outflowing from $\mathcal{L}_2$ acts to retard the angular momentum. In their notation $\frac{dJ}{dt} = -\epsilon \dot{M}_1 a_{L2}^2 \omega$ (compare $\frac{dJ}{dt} = -\beta \dot{M}_1 a_{L1}^2 \omega$ for RLOF through $\mathcal{L}_1$) for controllable parameters β and ϵ . The ambiguity of treatment leads to some codes to terminate at the onset of $\mathcal{L}_2$ outflow, such as POSYDON (Fragos et al., 2023). $\mathcal{L}_2$ outflow requires stars to be in reasonably tight orbits (~ 2 d), which we will approach, but none of our stars will enter this regime, and thus we do not afford it special treatment.

Accretion

Our accretion rates are taken from Bondi & Hoyle (1944); Hurley et al. (2002), and limits the accretion rate by the thermal timescale of the star. The prescription is that

$$\dot{M}_{\text{acc}} = \min \left(\dot{M}_{\text{RLOF}}, \frac{M}{T_{\text{KH}}} \mathcal{R} \left(1 - \frac{\omega}{0.8 \omega_{\text{crit}}} \right) \right), \quad (2.37)$$

where ω and ω_{crit} are the spin rate and critical spin rate of the star. We generally assume that the star is composed of individually rotating shells of variable thickness. If N is the number of meshpoints that the star is composed of (for our cases, normally $N = 199$), then the moment of inertia is given by

$$I = \frac{2}{5} M_c R_c^2 + \sum_{k=1}^N \frac{2}{3} (m_k - m_{k+1}) r_k^2, \quad (2.38)$$

where m_k is the mass internal to meshpoint k (sim. for r_k), and a subscript of c refers to the central meshpoint. It is assumed that the central meshpoint sits atop a solid sphere, hence $\frac{2}{5}$ as its prefactor.

Bondi-Hoyle accretion

Bondi-Hoyle accretion for a circular orbit is given by

$$\dot{M}_{\text{Bondi}} = \min \left[- \left(\frac{GM_2}{v_W^2} \right)^2 \frac{\alpha_W}{2a^2} \frac{1}{(1+v^2)^{3/2}}, 0.8 \right] \dot{M}_{\text{wind}}, \quad (2.39)$$

where we have assumed that the orbital velocity is given by

$$v^2 = \frac{G(M_1 + M_2)}{av_W^2}, \quad (2.40)$$

and that the wind velocity is given by

$$v_W = 2\beta_W \frac{GM_1}{R_1}. \quad (2.41)$$

Generalising this to an eccentric orbit causes the first argument in equation (2.39) to pick up a coefficient of $\frac{1}{\sqrt{1-e^2}}$. There are two free parameters to select: α_W and β_W . These ostensibly depend on stellar types, though we select for simplicity $\alpha_W = \frac{3}{2}$ (Boffin & Jorissen, 1988) and $\beta_W = 7$ (Lamers et al., 1995) for O-type stars. This parameterisation is likely incomplete; Hurley et al. (2002) argues that as per Lamers et al. (1995), β_W should vary between 7 for O-type stars to 0.5 for A and F-type stars, perhaps descending as low as $\beta_W \approx \frac{1}{8}$ for cool supergiants. We leave this implementation for future work under the assumption that our mass-loss rate is correct to within an order of magnitude.

2.5 | Post-Supernova

Wit nodded.

“Oh, is this the place where I talk?” Kaladin said.

“If you wish.”

“I don’t wish. Get on with the story.”

Rhythm of War, BY BRANDON SANDERSON

After a supernova, the star leaves behind its core as a compact remnant. Generally, these are either white dwarves, black holes, or neutron stars. The interior structure of each is either complicated or, in the case of black holes, impossible to observationally probe without violating causality. Their orbital motions are now deterministic from four initial parameters: m_1, m_2, a , and e . Motion is now wholly determined by the emission of gravitational radiation through (detectable) gravitational waves. The pertinent equations of motion are given by Peters (1964), namely,

$$\frac{da}{dt} = \frac{-\beta}{a^3 (1-e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4 \right), \text{ and} \quad (2.42)$$

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1-e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304}e^3 \right), \quad (2.43)$$

with the constant β defined as

$$\beta \equiv \frac{64}{5} \frac{G^3 m M (m + M)}{c^5}. \quad (2.44)$$

Now if one defines the constant c_0 through

$$c_0 = a_0 (1 - e_0^2) e_0^{-\frac{12}{19}} \left(1 + \frac{121}{304} e_0^2 \right)^{-\frac{870}{2299}}, \quad (2.45)$$

then the time from the second supernova to the conclusion of the inspiral, termed the *coalescence time*, will be given by

$$T_c(a_0, e_0) = \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2 \right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}} de. \quad (2.46)$$

The derivations for equations (2.42), (2.43) and (2.46) are given in appendix E. The integral has no closed-form solution, and a numerical integrator is typically used to compute the coalescence time. However, some approximations such as Mandel (2021) do exist and approximate the coalescence time rather well over the entire range of mass ratios.

2.5.1 | Merger Rates of compact binaries

The merger rate of a population of compact binaries at time t after formation can be formalised via the heuristic

$$m(t) = \text{number of binaries for which } T_c + T_{\text{rej}} + T_{\text{evo}} < t, \quad (2.47)$$

where T_c is the coalescence time, T_{rej} is the rejuvenation age (the age at which the primary star last accreted mass), and T_{evo} is the evolution time-scale (the length of time the primary spends on the main sequence). $m(t)$ is called the Delay-Time Distribution (DTD) of a population.

However, the underlying assumption in this is that the population was formed after an instantaneous burst of star-forming activity at time $t = 0$, which is clearly unphysical. We must account for the so-called Star Formation Rate Density (SFRD) $\psi(z)$, assumed to be taken from Madau & Dickinson (2014):

$$\psi(z) = 0.015 \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9} \right)^{5.6}} \text{ M}_\odot \text{ yr}^{-1} \text{ Mpc}^{-3}, \quad (2.48)$$

where z is the redshift. For the purposes of converting between lookback time t and redshift z , we assume a standard Λ CDM cosmology with $\Omega_M = 0.3, \Omega_\Lambda = 0.7, \Omega_k = 0, H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Of course, this ignores the effect of mixed-metallicity models, which we account for using the prescription of Langer & Norman (2006), whereupon the SFRD is multiplied by a factor χ :

$$\chi(z, Z) = \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot} \right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)}, \quad (2.49)$$

where $\hat{\Gamma}$ and Γ are the upper incomplete and complete Gamma functions respectively. This means our Stellar Formation History (SFH) – which takes into account both redshift and metallicity – denoted $\Psi(z, Z)$, is given by

$$\Psi(z, Z) = 0.015 \int_{z_1}^{z_2} \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}} \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot}\right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)} dz, \quad (2.50)$$

where z_1 and z_2 are the redshifts of our bin edges. The event rate as a function of time, $\mathcal{R}(t)$, is therefore given by

$$\mathcal{R}(t) = \Psi(z_t) * m(t), \quad (2.51)$$

where z_t is the redshift corresponding to lookback time t , given by the solution to the equation

$$t = t_H \int_0^{z_t} \frac{dz'}{(1+z') \sqrt{\Omega_M (1+z')^3 + \Omega_k (1+z')^2 + \Omega_\Lambda}}, \quad (2.52)$$

with t_H the Hubble time, and $\Omega_M, \Omega_k, \Omega_\Lambda$ the matter density, curvature parameter, and cosmological constant density parameter of the Universe. In general this integral does not have a solution, however in the case of Λ CDM with Hubble constant $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ as outlined above, an exact solution for redshift is

$$z = \left(\frac{a}{\sinh\left(\frac{b \operatorname{asinh}(a)+t}{b}\right)} \right)^{\frac{2}{3}} - 1 \quad (2.53)$$

where $a = \sqrt{\frac{7}{3}}$, $b = \frac{1}{105\sqrt{0.7}}$.⁷ Detail on how the convolution in equation (2.51) is computed may be found in Richards (2021).

2.6 | Numerical implementation

I think there is a world market for maybe five computers.

THOMAS WATSON, PRESIDENT OF IBM, 1943

Having now discussed the evolution of stars from a theoretical perspective, it becomes imperative to discuss the methods of modelling them. In this section, we provide an overview of the four codes we either use extensively or have developed to carry out the work of this thesis. These four codes are:

1. The Cambridge STARS code (section 2.6.1), originally developed by Peter Eggleton, which we use to solve the equations of stellar structure and evolution,
2. The TUI code (section 2.6.2), developed by Jan Eldridge, which performs our binary population synthesis,
3. The KAITIAKI code (section 2.6.3), developed for this thesis, which is our grid-manager code and allows modification of the input parameters to the Cambridge STARS code programmatically, and
4. The TAKAHE code (section 6.1.3), developed by Sean Richards for Richards (2021), which is our binary compact remnant evolution code.

⁷In general in a flat Universe (with $\Omega_k = 0$), $a = \sqrt{\frac{\Omega_\Lambda}{\Omega_M}}$, and $b = \frac{1}{H_0} \frac{2}{3\sqrt{\Omega_M}}$. This integral was computed via SYMPY.

2.6.1 | The Cambridge STARS code

But as constant dropping wears away a stone, constant progression,
comparatively imperceptible in its course, attains to immense
distances after the lapse of time.

Anno Domini 2000, or, Women's Destiny, BY SIR JULIUS VOGEL

Brief History

And through the night, behind the wheel, the mileage clicking west
I think upon Mackenzie, David Thompson and the rest
Who cracked the mountain ramparts and did show a path for me
To race the roaring Fraser to the sea.

Northwest Passage, BY STAN ROGERS

Stellar modelling became a field of active research in the 1950s. Prior to the invention of the computer, a model was treated as a mathematical process and thus was computed – often labouriously – by hand. For instance, in Eggleton (1967), Eggleton writes that

Since we have seen in Section 3 that the structure of the flame zone can be obtained independently (almost) of the rest of the star, it is reasonable to expect that instabilities of the type considered by Schwarzschild & Härm can also be examined in this way, so that the considerable labour of computing the structure of an entire star can be avoided.

– Eggleton (1967)

The Cambridge STARS code was one of the first detailed stellar evolution codes to be written, though initially it only modelled one star in detail. To model a binary system, tweaks were made whereupon the primary would be modelled in detail whilst approximating the secondary with the equations of Hurley et al. (2002) until the primary supernova, at which point the secondary would be evolved in detail. This had its merits; notably computation time was saved, and the results were generally accurate anyway.

However, Stancliffe & Eldridge (2009) introduced detailed modelling of *both* stars in a binary system into the STARS code – that is, the interiors of both stars were computed in detail, alongside the simultaneous computation of the orbital parameters. Since then, a wide array of improvements have been made to the numerical stability of the code, as well as introducing or refining new physics.

There are four variants of this code:

1. The Cambridge STARS code, which is the version originally developed by Peter Eggleton and originally described in Eggleton (1971). This version was originally only capable of solving the equations for a single star.

2. The ELDRIDGE STARS code described in Eldridge & Tout (2004b,a); Eldridge et al. (2017), which is the version used to generate the fiducial BPASS v2 grids. This version models the primary in detail while approximating the secondary using the equations of Hurley et al. (2002).
3. The BSTARS code, which is the unmodified version lately available from Cambridge, described in Stancliffe & Eldridge (2009). This version models both stars in detail throughout their lives.
4. The AOTEAROA STARS code, which is the version of the code incorporating the changes we have made.

Unless context otherwise requires, when we refer to “the STARS code” with no further qualification, we are referring specifically to the AOTEAROA STARS code.

Matrix Inversion

The heart of STARS is a matrix inversion algorithm. Although computing the inverse of a matrix is a polynomial time process, it can still take considerable time to invert. STARS implements a Henyey matrix (Henyey et al., 1964), which is of size $n = (4 + n_{\text{comp}})N_{\text{mesh}}$, with n_{comp} the number of equations to be solved and N_{mesh} the number of meshpoints in the model. An individual element of the Henyey matrix is given by

$$\mathbf{H}_{ij} = \frac{\partial (\text{equation } i)}{\partial (\text{variable } j)}. \quad (2.54)$$

For a “typical” binary computation, with $n_{\text{comp}} = 30$ equations and $N_{\text{mesh}} = 199$ meshpoints, the matrix is already 6766×6766 , for a binary which evolves up the AGB (the theoretically most complex calculation we perform) with $N_{\text{mesh}} = 999$, the matrix is 33966×33966 . Introducing k new equations into the code causes the runtime of the matrix inversion calculation to scale like $\mathcal{O}(n^3 + k^3 N_{\text{mesh}}^3)$.

The matrix inversion is simple Gauss-Jordan inversion, which is of complexity class (for an $n \times n$ matrix) $\mathcal{O}(n^3)$. Coppersmith & Winograd (1987) showed that this could be optimised to $\mathcal{O}(n^{2.376})$ and conjectured reducibility to $\mathcal{O}(n^2)$.⁸ Since our code is embarrassingly parallelisable, and can be executed in parallel simply by increasing the number of cores available, the payoff for decreasing the runtime by one polynomial order is inconsequential, and thus we use the straightforward implementation.

Initial Condition Selection

In this section we provide an account for the selection of our general initial conditions. Unless otherwise noted in this thesis, it is assumed that the values will be using the parameters described here and in appendix D.

In our implementation of Mixing Length Theory (MLT) (Vitense, 1953; Böhm-Vitense, 1958; Cox & Giuli, 1968), we use an MLT parameter of $\alpha_{\text{MLT}} = 1.985$, selected via calibration against a model of the Sun. This is effectively a dissipativity parameter, and represents the depth at which a parcel of matter penetrates into a lower layer before becoming fully dissipated. The value was selected by modelling a $1 M_{\odot}$ star and varying only the mixing length parameter until it had a luminosity of $1 L_{\odot}$ and a radius of $1 R_{\odot}$, at the present age of the Sun.

⁸The current “best” optimisation is $\mathcal{O}(n^{2.3727})$ (Williams, 2012). DeepMind’s AlphaTensor has claimed a faster result in Fawzi et al. (2022), though they do not state a complexity class. This is because “...this does not yield practical algorithms.”

Mesh spacing

A natural question that arises when modelling a star is where internally to model it – there is a counterbalance between the level of detail and the time cost of the matrix inversion. The obvious answer would be to distribute a mesh equally spaced within radius of the star, however the nonhomogeneity of a star, plus the potential for very thin shells of physically important material, necessitate a more intelligent approach.

Instead of being equally spaced *radially*, these mesh points are taken to be equally spaced within a function $\mathcal{Q}$, which is taken from Stancliffe et al. (2004) and given for a binary that does not have AGB-specific treatment enabled by

$$\begin{aligned} \mathcal{Q}(P, T, M, R) = & c_4 \ln P + c_5 \ln(P + c_H) + c_2 \ln(P + 3P_{\text{He}}) + c_2 \ln(P + 0.3P_{\text{He}}) + c_7 \ln\left(\frac{T}{T + 2 \times 10^4}\right) \\ & + \ln\left(\frac{c_6 M^{\frac{2}{3}}}{M + M^{\frac{2}{3}}}\right) - c_3 \ln\left(\frac{R^2}{c_8} + 1\right), \end{aligned} \quad (2.55)$$

where P, T, M, R are the total pressures, temperatures, masses, and radii of each star, the constants c_i are user controllable and given in appendix D, P_{He} is the pressure in the helium-burning shell, and

$$c_H = \begin{cases} 0.1P_H & X_H \geq 0.15 \\ 0.1\mathcal{P} & \text{otherwise} \end{cases} \quad (2.56)$$

where P_H is the pressures in the hydrogen-burning shell. Here, $\mathcal{P}$ represents the pressure in the hydrogen-burning shell when $X_H = 0.15$. The choice of mesh spacing coefficients $c_1 \dots c_{10}$ leads to different areas of the star being prioritised. Coordinates c_1, c_9 , and c_{10} are not used except in AGB-specific evolution, which is not required for most cases (and requires a change of parameterisation in the pressure-shell terms). c_5 is hardcoded to be 0.15, and c_2 to be 0.0. **[TODO: but why? “its what the code does” is a bad reason.]**

Placement of the mesh is controlled by the derivative $\frac{\partial \mathcal{Q}}{\partial k}$, where k is the meshpoint number (Stancliffe, 2016). Specifically, meshpoints are placed at locations where

$$\left. \frac{\partial \mathcal{Q}}{\partial k} \right|_k - \left. \frac{\partial \mathcal{Q}}{\partial k} \right|_{k+1} = 0. \quad (2.57)$$

The mesh is thus adaptive (insofar as it changes to place meshpoints where they are needed for evolution) and non-Lagrangian (insofar as the derivative is not at constant mass coordinate). If one desires to make the mesh viscous (i.e., partially Lagrangian), one can follow Stancliffe et al. (2004) and write, for $0 \leq \gamma \leq 1$,

$$\left(\left. \frac{\partial \mathcal{Q}}{\partial k} \right|_k - \left. \frac{\partial \mathcal{Q}}{\partial k} \right|_{k+1} \right) (1 - \gamma) + \gamma \left(\left. \frac{\partial m}{\partial t} \right|_k \right) = 0, \quad (2.58)$$

where now the parameter γ is a weighting coefficient such that if $\gamma = 1$, then the mesh spacing function is irrelevant and the parameters are spaced equally in mass, and if $\gamma = 0$, the original mesh spacing function is used. In between, if $\gamma = \gamma(k, dt)$, then parts of the mesh are allowed to be Lagrangian and parts non-Lagrangian; the mesh is said to be *viscous*. We typically keep the mesh non-viscous and non-Lagrangian, as a viscous mesh is

most useful for AGB evolution and evolving through thermal pulses.

More detail is provided in Eggleton (1971) for the original mesh spacing function, and Stancliffe et al. (2004) for the binary mesh spacing function.

Timestep control

The STARS code utilises an adaptive timestep control mechanism, whereby the timestep is allowed to shrink to model rapid changes or grow if the star is broadly inert and unchanging. The core computation is a parameter Δ , given by

$$\Delta = \frac{d_{\text{opt}}}{\sum_i \sum_j |\delta x_{ij}|}, \quad (2.59)$$

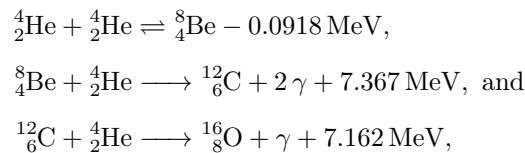
where d_{opt} is a pre-defined target timestep (typically taken to be ~ 1 , unless otherwise specified; see appendix D for our selections), and δx_{ij} is the change in variable x_i at meshpoint j . In this calculation, luminosity is explicitly excluded as this would cause the timestep to explode during certain points of the star's life-time. A new timestep, d' , is selected from the previous timestep d by

$$d' = \begin{cases} \min[d_1, \Delta] d & \Delta < 1 \\ d & \Delta = 1 \\ \max[d_2, \Delta] d & \Delta > 1 \end{cases}, \quad (2.60)$$

where d_1 and d_2 are pre-determined control parameters. Most commonly we take $d_1 = 0.1$ and $d_2 = 1.2$ unless otherwise specified.

Numerical instabilities

Stellar modelling is a nontrivial endeavour and there are many cases where the code can fail seemingly arbitrarily. The largest cause of numerical instability is the helium flash, which occurs in low-mass stars between about $0.8 M_{\odot} \lesssim M_{\text{ZAMS}} \lesssim 2.5 M_{\odot}$. In this band, the triple- α process, given by the reactions (Carroll & Ostlie, 2017)⁹



occurs within a highly degenerate helium core. As a result, the temperature of the star increases sharply during this process, leading to thermal runaway. This temperature increase in turn results in an increase in the luminosity (which peaks at $\sim 1 \times 10^{10} L_{\odot}$). The flash itself – being the period of thermal runaway – occurs on the order of a few minutes, though the spike in luminosity lasts approximately 7 Myr (for the $2.51 M_{\odot}$ star shown in figure 2.12).

⁹The reversible reaction is due to Beryllium-8 being unstable; its half-life is on the order of 1×10^{-16} s, and it decays back to $2 {}^4_2\text{He}$ if it does not react further.

A method termed “pseudoevolution” can be used to overcome this source of error. Briefly, a complete model (termed the *reference model*) of a $3 M_{\odot}$ star is evolved with a full model description saved at every timestep. When the helium flash occurs in a model (heuristically, $0.1 \leq M_{\text{He}}/M_{\odot} \leq 0.55$ and $\log T_c/\text{K} > 7.9$), we query the reference model for a model with the same helium core mass as the failed model. Using a custom subroutine in STARS, we turn off all nuclear and thermal burning and remove mass from the reference model to match the mass of the failed model. This model is then allowed to evolve as per normal, having overcome the helium flash. We note that this is not a *perfect* method, as this does induce discontinuities in the plots, as shown in figure 2.13, though these are essentially insignificant.

The other stage where low-mass stars are prone to fail is during a thermal pulse (Schwarzschild & Härm, 1965, 1967). The mass range for thermal pulses is broader, from $\sim 0.8 M_{\odot}$ to $\sim 8 M_{\odot}$. A thermal pulse occurs as a star ascends the AGB; as a Carbon-Oxygen core (CO core) or Oxygen-Neon core (ONe core) has developed, shells of burning helium and hydrogen (separated by a region of almost pure helium) sit atop the core (Pols et al., 2001). As the intershell region dissipates due to dredge up or helium burning, hydrogen burning reignites in the hydrogen shell. As both shells burn (at the same rate), the helium-burning shell becomes unstable and eventually a process of thermal runaway begins (similar to the helium flash, except off-core). This causes a spike in the luminosity, termed a thermal pulse. This can persist for many pulses – Stancliffe (2006) observes pulses before model termination, while Stancliffe (2005) observes 17 pulses for a model similar to those in figure 2.12.

The method to overcome this numerical instability is involved and not fully implemented in STARS. Briefly, the number of meshpoints must be drastically increases near the onset of thermal pulses; up to 999 instead of the typical 199/499. The timestep must also be forcibly decreased; thermal pulses typically last for around a year (Pols et al., 2001) so the timestep must be reduced to not miss them (and to not have the modulus of the change in parameters – $\sum_i \sum_j |\delta x_{ij}|$ in equation (2.59) – and thus the error, increase too much). This method was implemented to generate figure 2.12, though only two pulses were achievable.

Critical modifications

The major modifications we have made to the STARS code are as follows.

Firstly, the floating-point precision has been increased to double precision for all variables. This has minor effects on the evolved models, insofar as they are generally able to evolve to slightly further stages. This step has previously been undertaken, however not all variables were correctly converted. This did not invalidate any models previous versions of the STARS code has generated; rather it means they terminated slightly earlier than the point to which they may have been able to evolve.

As part of this, we also undertook a reasonably extensive refactor of the codebase. Whilst some of the code remains as originally written, it is now more compliant with modern programming practices and is in some sense easier to alter with new routines.

We have also improved mass loss interpolation in the MASSLOSS.F routine. Whilst not changing the underlying assumption that mass transfer is governed by de Jager et al. (1988); Vink et al. (2001) (section 2.4.3), we have improved the transition between typical O-type mass loss and WR mass loss rates by slowing turning on the WR mass loss rate as the surface hydrogen abundance drops below 0.4. The transition was previously abrupt and

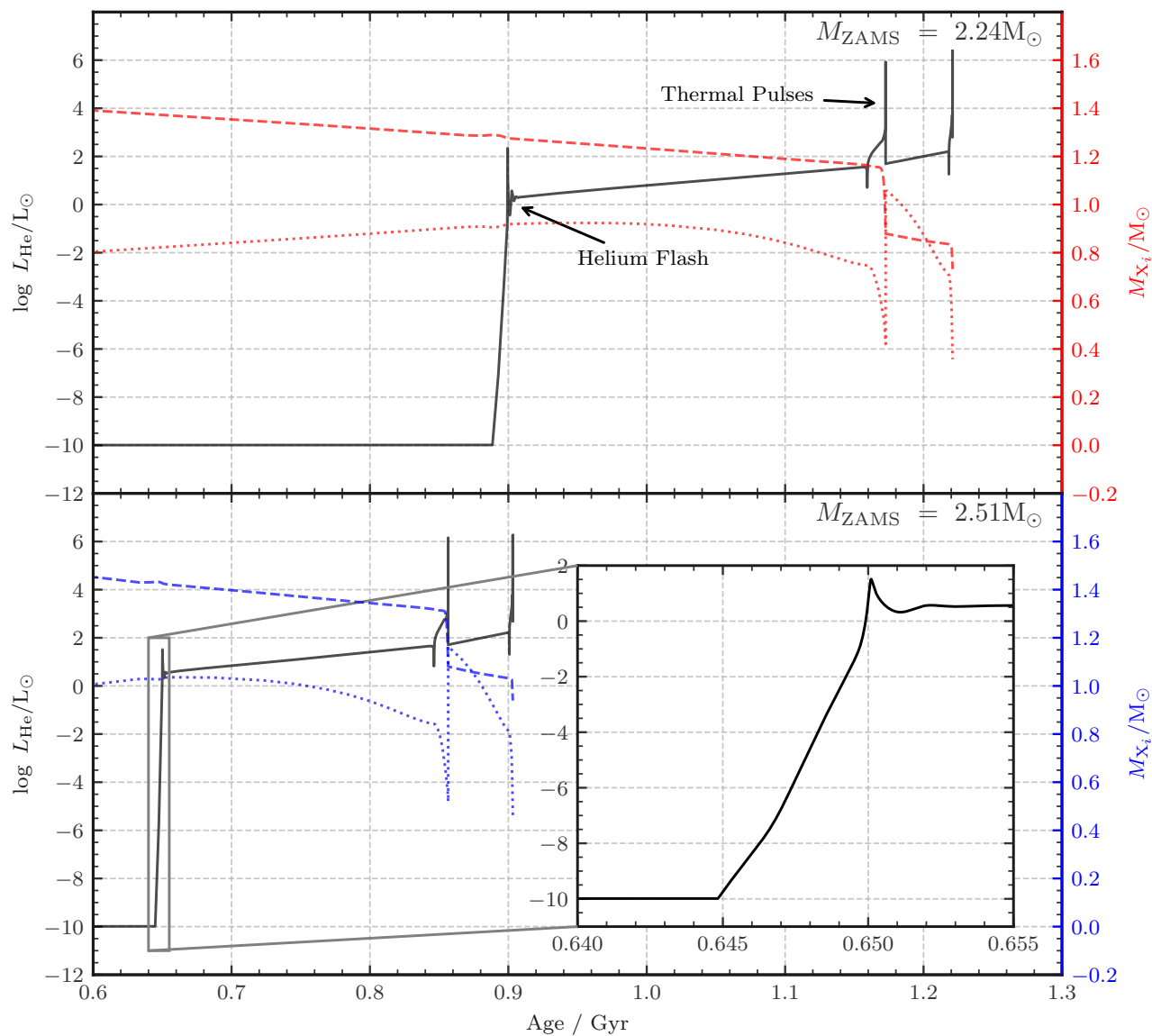


Figure 2.12: Two stars of a similar mass that undergo a helium flash and two thermal pulses. Shown in the inset is a zoomed-in helium flash that lasts ~ 7 Myr. Both stars undergo two thermal pulses before the code crashes due to the relative error in the calculation increasing too rapidly. The two stars could, in principle, undergo many more thermal pulses before their deaths.

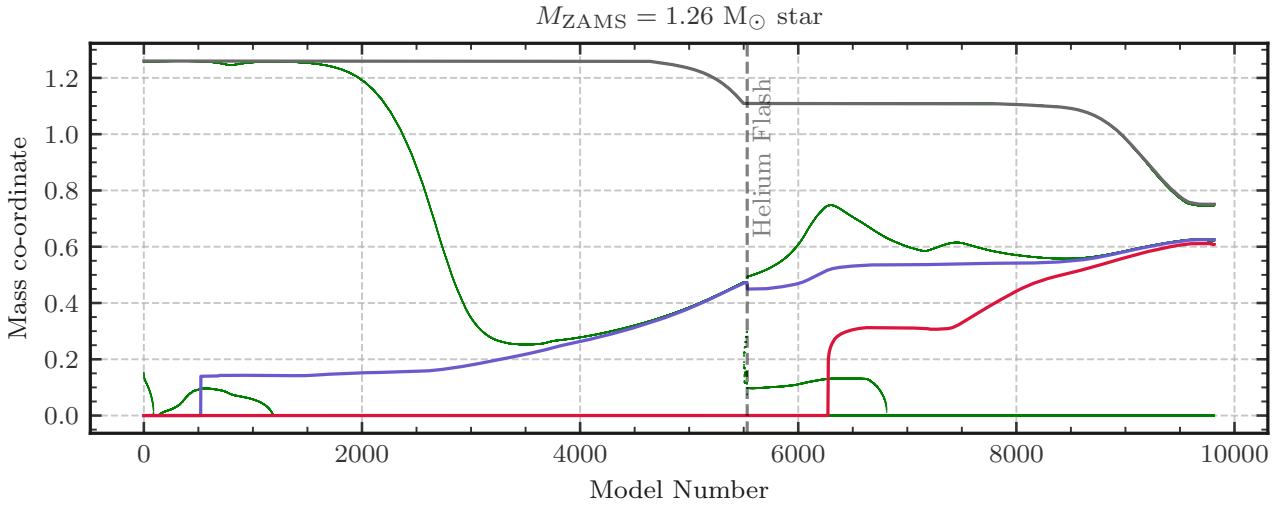


Figure 2.13: The helium flash (gray dashed line) shown on a Kippenhahn diagram. There is a slight discontinuity in the helium core mass (blue) due to a perfect match not existing in the $3 M_{\odot}$ reference model. The total mass is shown as the solid gray line, and the CO core mass is shown in red. The green lines show the boundaries of the convective regions of the star. **[TODO: vectorize]**

would cause the code to fail due to numerical instability if a non-WR star entered this regime.

Similarly to this, the code now manually reduces the timestep in the run-up to, during, and after the contact binary phase. Evolution during this phase breaks thermal equilibrium (i.e., the star evolves on a longer timescale than its Kelvin-Helmholtz time-scale) and hence the evolution can change rapidly as the star attempts to adjust back to thermal equilibrium. The contact phase, for example, typically last for between 100-200 kyr in our models.

We have also begun implementing an $\alpha - \lambda$ treatment of CEE, though we stress that this has not been rigorously tested yet, and is thus disabled for the majority of our models.

2.6.2 | The TUI code

TUI is the core population synthesis code within the BPASS project. TUI takes as input the output from the STARS code. TUI was first described in Ghodla et al. (2023), and was further described in Stevance et al. (2023), though we note that their version is technically a refactor of the version we use. The predominant function we use TUI for is the matching of pre- and post- supernova models for chapter 6, and in this section we provide a brief account of that functionality of TUI (though the control of the population synthesis models is beyond the scope of this thesis).

TUI has precompiled “input_” files which report a STARS plot file, an IMF weight, and a model type (merger/normal primary/normal secondary/single model/quasi-homogeneous evolution binary/single star). After loading in a model, TUI applies a given kick and computes the new orbital parameters from Tauris & Takens (1998). Crucially, a core assumption of BPASS is that the orbit remains circular after the first supernova and is allowed to be eccentric after the second. Of course, nature cares little for our assumptions, and the supernova may cause the binary to become eccentric after the first supernova; TUI accounts for this by setting the separation after the first supernova to be the semi-latus rectum of the eccentric binary. TUI repeats this process after the second

supernova.

For the purposes of chapter 6, if the binary remains bound (has $0 \leq e < 1$) after the second supernova, we record the following parameters:

1. The mass of the primary,
2. The mass of the secondary,
3. The orbital separation,
4. The eccentricity,
5. A weight term proportional to the IMF weighting,
6. The rejuvenation age, which is the time since the companion last accreted material, if it accreted more than five per cent of its initial mass (see Richards et al., 2023), and
7. The evolution age, which is the length of time from the secondary’s ZAMS until its death.

This provides us with the post-secondary supernova orbital parameters which determine the properties of the merger, such as the event rate $\mathcal{R}(z)$ and its path through period-eccentricity space.

2.6.3 | The KAITIAKI grid-watcher code

KAITIAKI is a grid-watcher code we have developed that sits atop the STARS code and manages its execution. It was developed primarily to manipulate the fixed-width text files that STARS uses, in contrast to the “typical” manipulation done through the command line. However development proceeded in such a way that it now provides an interface to all of the (relevant to this thesis) output and input files. Further, it also now allows the user to control jobs through a PYTHON interface, greatly simplifying grid calculations.

The core of KAITIAKI is the `STARSController()` interface. The most naïve run would be to initialise a model of, say, $8 M_{\odot}$ and evolve it until it evolves up the AGB. A sample of code for this is shown in code sample 2.1 which evolves a $10 M_{\odot}$ star.

Code Sample 2.1: A KAITIAKI script to evolve a $10 M_{\odot}$ star

```

1 import kaitiaki
2
3 # First, instantiate the STARS code. run_bs is the location of the run_bs
4 # bash script, which sets up the symlinks and actually runs the code.
5 STARS = kaitiaki.STARS.STARSController(run_bs='..')
6
7 # Loads the default data file and solar metallicity opacity table
8 STARS.blit('z020')
9
10 # Loads a 1 Msun model
11 STARS.load_default_modin()
12
13 # Sets the code up to target a ZAMS mass of 10 Msun.
14 STARS.setup_zams_inflation(10)
15
```

```

16 # Executes the code, such that the final modout will be a 10 Msun model.
17 STARS.run()
18
19 # Equivalent to "tail -399 modout > modin" -- transfers the last 2*199+1
20 # lines from modout (the last converged model) to modin
21 STARS.modout_to_modin()
22
23 # Again, to reload data
24 STARS.blit()
25
26 # Set up the key parameters for our run
27 STARS.configure_parameters({
28     'DT1': 0.1, # Fraction of timestep to allow for lower variation
29                # (if the timestep is cut, it can be cut to 10% of its
30                # present value)
31     'DDD': 2,   # Modulus of total desired timestep
32     'IML1': 5,  # WR mass-loss rates
33     'RML': 0,   # This was set to 10 Msun during inflation -- should be
34                # zero so no mass is added except via the usual ML channels
35                # (this parameter, formally, sets a constant ML rate so
36                # that in the absence of other mechanisms, the mass would
37                # always decrease)
38     'IMODE': 1, # Single star mode
39     'IX': 1,   # Hydrogen burning
40     'IY': 1,   # Helium burning
41     'IZ': 1,   # Metal burning
42     'ITH': 1,  # Thermal energy generation
43     'ISTART': 1 # Reset age, NMOD, and the timestep (to 10% of T_KH)
44 })
45
46 # Execute the model, run until the AGB.
47 STARS.run()

```

2.7 | Chapter Summary

If in the first act you have hung a pistol on the wall, then in the following one it should be fired. Otherwise don't put it there.

ANTON CHEKHOV

In this Chapter, we have explored the evolution of stars from pre-main sequence evolution to supernova, accounting for the effects of binary interaction. This represents the state of the field as at the commencement of this work. In future chapters we will detail the relevant changes we have made to these prescriptions, motivating the work we have performed.

There are more things in Heaven and Earth, Horatio,
than are dreamt of in your philosophy.

– HAMLET

Chapter

3

Detailed Models of Overcontact Binary Stars

AS PREVIOUSLY DISCUSSED, STARS undergoing mass transfer will generally experience RLOF during their lifetimes. In a binary where both stars are undergoing mass transfer, both stars may undergo RLOF simultaneously, resulting in a contact binary system in which both stars physically touch. Contact binaries are not uncommon – they form one in every approximately 100 main-sequence stars (Rucinski, 1998) and 39 in every 100 binary systems (Green et al., 2023).

The evolution is also complex. Sufficiently close binaries may overflow their $\mathcal{L}_2$ points, or create mergers on exceptionally short timescales – typically $P_{\text{orb}} \lesssim 2$ d, but periods as low as $P_{\text{orb}} = 85$ min (El-Badry et al., 2022) are possible in theory or ~ 167 min in practice (Koen, 2022).

3.1 | Observations of contact binaries

Animals can talk.

THIRD OF THE SIX IMPOSSIBLE THINGS

Observations of contact binaries date back to the 1940s (Kuiper, 1941), and in the past eighty years the catalogue has grown to somewhere between $\sim 42\,000$ (Petrosky et al., 2021) and $\sim 86\,000$ (Qian et al., 2020) contact binaries, representing between 26.7 and 54.8 per cent of all observed binaries (those from the Washington Double Star catalogue; Mason et al. 2001). The binaries we study in this chapter are summarised in table 3.1, and were originally compiled in Abdul-Masih et al. (2022). Their sample focused on measuring the period stability of contact binaries and is thus subset of the known sample. They rejected a handful of other non O+O overcontact binaries, for reasons that include short-period von Zeipel-Kozai-Lidov oscillations, incomplete photometric / radial velocity fits, non-Main Sequence (MS) binaries, unknown separations, and irregular light curves.

Ages for the six binaries have not been published, though a soft restriction can be placed on their evolutionary status; Abdul-Masih et al. (2022) compiles specifically MS contact binaries, meaning their interactions must take place whilst the progenitor stars are on their respective main sequences. Nevertheless, we present inferred ages based on matches to the BPASS v2.2 models in table 3.2.

Binary name	M_1	M_2	P	Z	$\log T_1$	$\log L_1$	$\log T_2$	Ref.
LSS 3074	14.8 ± 2.2	17.2 ± 4.2	2.183 ± 1	0.020^a	4.6 ± 0.2	5.15 ± 0.2	4.53 ± 0.5	A
MY Cam	37.7 ± 3.2	31.6 ± 4.2	$1.175^{+6}_{-1.175}$	0.020^a	4.62 ± 0.2	5.2 ± 0.5	4.59 ± 0.5	B
SMC 108086	16.9 ± 1.2	14.3 ± 5.1	0.883 ± 0.88	0.004^b	4.53 ± 0.1	4.608 ± 0.4	4.48 ± 0.5	C
TU Mus	16.7 ± 1.6	10.4 ± 1.4	1.387 ± 1	0.020^c	4.57 ± 0.2	5.02 ± 0.4	4.54 ± 0.5	D
V382 Cyg ^d	26.1 ± 3.2	19.0 ± 4.2	1.886 ± 1	0.020^e	4.57 ± 0.2	5.47 ± 0.4	4.58 ± 0.5	E
VFTS 352	28.6 ± 3.6	28.9 ± 1.8	$1.124^{+2}_{-1.124}$	0.010^f	4.63 ± 0.45	5.19 ± 0.5	4.61 ± 0.5	F

Table 3.1: Search parameters for the six close-contact binaries compiled in Abdul-Masih et al. (2022), as used in our fitting to the BPASS v2.2.1 grid. The uncertainties presented are not the physical uncertainties on their parameters, but instead represent the size of the range that we search within, e.g., $P = 1.175^{+6}_{-1.175}$ indicates that we match on all periods between 0 and 7.165 d inclusive. Masses and Luminosities are measured relative to solar, periods are measured in days, and temperatures measured in Kelvin. **Notes:** ^aSource paper assumes solar metallicity with no further qualification ^bUsing the mean metallicity of the SMC ^cBased on Andersen & Grønbech (1975) claiming 0.021 and Terrell et al. (2003) confirming a comparison with synthetic spectra “does not support a carbon content, and therefore metallicity, far from the solar value” ^dA mass ratio inversion occurred for this binary (see Martins et al., 2017) and hence we use the definition of “primary” from within the paper when reporting in this table. ^eUsing the mean metallicity of the Milky Way ^fAs Crowther et al. (2018) reports “half-solar” for the Tarantula Nebula, the host nebula of VFTS 352. **References:** A: Raucq et al. (2017), B: Lorenzo et al. (2014), C: Hilditch et al. (2005), D: Penny et al. (2008), E: Martins et al. (2017), F: Almeida et al. (2015)

3.1.1 | Progenitor history

We now examine each of the overcontact systems in detail, providing observational context to each system.

LSS 3074

The binarity of LSS 3074 (Raucq et al., 2017) was first observed in 1985 by Morrell & Niemela (1990) with an observed period of 2.19 d, with its HeI and HeII spectra causing the authors to conclude that it was an O4f+ and O6-7 binary, with inclination-weighted masses of $M_1 \sin^3 i = 8 M_\odot$ and $M_2 \sin^3 i = 9 M_\odot$. With later measurements of the inclinations to be $i \approx 24^\circ$, these masses were refined to $M_1 = 17.2 M_\odot$ and $M_2 = 14.8 M_\odot$. It is located behind the Coalsack Nebula, at a distance of 2256 pc, with a B-band magnitude of 12.81 mag (Stephenson & Sanduleak, 1971). It has a fillout factor $f = 1.05$.¹

MY Camelopardalis

MY Camelopardalis (MY Cam; Lorenzo et al., 2014) is a binary in the cluster Alicante 1, and its binarity was established in 2003 by Negueruela & Marco (2003). It is likely to be a very massive merger progenitor, though is barely in contact with $f = 1.01$ (Abdul-Masih et al., 2022).

¹A fillout factor is a measurement of the degree of which a system overfills its Roche lobes, with $1 < f < 2$ denoting an overcontact binary, and a high fillout factor representing a more swelled and in-contact system.

SMC 108086

OGLE SMC-SC10 108086 (SMC 108086; Hilditch et al., 2005) is a O9 + O9.5/B0 binary. It is a system deeply in contact, with $f = 1.70$ (Hilditch et al., 2005; Abdul-Masih et al., 2022) – the deepest known – and has a short period of $P = 0.88$ d. It is located at a distance of 56.5 kpc, in the Small Magellanic Cloud (SMC).

TU Muscae

TU Muscae (TU Mus; Penny et al., 2008) is an O8V + B0V binary discovered by Oosterhoff (1928, 1930). Its binarity was confirmed as late as 1971, in Knipe (1971). The system was profiled in Andersen & Grønbech (1975), who concluded that, based on comparison to theoretical models by Stothers (1972, 1973), the system was of solar composition ($Z = 0.021 \approx Z_{\odot}$) and on the main sequence with $X_{\text{H}} = 0.739$. It has a fillout factor of $f = 1.12$.

V382 Cygni

V382 Cygni (V382 Cyg; Martins et al., 2017) is an O6.5 + O6 binary whose components are oblate spheroids (deduced from the β -Lyrae light curves; Pearce 1952). It was originally discovered by Morgenroth (1935) and A. A. Petrov in 1946 (Landolt, 1964; Yaşarsoy & Yakut, 2012).² It has a fillout factor of $f = 1.10$.

VFTS 352

VFTS 352 was detected in the VLT-FLAMES survey of the Tarantula Nebula (Evans et al., 2011) and is the most recent detection in our sample. The Tarantula Nebula / 30 Doradus is a large H(II) region in the Large Magellanic Cloud (LMC). The progenitor of VFTS 352 is likely an equal mass, short period binary with no strong helium or nitrogen surface abundances (Abdul-Masih et al., 2019), which may be due to biases in the CNO abundances, or similar, in that atmospheric study. It has a fillout factor of $f = 1.28$.

3.1.2 | Data matching with BPASS v2.2

We perform our data matching using the HOKI code (Stevance et al., 2020). The search parameters are shown in table 3.1. We query the BPASS v2.2.1 dataset and record whether or not they enter the (widened) parameter values. Parameters are widened as some of the periods are tighter than the BPASS bins and thus can be missed in the search.

The matching values HOKI finds are shown in table 3.2. The primary models are also shown on ???. We also report in the table the inferred ages at match, which lie between $\mu_{\log(\text{age})} = 6.13$ (for MY Cam) and $\mu_{\log(\text{age})} = 7.14$ (for TU Mus).

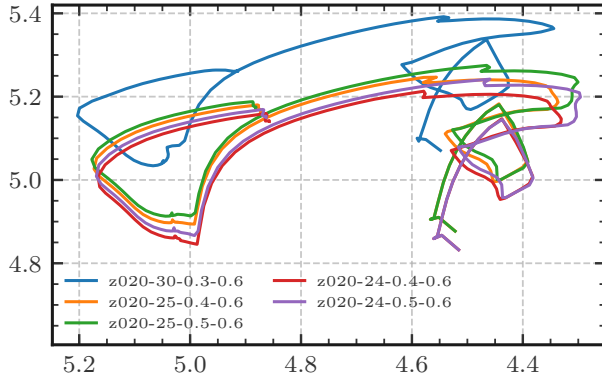
3.2 | Forward evolution

Having now established the ZAMS parameters for our binaries, we can take these as input to our STARS code to determine their future history. We set up a grid, with $m_1 \in [5, 60]$, $m_2 \in [5, m_1]$, and $\log P \in [0.6, 1.2]$. We take

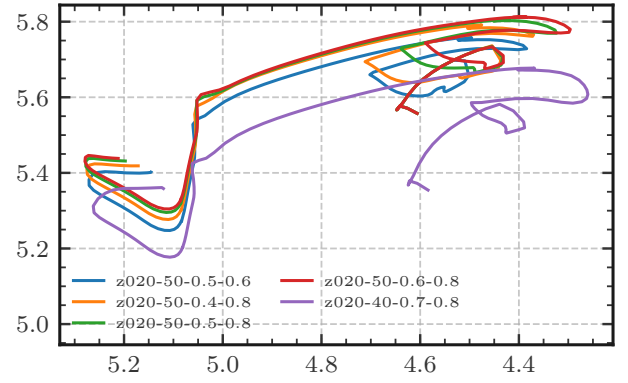
²Petrov’s paper in *Peremennye Zvezdy* is not available digitally, and given the ongoing Russia-Ukraine conflict, the author is unable to access a physical copy to verify the citation and cite this statement.

Binary	BPASS v2.2.1 matches	$M_{1,ZAMS}$	$M_{2,ZAMS}$	q	P_{ZAMS}	$\mu_{\log(\text{age})}$	$\sigma_{\log(\text{age})}$
LSS 3074	sneplot-z020-30-0.3-0.6	30.0	9.0	0.3	3.98	6.54	0.64
	sneplot-z020-25-0.4-0.6	25.0	10.0	0.4	3.98	6.80	0.12
	sneplot-z020-25-0.5-0.6	25.0	12.5	0.5	3.98	6.80	0.13
	sneplot-z020-24-0.4-0.6	24.0	9.6	0.4	3.98	6.83	0.09
	sneplot-z020-24-0.5-0.6	24.0	12.0	0.5	3.98	6.83	0.09
MY Cam	sneplot-z020-50-0.5-0.6	50.0	25.0	0.5	3.98	6.35	0.67
	sneplot-z020-50-0.4-0.8	50.0	20.0	0.4	6.31	6.31	0.72
	sneplot-z020-50-0.5-0.8	50.0	25.0	0.5	6.31	6.21	0.81
	sneplot-z020-50-0.6-0.8	50.0	30.0	0.6	6.31	6.13	0.88
	sneplot-z020-40-0.7-0.8	40.0	28.0	0.7	6.31	6.43	0.66
SMC 108086	sneplot-z004-24-0.5-0.4	24.0	12.0	0.5	2.51	6.47	0.77
	sneplot-z004-23-0.4-0.4	23.0	9.2	0.4	2.51	6.63	0.63
	sneplot-z004-23-0.5-0.4	23.0	11.5	0.5	2.51	6.62	0.65
	sneplot-z004-22-0.4-0.4	22.0	8.8	0.4	2.51	6.70	0.57
	sneplot-z004-22-0.5-0.4	22.0	11.0	0.5	2.51	6.70	0.57
	sneplot-z004-21-0.4-0.4	21.0	8.4	0.4	2.51	6.77	0.50
	sneplot-z004-21-0.5-0.4	21.0	10.5	0.5	2.51	6.76	0.51
	sneplot-z004-20-0.4-0.4	20.0	8.0	0.4	2.51	6.79	0.49
	sneplot-z004-20-0.5-0.4	20.0	10.0	0.5	2.51	6.78	0.49
	sneplot-z004-19-0.4-0.4	19.0	7.6	0.4	2.51	6.80	0.49
TU Mus	sneplot-z020-25-0.3-0.4	25.0	7.5	0.3	2.51	6.69	0.56
	sneplot-z020-24-0.3-0.4	24.0	7.2	0.3	2.51	6.71	0.56
	sneplot-z020-23-0.3-0.4	23.0	6.9	0.3	2.51	6.77	0.52
	sneplot-z020-16-0.6-0.4	16.0	9.6	0.6	2.51	7.14	0.05
V382 Cyg	sneplot-z020-50-0.2-0.6	50.0	10.0	0.2	3.98	6.44	0.56
	sneplot-z020-40-0.3-0.6	40.0	12.0	0.3	3.98	6.52	0.54
	sneplot-z020-40-0.4-0.6	40.0	16.0	0.4	3.98	6.51	0.55
	sneplot-z020-35-0.4-0.6	35.0	14.0	0.4	3.98	6.55	0.56
	sneplot-z020-30-0.4-0.6	30.0	12.0	0.4	3.98	6.60	0.57
	sneplot-z020-30-0.5-0.6	30.0	15.0	0.5	3.98	6.59	0.57
VFTS 352	sneplot-z010-40-0.5-0.6	40.0	20.0	0.5	3.98	6.55	0.53

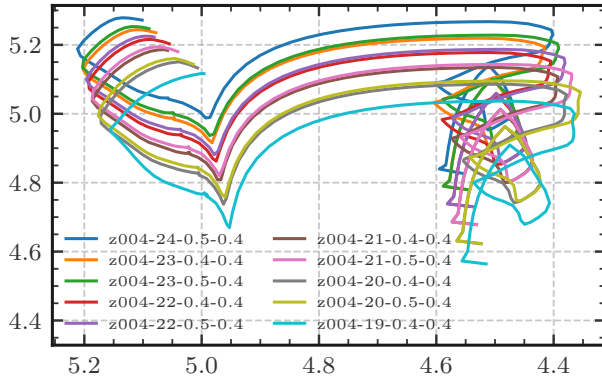
Table 3.2: Matches from the HOKI (Stevance et al., 2020) software. Masses are measured in solar masses, periods in days, and ages in years. Ages were computed as the mean of the ages in log-space where a match occurs, and similar for the standard deviations.



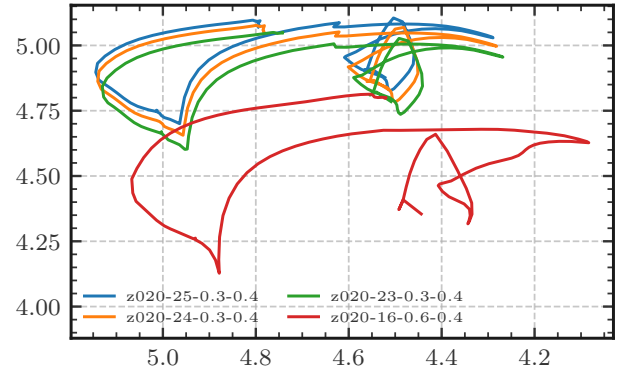
(a) LSS 3074



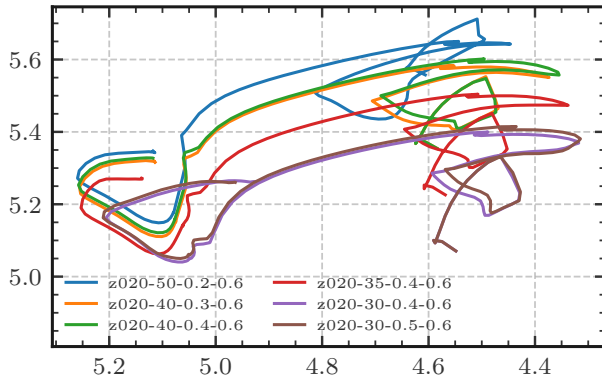
(b) MY Camelopardalis



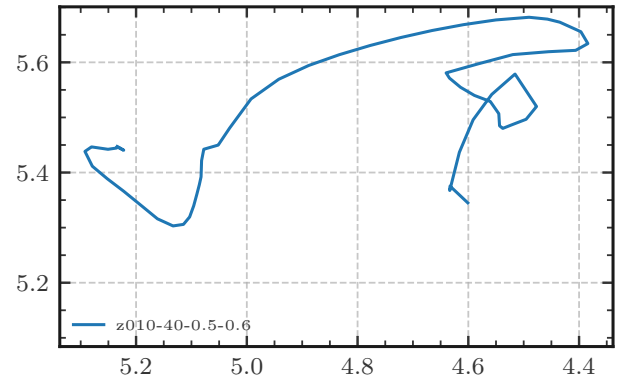
(c) SMC 108086



(d) TU Muscae



(e) V382 Cygni



(f) VFTS 352

Figure 3.1: BPASS v2.2.1 primary models for the matches obtained by HOKI for the parameters in table 3.1. The legend key is the BPASS stellar model file name (omitting the “sneplot-” prefix common to every entry), and is broken down into its parts in table 3.2.

steps of $\delta m = 5 M_{\odot}$ and $\delta \log P = 0.1$ dex. The mass space is set up to enforce a mass ratio of $q < 1$. There are two reasons for this:

1. We do not consider $q = 1$ – equal mass binaries – as equal masses induce extra effects on the MS that STARS is unable to effectively model,
2. We do not consider $q > 1$, as this is equivalent to a relabelling of $m_1 \rightarrow m_2$ and $m_2 \rightarrow m_1$; that is, binary evolution is symmetric under the exchange of mass.

We also compute the grid at three metallicities: z020, z010, and z004, corresponding to the three metallicities in table 3.1 and table 3.2. An example model run for a $Z = 0.020 = Z_{\odot}$, $m_1 = 30 M_{\odot}$, $m_2 = 10 M_{\odot}$, $\log P = 0.6$ d is shown in figure 3.2. This model corresponds to a potential progenitor of LSS 3074.

In figure 3.3 we also display the coverage map in $\log T_{\text{eff}} / \log L$ space of our grid at each metallicity as a density map.

3.3 | Matches to observed close contact models

From our coarse grid, we are able to identify matches to the observed systems. We first interpolate the matches found by HOKI onto our coarse grid. Specifically, in order to preserve the mass ratio as best as possible, we interpolate m_1 onto the grid, and then calculate m_2 as

$$m_2 = \text{round}(q \times m_1) \quad (3.1)$$

where the round function interpolates back onto the grid. Our matches for the systems LSS 3074, MY Cam, V382 Cygni, and VFTS 352 are shown in figure 3.4. We observe that on this coarse grid, we match V382 Cyg on two models:

1. $m_1 = 30 M_{\odot}$, $m_2 = 15 M_{\odot}$, $\log P = 0.6$ d, from an age of 63 606 358 yr to 63 611 433 yr (i.e., a matching time of 5075 yr), and
2. $m_1 = 40 M_{\odot}$, $m_2 = 10 M_{\odot}$, $\log P = 0.6$ d, from an age of 34 366 921 yr to 34 373 848 yr (i.e., a matching time of 6927 yr).

We also note an absence of matches for two of our systems: TU Mus and SMC 108086

3.4 | Exotic binaries arising from contact systems

3.4.1 | Double contact systems

3.4.2 | Reverse mass transfer

3.4.3 | Overflow through $\mathcal{L}_2$

3.5 | Chapter Summary

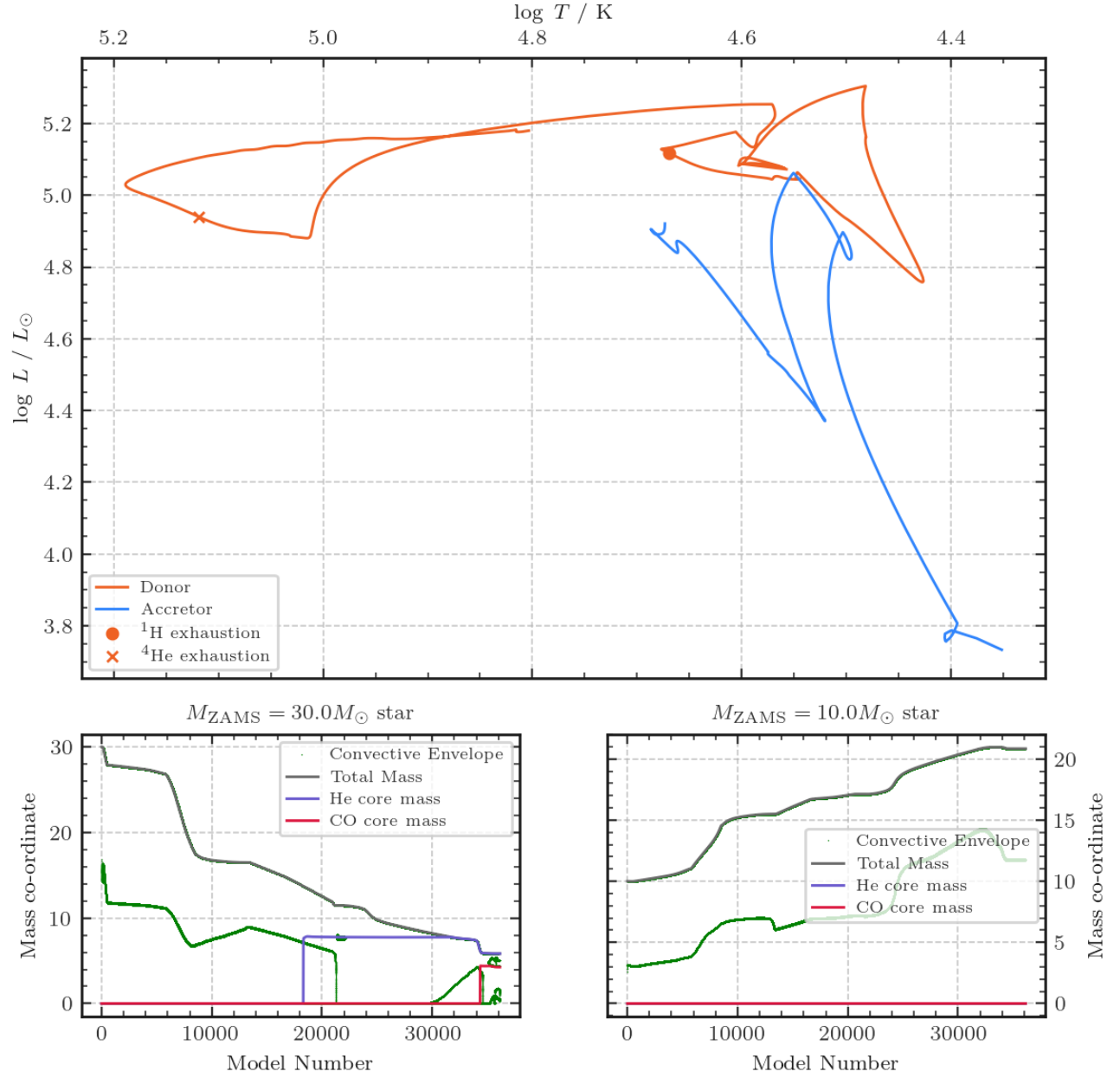


Figure 3.2: A sample grid execution run, matching the ZAMS parameters of LSS 3074. The top panel is an HR diagram of both stars, with core abundance exhaustion marked as symbols on the figure. The bottom two panels are Kippenhahn diagrams showing both the convective zones of the stars, and the cores.

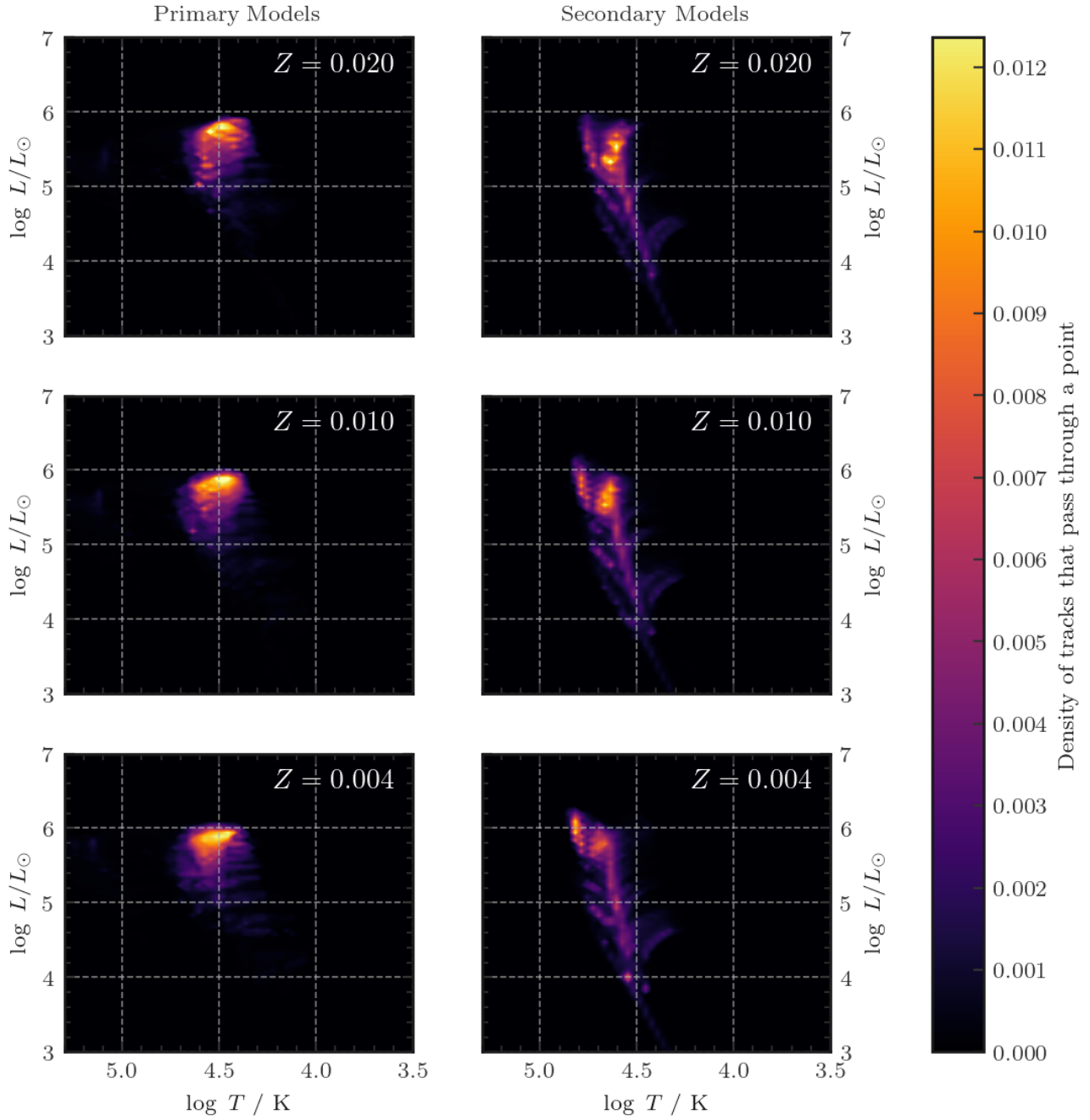


Figure 3.3: The location in HR-space that the grid we compute occupies, for the three metallicities we presently cover.

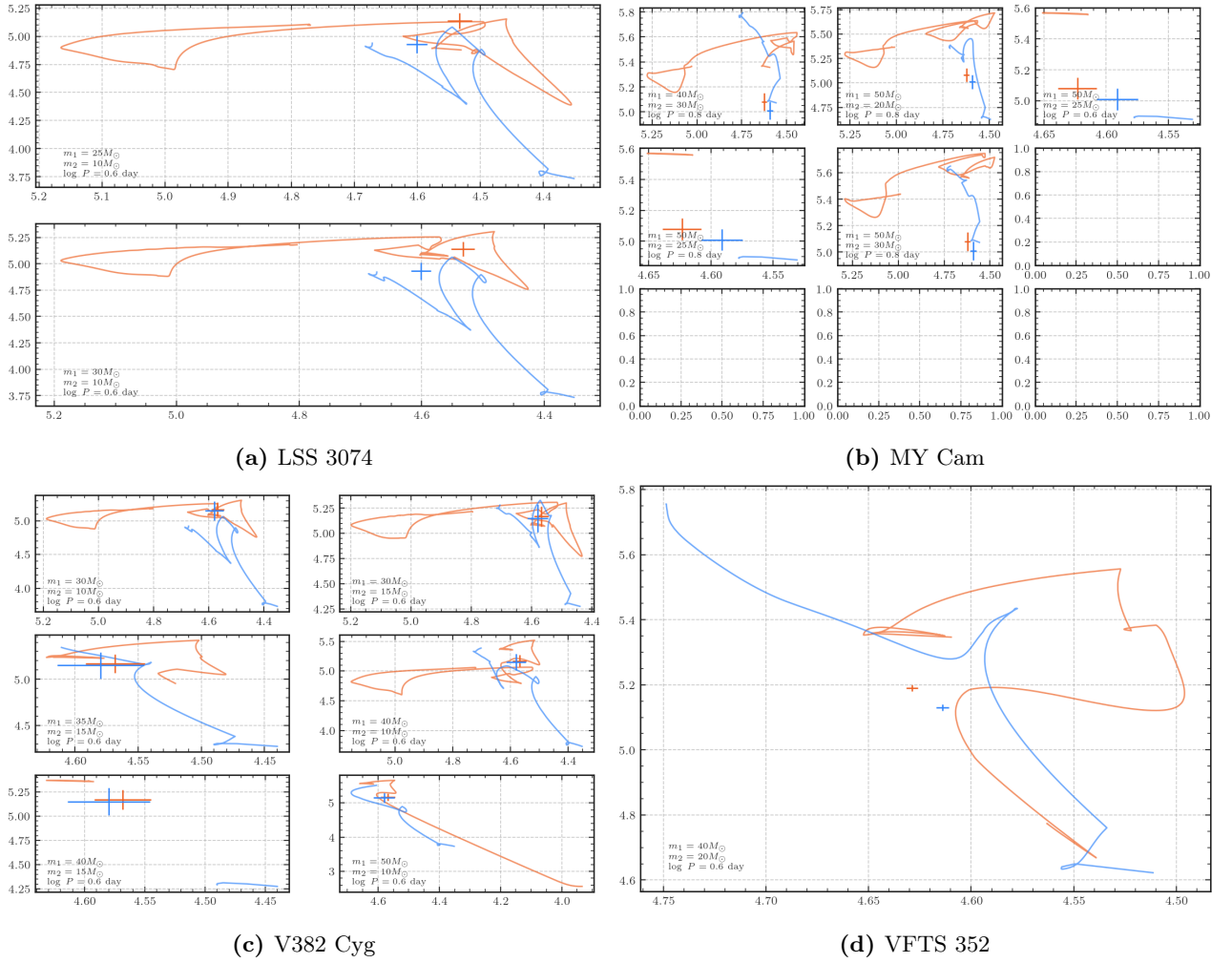


Figure 3.4: Detailed models of the close contact binaries found in section 3.1.2. Not shown are SMC 108086 and TU Mus, which both only have matches outside of the coarse grid. Also not shown is a match for MY Cam, which has a primary which is more massive than the grid.

Isn't it enough to see that a garden is beautiful without
having to believe that there are fairies at the bottom of
it too?

– DOUGLAS ADAMS

Chapter

4

Helium Rich Mass Transfer

SINCE THE FORCE BETWEEN TWO objects due to gravity scales like $F \sim \frac{1}{r_{\text{sep}}}$, another question naturally arises during the consideration of contact binaries. For sufficiently close binaries, one can imagine that mass transfer through RLOF or tidal effects could become reasonably strong, stripping the hydrogen envelope of a main sequence (or shortly post-main sequence) star and exposing the helium-enriched core underneath. If this stripping does not cause the star to retreat within its Roche lobe, then this core material would continue to be accreted, creating helium-enriched matter streams accreting onto its companion.

As discussed in section 2.4, the presence of a companion measurably alters the evolution of a star. Typically this manifests as allowing matter to be exchanged between the components or the gravity of one star warping the photosphere of the other. In this chapter we focus on the first instance, where mass transfer occurs between two stars. In particular, we consider a binary system with ZAMS masses of $30 + 10 M_{\odot}$, and an initial period of $10^{0.6} \approx 3.98$ d. We specifically explore this case as a case study on the effect of helium enriched mass transfer, and in chapter 5 we will explore the effect that this has on a grid of close binaries.

The general background for this section remains as elaborated in chapter 2, however our parameterisation requires us to delineate between two cases: that when variable composition accretion is enabled and the case where it is disabled. These correspond to setting the parameter IVARACC in appendix D to 1 or 0 respectively.

Stage	Letter	Section described
Model begins	a	section 4.1.1
RLOF begins	b	section 4.1.2
Contact begins	c	section 4.1.3
Contact ends	d	section 4.1.4
RLOF ends	e	section 4.1.5

Table 4.1: The five major stages of the evolution of our binary.

4.1 | Evolution overview

The major stages of the binary’s evolution can be broken up into five main stages, explored in the following section. Each stage is assigned a label, which is delineated in table 4.1. Unless otherwise indicated, at every stage we shall be discussing the non-variable composition accretion case. The full evolution for the variable (non-variable) accretion case is shown in figure 4.1 (figure 4.2), with subplots showing the behaviour around the contact phase.

4.1.1 | Pre-interaction (stages A–B)

The pre-interaction phase of both stars is identical in both cases, as the key change enabling this work only begins when RLOF occurs. In both cases, it evolves as a fully detached binary for 4.51 Myr, with the donor almost completing its main sequence evolution. Hydrogen is exhausted in the core, and a helium core develops. In both cases, as the donor nears the Hertzsprung gap hook, it inflates to fill its Roche lobe, causing matter to begin to be exchanged via RLOF and kick-starting the pre-contact phase.

4.1.2 | Pre-contact (stages B–C)

When the donor swells to fill its Roche lobe, it begins to efficiently transfer material to its companion. The increased drag causes the orbit to become hardened; shrinking by approximately 23 per cent. The accretor begins to accrete material, bleeding angular momentum into the system and causing the accretor to spin up. As the contact phase nears, the accretor approaches 40 per cent of the critical rotation rate, as shown in figure 4.3. It reaches approximately 20 per cent when the contact phase begins.

4.1.3 | Contact phase (stages C–D)

4.1.4 | Post-contact (stages D–E)

4.1.5 | Post-interaction (stage E–onwards)

4.1.6 | Future History

4.2 | Atypical abundance profiles

Because the stars are quite close together and have a mass ratio of $1/3$, when RLOF starts it is quite voracious, and becomes even more so during the contact phase. The donor star sheds its hydrogen-rich surface and material from the helium-rich core is exposed, which too becomes part of the accreted material. The material stream becomes increasingly helium-rich, depositing this material on top of a hydrogen-rich surface, causing the accretor to have an atypical abundance profile, as shown pictorially in figure 4.4. The net effect is that there is a helium-rich surface sitting atop a hydrogen-rich layer sitting atop a helium-rich core. At this stage thermohaline mixing is projected to occur, as there is an unstable composition gradient (owing to helium having a higher mean molecular weight than hydrogen).

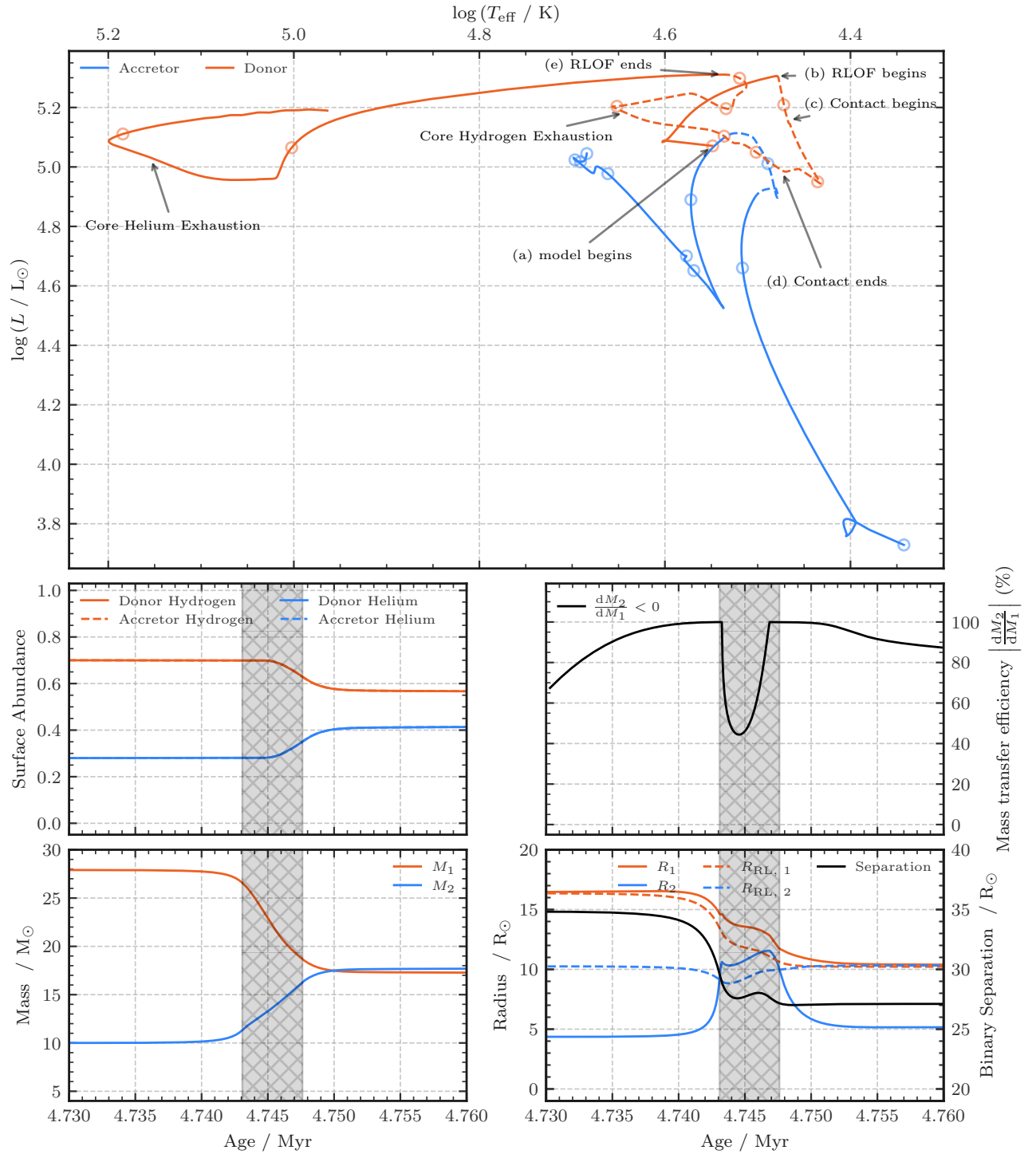


Figure 4.1: The main evolution of a $30 + 10 M_{\odot}$ binary with variable composition accretion enabled. Clockwise from top: the HR diagram with stages in figure 4.4 and table 4.2, the mass transfer efficiency, the separation / radius, the mass, and the surface abundances. In the lower four plots, the gray region shows the contact phase, and the plots are cropped to show the ~ 13 kyr on each side of the contact phase. The dashed line on each HR diagram track shows when the relevant star is overfilling $\mathcal{L}_1$, and contact is when both stars are dashed.

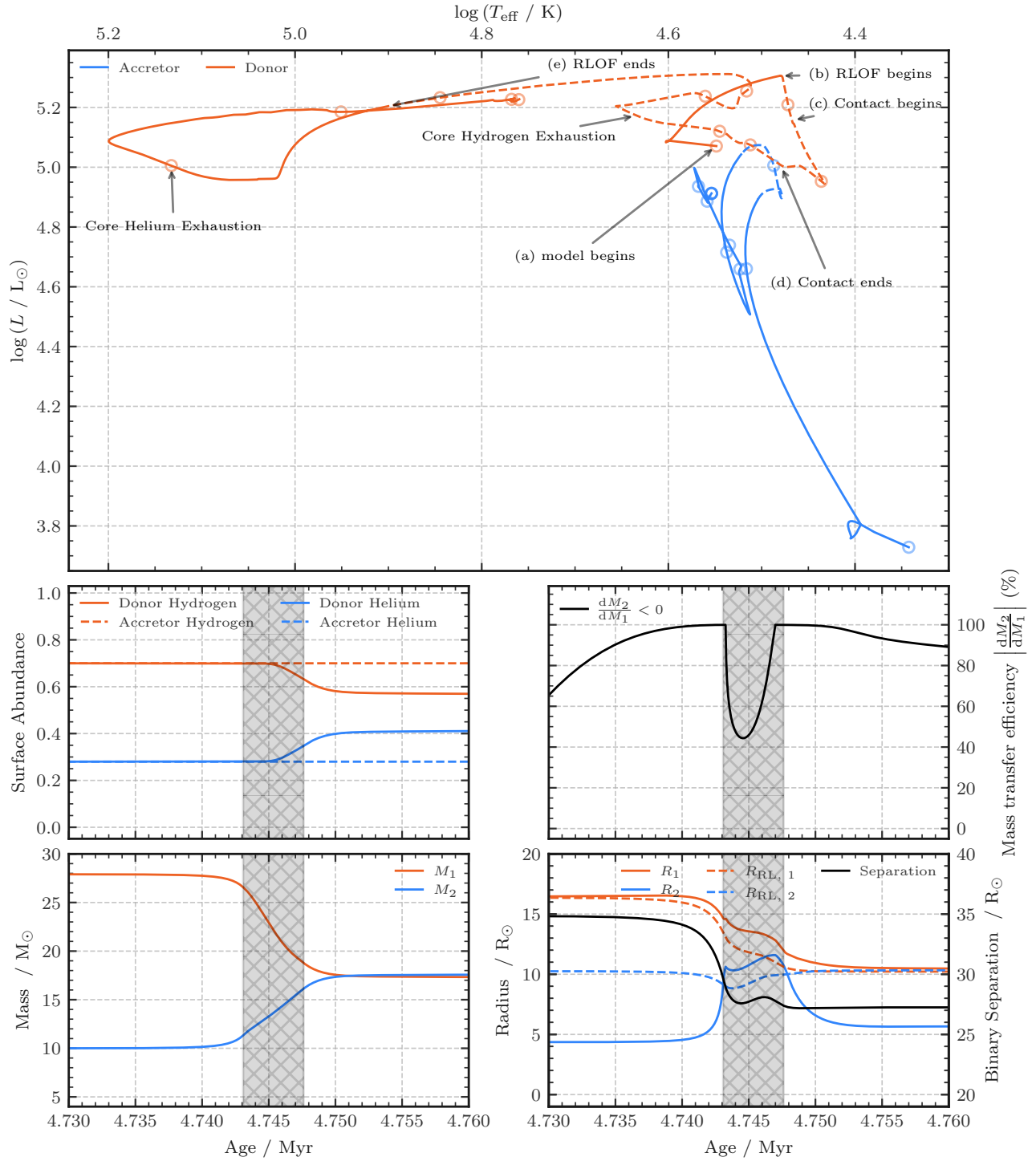


Figure 4.2: As for figure 4.1, but absent variable composition accretion.

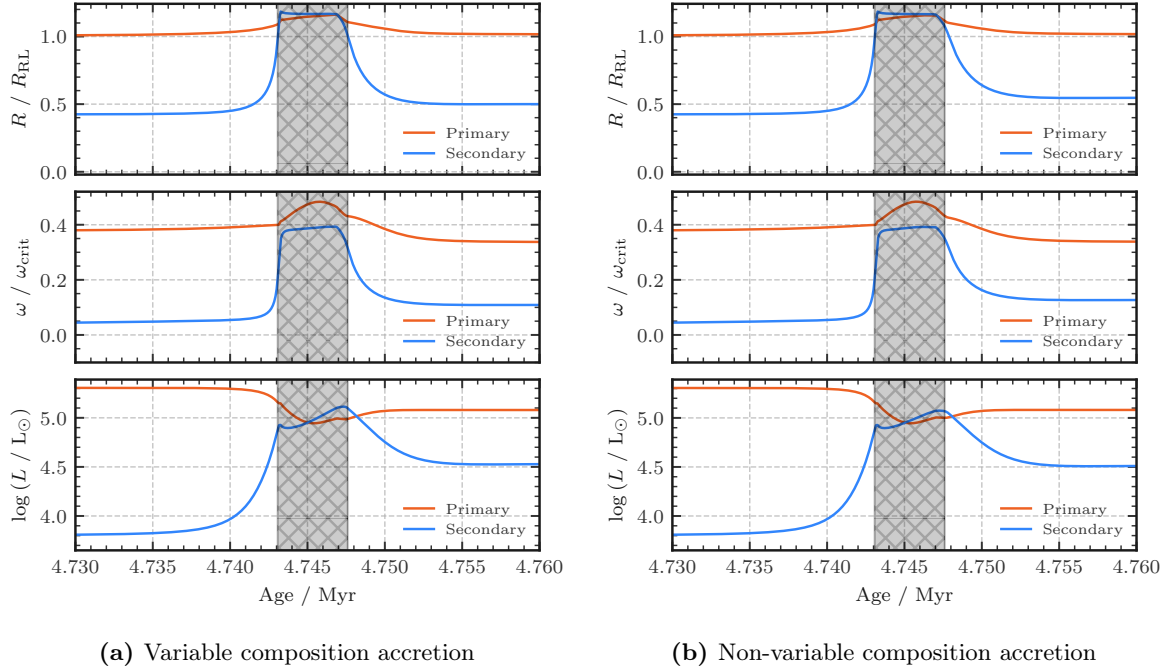


Figure 4.3: The evolution of the radius, rotation rate, and luminosity in the presence of variable composition accretion, shown around the contact phase.

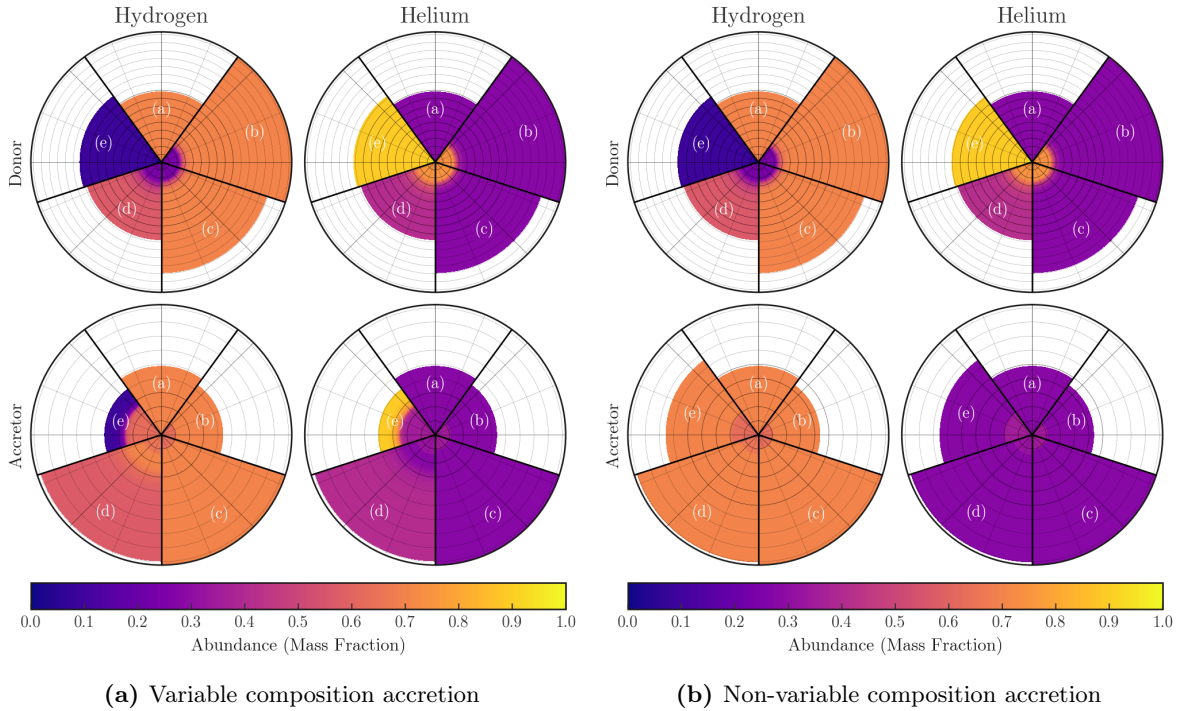


Figure 4.4: The chemical composition of each star as a function of the stellar radius for the donor and accretor at the five key stages of the binary's lifetime as detailed in table 4.2. Letters are as given in the same table. Each grid marking in the radial direction represents $1 R_{\odot}$.

		$t = 0$ (a)	Star 1		Star 2		Star 1		Star 2	
			Begin	End	Begin	End	Begin	End	Begin	End
			(b)	(e)	(c)	(d)	(b)	(e)	(c)	(d)
Donor	$M / M_{\odot}$	30.00	27.93	7.56	26.91	17.43	27.93	7.57	26.91	17.27
	$R / R_{\odot}$	8.82	16.32	10.20	13.88	9.76	16.32	10.15	13.88	9.77
	$\log L / L_{\odot}$	5.07	5.30	5.25	5.16	4.94	5.30	5.25	5.16	4.98
	X_{surf}	0.70	0.70	0.09	0.70	0.57	0.70	0.09	0.70	0.57
	Y_{surf}	0.28	0.28	0.89	0.28	0.41	0.28	0.89	0.28	0.42
Accretor	$M / M_{\odot}$	10.00	9.99	20.97	11.00	14.57	9.99	20.77	11.00	14.52
	$R / R_{\odot}$	4.87	4.36	4.02	9.23	8.98	4.36	6.53	9.23	9.02
	$\log L / L_{\odot}$	3.73	3.81	4.89	4.90	5.06	3.81	4.73	4.90	4.95
	X_{surf}	0.70	0.70	0.09	0.70	0.57	0.70	0.70	0.70	0.70
	Y_{surf}	0.28	0.28	0.89	0.28	0.41	0.28	0.28	0.28	0.28
Binary	$P_{\text{bin}} / \text{d}$	3.98	3.85	4.41	3.14	2.52	3.85	4.38	3.14	2.54
	$a / R_{\odot}$	36.14	34.74	34.56	30.28	24.71	34.74	34.33	30.28	24.78
	Age / kyr	0.00	4707.16	6450.23	4712.75	4716.42	4707.16	6448.11	4712.75	4716.70

(a) Variable composition accretion
(b) Non-variable accretion

Table 4.2: The key evolutionary parameters of both stars and the binary configuration at five critical points: the initial parameters, and the beginning and ends of both epochs of RLOF. All values are rounded to 2 decimal places. The letter in parentheses represents which segment of Figure 4.4a (for Table 4.2a) or Figure 4.4b (for Table 4.2b) the abundances are given in.

4.3 | Comparison between variable and non-variable composition accretion

In figure 4.5 we display a comparative HR diagram between the cases of variable and non-variable composition accretion. In each case an $30 + 10 M_{\odot}$ binary was evolved with the only difference between the two being whether material was allowed to have variable composition accretion or the standard accretion prescription. In the bottom panel only the companion is shown, as the donor has identical evolution in both cases.

The first and most striking conclusion that we can draw from this result is that the companion is, in the variable composition accretion case, hotter and more luminous than the non-variable case.

4.4 | Implications

4.4.1 | Spectroscopic masses

Cats can disappear.

FOURTH OF THE SIX IMPOSSIBLE THINGS

Throughout the history of binary evolution theory, a consistent thread has been the consistent underprediction of the mass predicted by models (the “evolutionary mass”) vs the mass inferred from spectroscopic studies of stars (the “spectroscopic mass”). Summaries are found in works such as Herrero et al. (1992); Massey et al. (2009); Bouret et al. (2013); Massey et al. (2013); McEvoy et al. (2015); Markova et al. (2018); Ramachandran et al. (2018); Bestenlehner et al. (2020). The reason has been left as either deficiencies in the physics of model atmospheres (Herrero et al., 1992) or non-included or not well-understood physics (Bestenlehner et al., 2020).

This phenomenon is not unique to one code or methodology. To name but four, it has been observed in:

1. BONNSAI (work in Bestenlehner et al. 2020, BONNSAI in Schneider et al. 2014),
2. the work of Maeder (1990) (Herrero et al., 1992),
3. the Geneva tracks (work by Markova et al. 2018, Geneva code by Ekström et al. 2012), and
4. the PoWR code (work by Ramachandran et al. 2018, PoWR code in Gräfener et al. 2002; Hamann & Gräfener 2003; Sander et al. 2015).

If luminosities are systematically incorrect, as discussed above, then there would be a corresponding systematic incorrect calculation of the mass; that is, a traditional mass-luminosity relationship may not hold. We show in figure 4.6 the relationship between the evolutionary mass and luminosity for both variable and non-variable composition accretion cases, as evolved in subsection 4.1.6. We also display this for integer mass single-star models. Unsurprisingly, the non-variable accretion case evolves like a $24 M_{\odot}$ model – the initial mass of the model when this single-star evolution began was $23.71 M_{\odot}$, and the abundance profile typifies a single star; since the mass transfer was hydrogen-dominated, the profile is not inverted and thus resembles a single star.

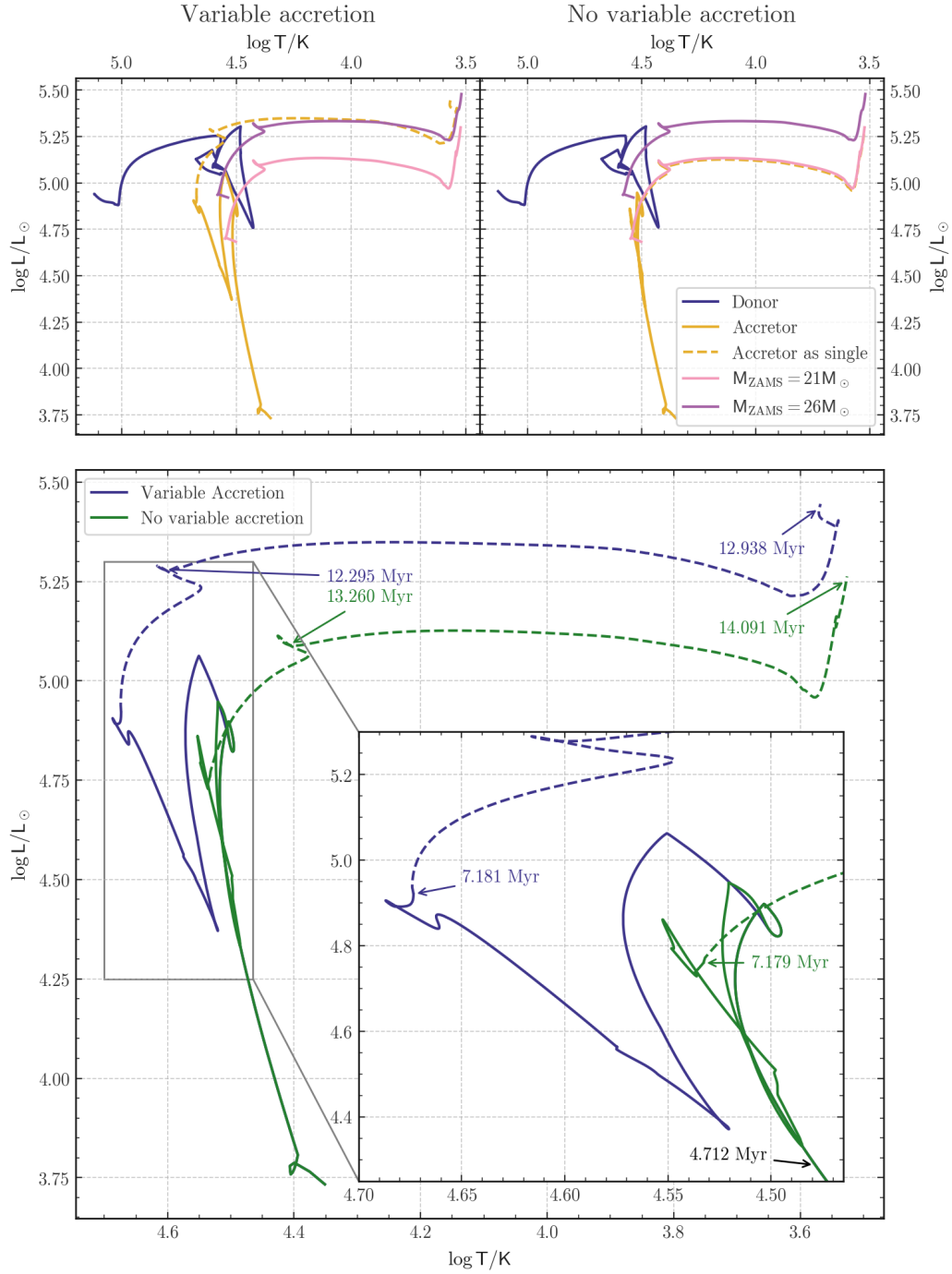


Figure 4.5: An HR diagram showing the evolution of our model both with (top left) and without (top right) the variable composition accretion prescription. The bottom panel shows only the evolution of the secondary. The point of divergence between the two models is shortly into the contact phase. The two dashed lines on the bottom HR diagram represent models evolved as the final model of the binary secondary, as a single star. On the top HR diagrams, the pink and brown models are single stars evolved with ZAMS masses of 21 and 26 $M_{\odot}$ respectively.

On the other hand, the variable accretion case would (if observed) initially appear to be more massive than a $24 M_{\odot}$ star (despite being $23.66 M_{\odot}$ initially), eventually appearing to be approximately $21 M_{\odot}$ – i.e., $\sim 3 M_{\odot}$ less massive than it actually would be.

Consequently, any masses inferred from spectroscopy or luminosity measurements may be an underestimate of the true (evolutionary) mass. This would affect, for example, the inferred masses of compact objects and thus may have implications for our understanding of the progenitors of gravitational waves.

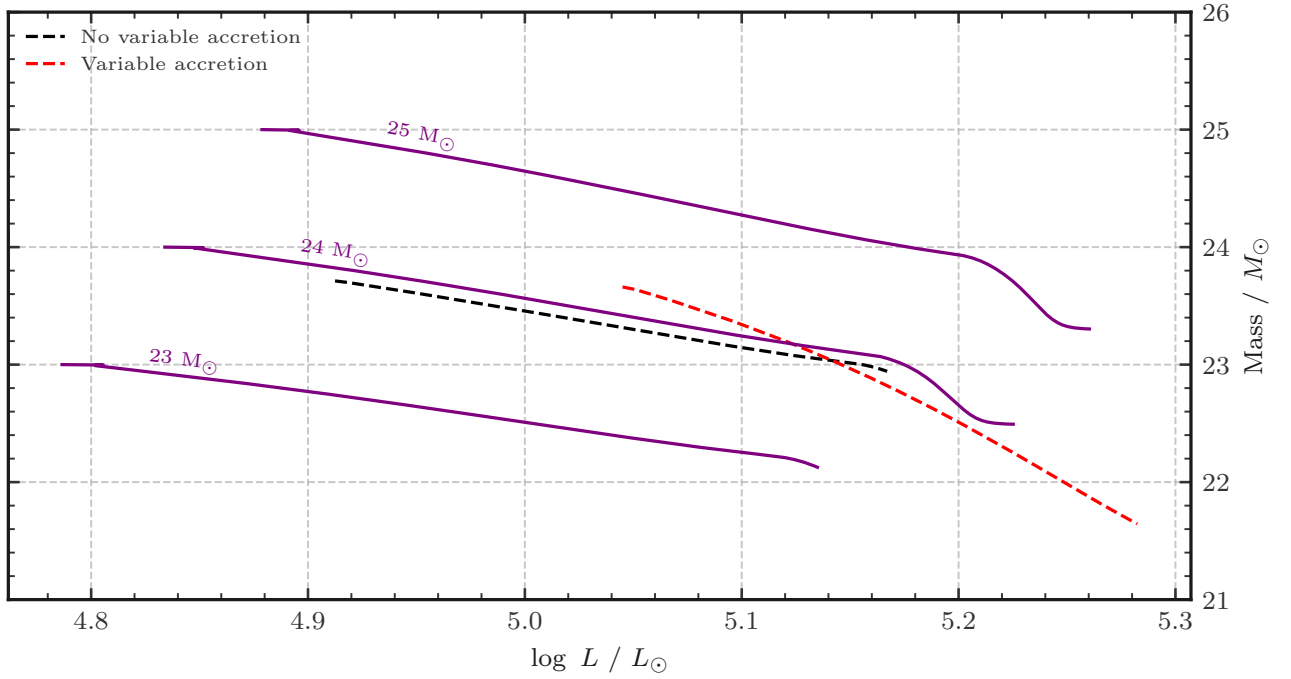


Figure 4.6: The (evolutionary) mass-luminosity relationship in the presence and absence of variable composition accretion. Also shown are the curves for reference models of between $M_{\text{ZAMS}} = 21 M_{\odot}$ and $26 M_{\odot}$, which were evolved as a single star.

4.5 | Chapter Summary

I also don't know if any of it makes sense I kind of entered
a trance

– ARTHUR YOUNG

Chapter

5

Population properties of overcontact binaries and helium rich mass transfer

HAVING ESTABLISHED THE NATURE and specific cases of both overcontact binaries and helium-rich mass transfer, we are now in a position to discuss the implications of these phenomena into a larger population. We shall build to this by discussing four core aspects of the work performed so far, and observing the regions of the grid which allow these kinds of interactions.

The four core phenomena we examine are the distributions of the lengths of the contact phase (cf. chapter 3), multiply-interacting systems (cf. section 3.4.1), reverse mass transfer (cf. section 3.4.2), and helium-rich mass transfer (cf. chapter 4).

5.1 | Helium-Rich mass transfer

To determine the prevalence of helium-rich mass transfer, for every binary in our grid we track whether it becomes helium-dominant ($X_{\text{He}} > 0.5$) whilst the donor is filling its Roche lobe. If this occurs, we assume that the streamed material is becoming helium rich and will lead to a binary of the type studied in chapter 4. The results of this are shown in figure 5.1. There is somewhat of a critical value of mass ratio, where for systems with $q < 0.5$ initially, helium rich mass transfer happens essentially independently of the initial period (within our grid range). There are still a handful of systems above this critical mass ratio, however they are rarer, and tend to be more metal-poor.

This metallicity dependence is not unsurprising, as the Vink et al. (2001) mass-loss scheme does contain a dependence on the metallicity (cf. equation (2.25)) – thus metal-poor binaries have weaker mass loss rates, allowing interaction to start later, and allowing the models with mass ratios closer to unity to exchange helium.

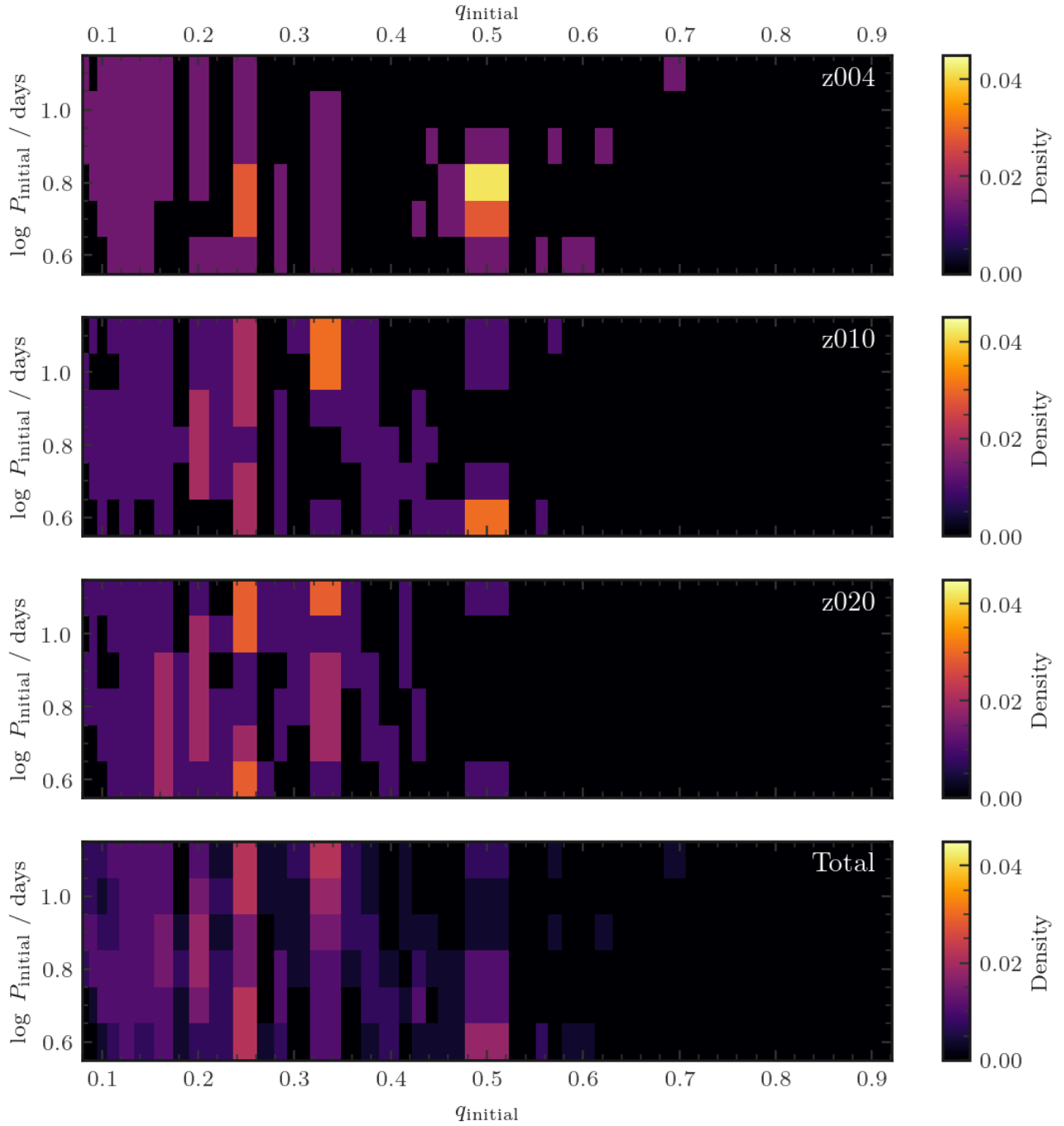


Figure 5.1: The density of systems which undergo helium-rich mass transfer across our grid. There is effectively a cutoff point at $q_{\text{initial}} = 0.5$ where systems above this mass ratio undergo helium-rich mass transfer significantly more rarely than systems below this.

5.2 | Mass transfer reversal

5.3 | Multiply-interacting contact binaries

5.4 | Contact phase length distributions

And the memories were lost long ago
 But at least you have beautiful ghosts.
 – TAYLOR SWIFT

Post-Supernova Kinematics: The Natal Kick

THE TERMINUS OF A MASSIVE STAR'S life is a core-collapse supernova, when the hydrostatic equilibrium between the pressure of the star and its self-gravity is broken, and the stellar core collapses in on itself. Once the star reaches nuclear densities, neutron degeneracy is able to stabilise the remnant, causing a shockwave to propagate outwards from the core. The energy released is roughly an order of magnitude greater than the binding energy of the star (Eldridge & Tout, 2018), resulting in the ejection of an amount of mass we denote M_{ej} and leaving a remnant of mass M_{rem} behind. If the ejection is asymmetric, the collapse endows the nascent remnant with a velocity v_{k} whose overall direction is antialigned to the more massive ejecta. This motion is termed a natal supernova kick.

Given an observation of the proper velocity of a compact remnant, it is in principle possible to trace its motion back and ascertain its kick velocity. However, this is not necessarily a straightforward endeavour. If the proper velocity is high and the remnant old, it has travelled quite a distance from its explosion site to be located at the point the observation has been taken. In addition, the star has potentially had its proper velocity modified by gravitational attraction from either other runaway stars or its host galaxy's potential. One would require the exact location in 3D space as well as the 3D velocity and remnant age in order to ascertain the motion, but from there the inverse problem becomes straightforward.

6.1 | Overview

As a consequence of conservation of momentum, it can be shown that if the ejecta is assumed to eject along a preferred axis, the velocity of the kick can be written as

$$v_{\text{k}} = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta \left(\frac{M_{\text{NS}}}{M_{\text{rem}}} \right), \quad (6.1)$$

for free parameters α and β (both measured in km s^{-1}). Here, M_{NS} is the mass of a NS; if the kick is applied to a NS, then $M_{\text{NS}} = M_{\text{rem}}$ and the coefficient reduces to unity. If the kick is applied to a BH, then we can take $M_{\text{NS}} = 1.4 M_{\odot}$, which serves to convert the distribution from one in velocity-space to one in momentum-space; see Bray & Eldridge (2016, 2018); Richards et al. (2023) for further detail.

This parameterization introduces, effectively, four free parameters in $\alpha, \beta, \theta, \phi$ (the latter two being azimuthal and equatorial angles of the ejecta mass). We assume that the preferential axis is sampled uniformly across the 4π steradians of the star, and that our only two free parameters are α and β . It is pertinent to note two things due to this assumption: firstly, we are not assuming that the ejection occurs along one axis and that the kick is aligned along a different, random, axis. Rather, we are assuming that the asymmetric ejection occurs along a randomly sampled axis and that the kick is antialigned to that. The second note, explored further in section 6.1.2, is that there is considerable uncertainty around the nature of the kick and the precise mechanism that causes it to eject along a given axis. This is the reason we sample the kick direction uniformly, though we will simulate sufficiently many systems (on the order of $\mathcal{O}(10^{12})$) that this will be marginalised over.

The existence of two free parameters, α and β , mean that a choice on the values must be made. The “traditional” way to choose these parameters, as outlined in Bray & Eldridge (2016, 2018), is to refine them based on isolated pulsar velocities. This methodology, however, has significant drawbacks. Notably, the reliance on one observation can engender biases in the result at the expense of no longer predicting accurate results for other observables. For instance, in the limit of ultra-stripped supernovae (USSNe) where the ejecta-remnant mass ratio becomes small, the fiducial kick in Bray & Eldridge (2018) would produce kick velocities that are strongly negative (indicating a strong kick aligned with the direction of motion of the ejecta), contrasted with the weak kicks we expect from these supernovae (e.g., Willcox et al., 2021). It becomes evident that a new series of empirical tests are needed, whereupon more constraints than simply pulsar velocities are tested. In this Chapter, we outline four such tests that form the basis of the work of Richards et al. (2023). These tests – using observed LVK Collaboration merger rates, period-eccentricity distributions of Binary Neutron Stars (BNSs), isolated pulsar velocities, and production of USSNe – allow us to narrow the parameter space significantly, with the promise of future refinement with more observations (such as through LVK Collaboration O4).

6.1.1 | Parameter space

We perform our computation on three populations, termed FINEGRID, WIDEGRID, and HOBBS, whose properties are outlined in table 6.1. These populations simulate 1,000 kicks per system sampled. The aim of the three populations is twofold. Firstly, by generating FINEGRID and WIDEGRID, we are able to look both at fine detail in the region at which the most likely value is expected to lie and also the large-scale behaviour over wider parameter ranges.

Population Name	α_{low} [km s ⁻¹]	α_{high} [km s ⁻¹]	β_{low} [km s ⁻¹]	β_{high} [km s ⁻¹]	Step Size [km s ⁻¹]
FINEGRID	0	400	-200	200	10
WIDEGRID	0	700	-350	350	50
HOBBS	-	-	-	-	-

Table 6.1: The populations under study. HOBBS is a fiducial kick prescription as given by Hobbs et al. (2005), which is a Maxwellian distribution with distribution parameter $\sigma = 265 \text{ km s}^{-1}$, based on an empirical fit to the velocities of 233 pulsars. All three prescriptions simulated 1,000 kicks for each system, with the resultant kick being the mean velocity sampled, and the direction of the kick being sampled uniformly across the 4π steradians of the orbit.

Secondly, by using Hobbs et al. (2005), we are able to create a population governed by a kick that has been well-tested in literature (see, for example, Bouffanais et al., 2021; Broekgaarden et al., 2021; Fragione, 2021; Igoshev et al., 2021; Mandel & Broekgaarden, 2022) which serves as a fiducial comparison. From a Bayesian perspective, we are essentially choosing the HOBBS distribution as our prior for the express purpose of a Bayesian test in section 6.3.

The choice of extremal values for α and β are somewhat arbitrary. figure 6.1 shows a wider parameter space with $\alpha \in [0, 1000]$, $\beta \in [-500, 500]$. Regions in the upper right (bottom left) corner of the plot have quite high (low) mean kick velocities. Although both grids encroach on these regions somewhat, there is a tradeoff between exploring the grid space fully and using computational resources efficiently. Our mean kick velocities should not be too extreme; the mean for a Hobbs et al. (2005) kick is $\sigma = 265 \text{ km s}^{-1}$, for a Verbunt et al. (2017) is $\sigma_1 = 75 \text{ km s}^{-1}$ and $\sigma_2 = 316 \text{ km s}^{-1}$, and for the observed Willcox et al. (2021) kick $\sigma_1 \approx 30 \text{ km s}^{-1}$, $\sigma_2 \approx 230 \text{ km s}^{-1}$. These¹ are shown on the grid as black lines.

Whilst the exact parameters can be shifted somewhat and maintain the statistical integrity given these constraints, there are a few additional heuristic constraints we place on the regions to select. First, we wish to capture low-velocity USSN kicks (as will be discussed in section 6.5), necessitating a region near $\alpha = \beta = 0$ (where the kick would be identically zero for all ejecta-remnant mass ratios). Secondly, the fiducial work of Bray & Eldridge (2018) found $\alpha = 100$, $\beta = -170$. Although a goal was to (ideally) remove $\beta < 0$ as a possible value (for USSN again), to impose that as a prior when the fiducial value sat within that range would have gone contrary to typical Bayesian practices. Thirdly, we expected but could not guarantee that the magnitude of each parameter would shrink with further tests, given the large uncertainty bands that Bray & Eldridge (2018) found (hence broadening our search area beyond $|\alpha| = 100$, $|\beta| = 170$). Finally, we elected to increase the grid search region by 75 per cent for WIDEGRID as this would allow us to just reach $\bar{v}_k = 700 \text{ km s}^{-1}$, which is where the Cumulative Distribution Function (CDF) of the Hobbs et al. (2005); Verbunt et al. (2017); Willcox et al. (2021) distributions all (approximately) begin to level out (signifying few velocities above those), as shown in figure 6.7.

6.1.2 | Natal Kick Prescriptions

There's a place called Wonderland.

FIFTH OF THE SIX IMPOSSIBLE THINGS

There are conflicting accounts as to the cause of the kick (see, for example, Fryer et al., 1998; Arzoumanian et al., 2002; Bombaci & Popov, 2004; Hobbs et al., 2005; Bray & Eldridge, 2016; Verbunt et al., 2017). There are conflicting accounts as to the form and distribution of the kick too, as discussed in, for example, Blaauw (1961); Gunn & Ostriker (1970); de Loore et al. (1975); Sutantyo (1978); Hobbs et al. (2005); Baker et al. (2008); Wongwathanarat et al. (2013); Bray & Eldridge (2018); Giacobbo & Mapelli (2020); Mandel & Müller (2020).

We stress that we are not attempting to prescribe *which* prescription is somehow “best”, but seek an easy-to-configure kick which allows us to configure kicks without changing any of the physics of how our population was formed. This can be done by modifying the kick of Bray & Eldridge (2016), which takes two free parameters

¹Due to the grid size and the random sampling, a true contour is difficult to observe due to sampling noise. The contours shown are thus linear fits to the actual contours.

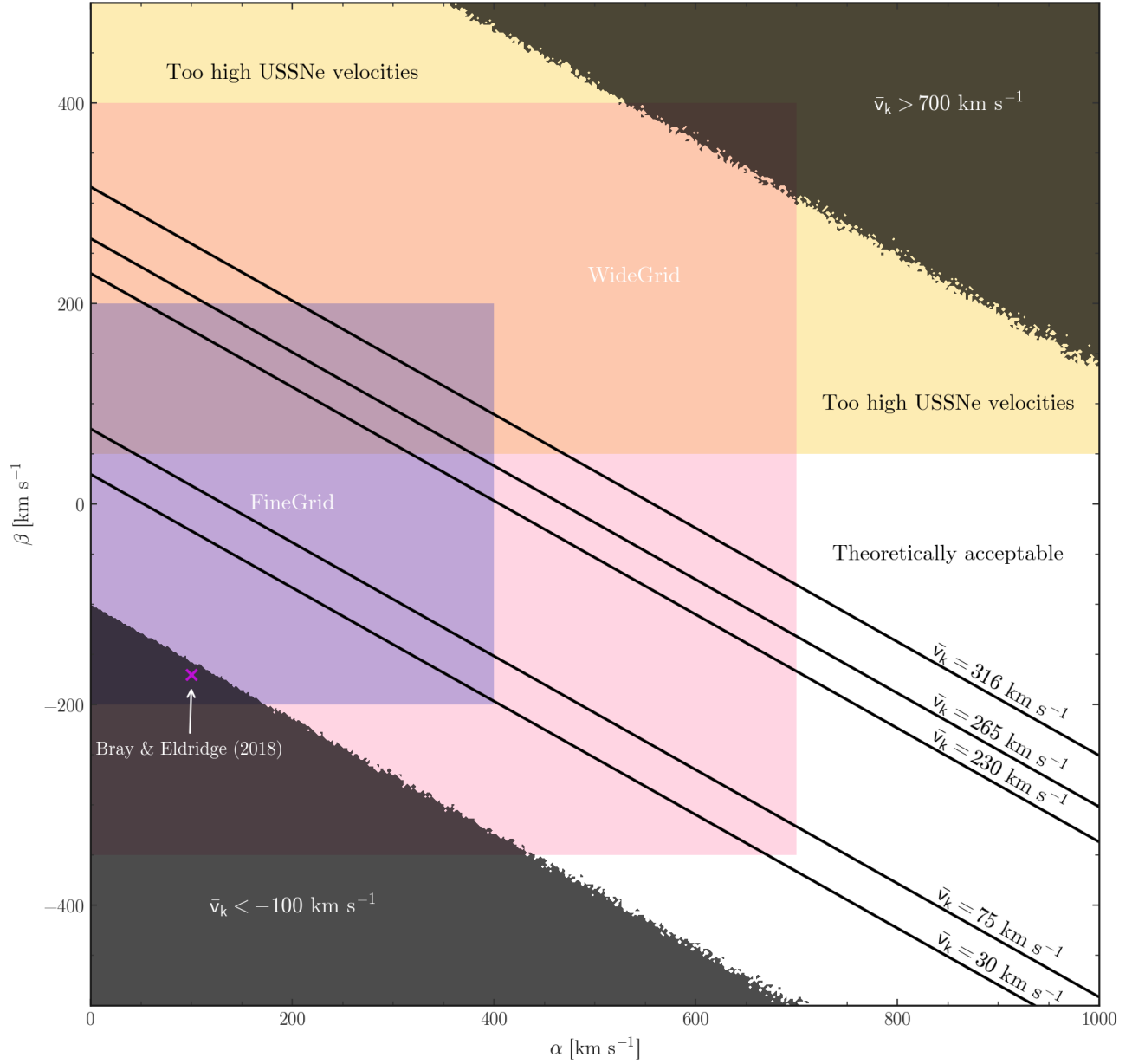


Figure 6.1: A schematic of how FINEGRID and WIDEGRID are related. The mean kick velocity v_k was computed by averaging the kick velocities of 1,000 simulated kicks for each combination of α and β , with $m_{\text{rem}} \in [0.1, 2.5]$ and $m_{\text{ej}} \in [0.1, m_{\text{rem}}]$ sampled uniformly. The pink cross represents the point $\alpha = 100, \beta = -170$, which are the fiducial kick parameters from Bray & Eldridge (2018).

which can be changed without imposing constraints on physics. This prescription is hereinafter referred to as the “Bray kick”. Similarly, we require for the purposes of computing log-Likelihoods in section 6.3 a comparison kick, for which we select the kick of Hobbs et al. (2005), a well-known empirically derived kick. This prescription is hereinafter referred to as the “Hobbs kick”.

The Bray kick velocity is prescribed by equation (6.1). The coefficient to the β term, $\frac{M_{\text{NS}}}{M_{\text{rem}}}$, serves to generalise the kick from purely NSs to BHs as well. When applied to a BH, M_{NS} is taken to be $1.4 M_{\odot}$, and conversely when applied to a NS, M_{NS} is by definition the remnant mass, so the kick velocity becomes

$$v_{\text{k}} = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta. \quad (6.2)$$

On the other hand, the Hobbs kick velocity is given by the Probability Density Function (PDF)

$$P(v_{\text{k}}) \, dv_{\text{k}} = \sqrt{\frac{2}{\pi}} \frac{v_{\text{k}}^2}{\sigma^3} \exp \left(-\frac{v_{\text{k}}^2}{2\sigma^2} \right) \, dv_{\text{k}}, \quad (6.3)$$

where $\sigma = 265 \text{ km s}^{-1}$ is the distribution parameter found by fit in Hobbs et al. (2005) to the velocities of 233 pulsars.

6.1.3 | The TAKAHE orbital evolution code

As part of this work, we have developed the TAKAHE binary evolution code, which solves the coupled equations of motion equation (2.42) and equation (2.43). Originally, to facilitate the equations being solved rapidly, we wrote the code as a PYTHON wrapper to a JULIA (Bezanson et al., 2017) integrator. However, the PYJULIA library (Arakaki et al., 2023) proved to be fragile² and is not in fact supported officially on recent Python versions. To facilitate the potential interoperability between KAITIAKI and TAKAHE (and to remove a prolonged loading time when the JULIA runtime was loaded, given that PYJULIA precompiles our modules), we elected to rewrite the core integrator into PYTHON and to optimise it using traditional PYTHON methods instead.

There are two relevant functions in TAKAHE. The remaining functions are utility functions that allow us to (for instance) convert between lookback time / redshift, and period / separation. These functions are `integrate_eoms()` (code sample A.1), which solves the coupled differential equations for $\frac{da}{dt}$ and $\frac{de}{dt}$ (equation (2.42) and equation (2.43)), and `integrate_timescale()` (code sample A.2), which solves equation (2.46) for the coalescence time of the binary. With sufficient performance optimisations,³ `integrate_timescale()` is able to achieve speeds of approximately 18,000 numerical integrations per second on an office PC⁴ on a sample of Neutron Star - Black Hole Binary (NS-BH) binaries in the LISA dataset (Tang, priv. comm.).

Algorithmically, both codes employ a simple Euler integration routine (where $f_{n+1} = f_n + \frac{df_n}{dt} dt$) and the choices work as follows:

1. When `integrate_eoms()` takes a step in time, it evaluates da and de . It then computes the relative changes in both e and a that arise by taking this timestep (through $\delta f = \left| 0.01 \frac{f}{f} \right|$), and takes a timestep

²It requires custom installation procedures (so cannot be included as a dependency in e.g. a PYTHON package), would hang with no discernible output or ability to ascertain why, and could not handle KeyboardInterrupt exceptions in PYTHON

³Primarily, just-in-time compilation with the NUMBA (Lam et al., 2015) library.

⁴A Dell OptiPlex 7480 AIO with a 12-core 3.1 GHz Intel Core i5-10500 CPU, running no other programs except for during the final few iterations

corresponding to the smallest change between a and e , or half of the width of the smallest BPASS time bin.

2. `integrate_timescale()` breaks up equation (2.46) into four cases dependent on the initial eccentricity. Let τ be the coalescence time of a circular binary ($\tau = \frac{a_0^4}{4\beta}$), then the cases are:

$$t_c = \begin{cases} \tau & e_0 = 0 \\ \frac{c_0^4}{4\beta} e_0^{\frac{48}{19}} & e_0 < 0.01 \\ \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}} de & 0.01 \leq e_0 \leq 0.99 \\ \tau \frac{768}{425} (1 - e_0^2)^{\frac{7}{2}} & 0.99 < e_0 \end{cases}, \quad (6.4)$$

where all symbols have the same meaning as in section 2.5. The integral in the intermediate case is evaluated numerically, taking a fixed timestep of $de = \frac{e_0}{1000}$ such that the discretised sum becomes nothing more than

$$t_c (0.01 \leq e_0 \leq 0.99) = \frac{e_0}{1000} \frac{12}{19} \frac{c_0^4}{\beta} \sum_{e=0}^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}}, \quad (6.5)$$

and dependence of the coalescence time t_c on the initial separation a_0 is handled through the c_0 term. Since the step size is an arbitrary choice, a tradeoff can be made between accuracy and runtime. figure 6.2 shows the effect of changing the step size ($\frac{e_0}{1000}$ in equation (6.5)) on convergence to a final result, and figure 6.3 shows the runtime as a function of step size on an office computer.

6.2 | Merger Rates of BNS systems

The first test is to match the observed merger rates of BNS systems. To do this, we use the LVK Collaboration catalogue of results from Operating Run 3 (O3). The simplest test is to match the merger rate today, which we denote as $\mathcal{R}_0$, to that observed by LVK Collaboration (LIGO-Virgo, 2021; LIGO-Virgo-KAGRA, 2021). We display this in figure 6.4 alongside the previously reported fiducial values of $\alpha = 100 \text{ km s}^{-1}$ and $\beta = -170 \text{ km s}^{-1}$, the theoretical merger rate calculation performed by Eldridge et al. (2019), and the observed merger rates from LIGO-Virgo (2021) and LIGO-Virgo-KAGRA (2021). We term the contour defined by LIGO-Virgo (2021) (in black) the “LIGO contour”.

Selecting the value to calibrate to is not a straightforward task. Although the LIGO-Virgo (2021) contour is superseded by the LIGO-Virgo-KAGRA (2021) contour, the separation into different inference models means that the merger rates for BNS systems have intractably large error bars covering the entire plane; for instance, their MULTISOURCE (MS) model alone spans $2.11 \leq \log \mathcal{R}_0 \leq 3.23$ which covers FINEGRID entirely and almost entirely covers WIDEGRID. In addition to the lack of constraint, we would be imposing additional parameters into the model insofar as the selection of model – whether, for example, PDB was taken as truth over BGP, MS, or one of the other population models. Whilst an interesting and assuredly insightful endeavour, the aim of this work is instead to develop a simple test that can be used in future on extra populations such as these. A simple criterion which can be extended is valuable, and future work may extend these tests in multiple directions.

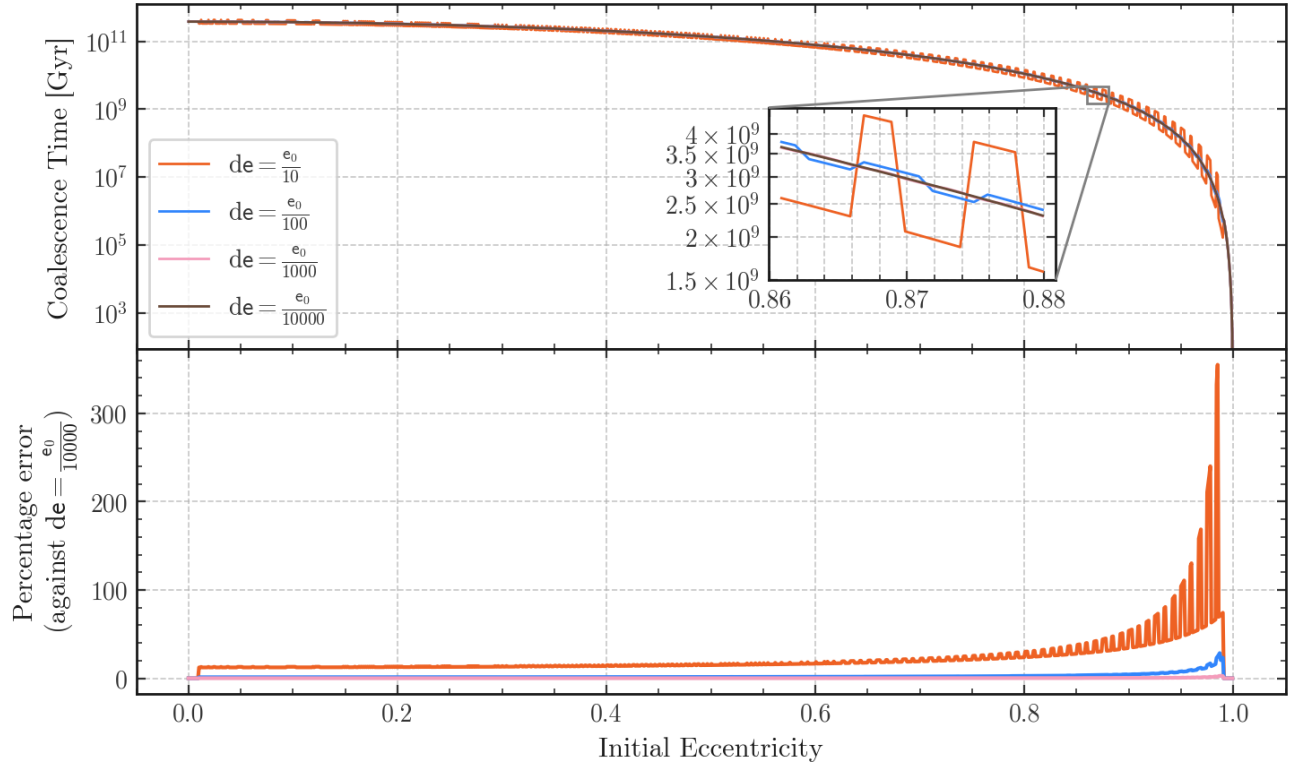


Figure 6.2: The effect of varying de in equation (6.5) on the coalescence time. Although hard to see, the pink line does oscillate above and below the brown line. This binary was assumed to have $m_1 = m_2 = 1.44 M_\odot$ and $a_0 = 1970.5 R_\odot$ (the mean separation of a sample of binaries with $\alpha = 100 \text{ km s}^{-1}$, $\beta = -170 \text{ km s}^{-1}$).

6.3 | Period-Eccentricity Distributions

Since in the absence of any mass transfer the motion of two compact binaries is wholly dependent on their initial separation and orbital eccentricity, they trace out paths in a two-dimensional phase space. In general, since binaries circularise over time, and by the emission of gravitational waves their orbits shrink, a binary will start in the high-period, high-eccentricity region of period-eccentricity space, and evolve towards the low-period, low-eccentricity region. As a result, given a population of binary stars, we can design a density plot, where the density is effectively the likelihood of a binary selected at random having the parameters we sample.

Computationally, for each binary in our sample, we trace its path through period-eccentricity space, and at every timestep we add $w \times dt$ to the density plot, where dt is the timestep, and w is the weight term output from the TUI code proportional to the IMF weight (where the constant of proportionality is related to the uniformly sampled kick direction). The result of this computation, for the parameter values that lie along the LIGO contour, are shown in appendix C. We also show the same result in figure 6.5. Given a catalogue of N (observed) binary stars, we can then compute the likelihood of the model as

$$\ln \mathcal{L} = \sum_{n=0}^N \ln p(\ln P_n, e_n), \quad (6.6)$$

where $p(\ln P_n, e_n)$ is the content of the density plot at the location of the n^{th} observed BNS system. Then,

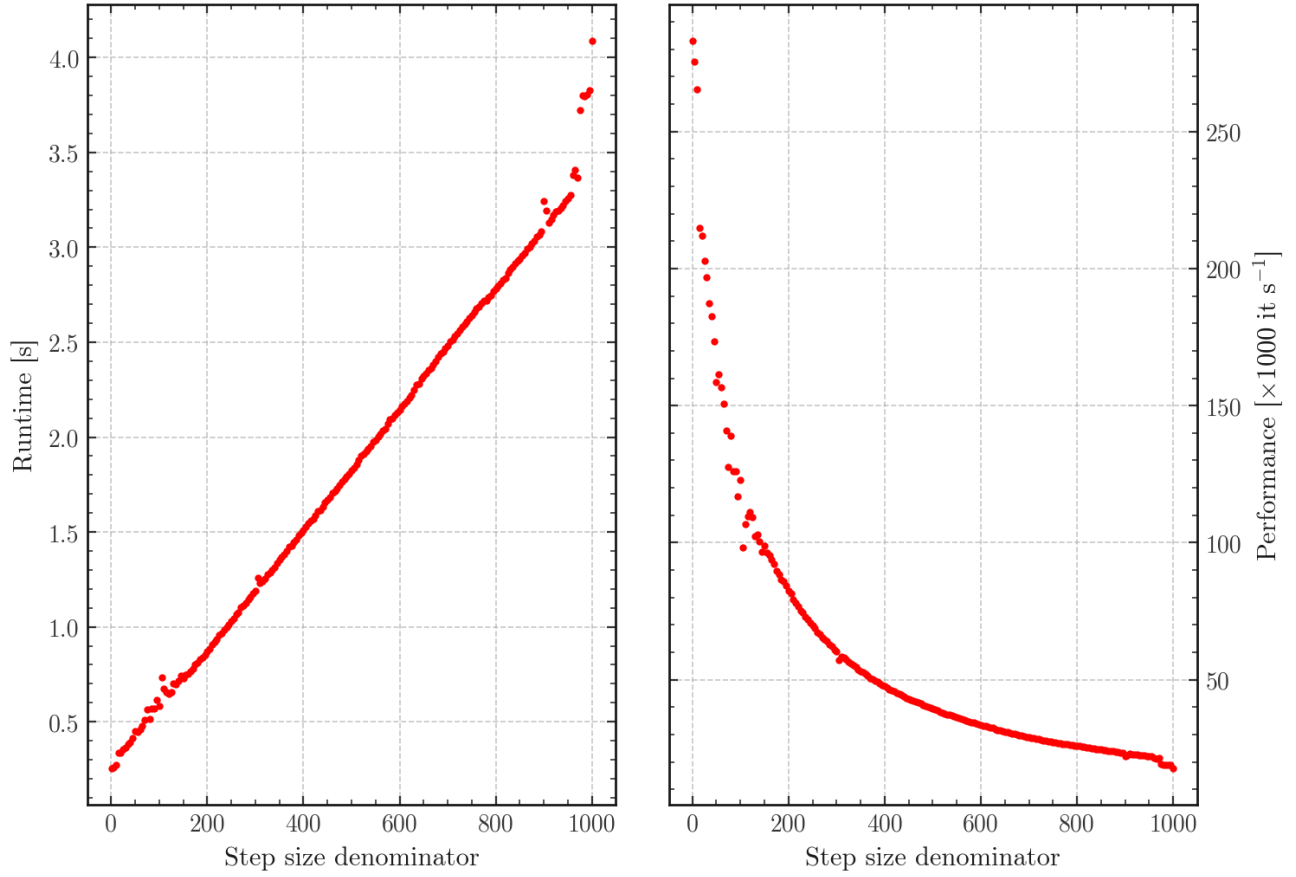


Figure 6.3: The effect of varying de in equation (6.5) on the runtime, which is demonstrably $\mathcal{O}(n)$, and the performance of the algorithm. The input data to this algorithm were the LISA binary sample from Tang (priv. comm.). The performance is shown in units of thousands of iterations per second.

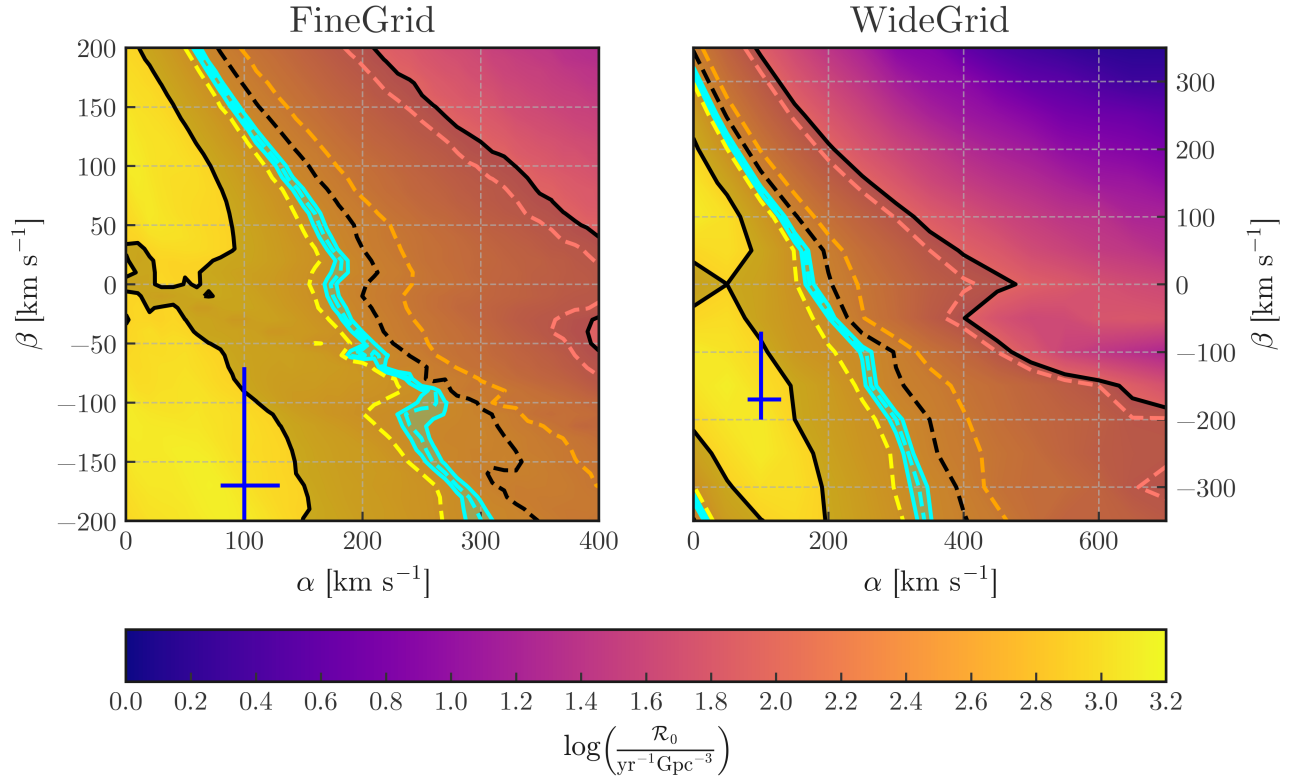


Figure 6.4: The merger rate as a function of kick parameters α and β . The cyan contour is the theoretical merger rate as reported by Eldridge et al. (2019). The black contour is the merger rate observed in LIGO-Virgo (2021). The remaining contours are the merger rates reported in LIGO-Virgo-KAGRA (2021) (pink: BGP, yellow: MS, orange: PDB [ind], see LIGO-Virgo-KAGRA (2021) for an explanation of each model). Shaded regions represent uncertainties, and uncertainties are not reported for the LIGO-Virgo-KAGRA (2021) results as they would cover the entire plane. The blue cross is the fiducial value of α and β from Bray & Eldridge (2018). This value lies off the previous work's contours as the rejuvenation age calculation has been improved in TUI in preparation for the next major BPASS release.

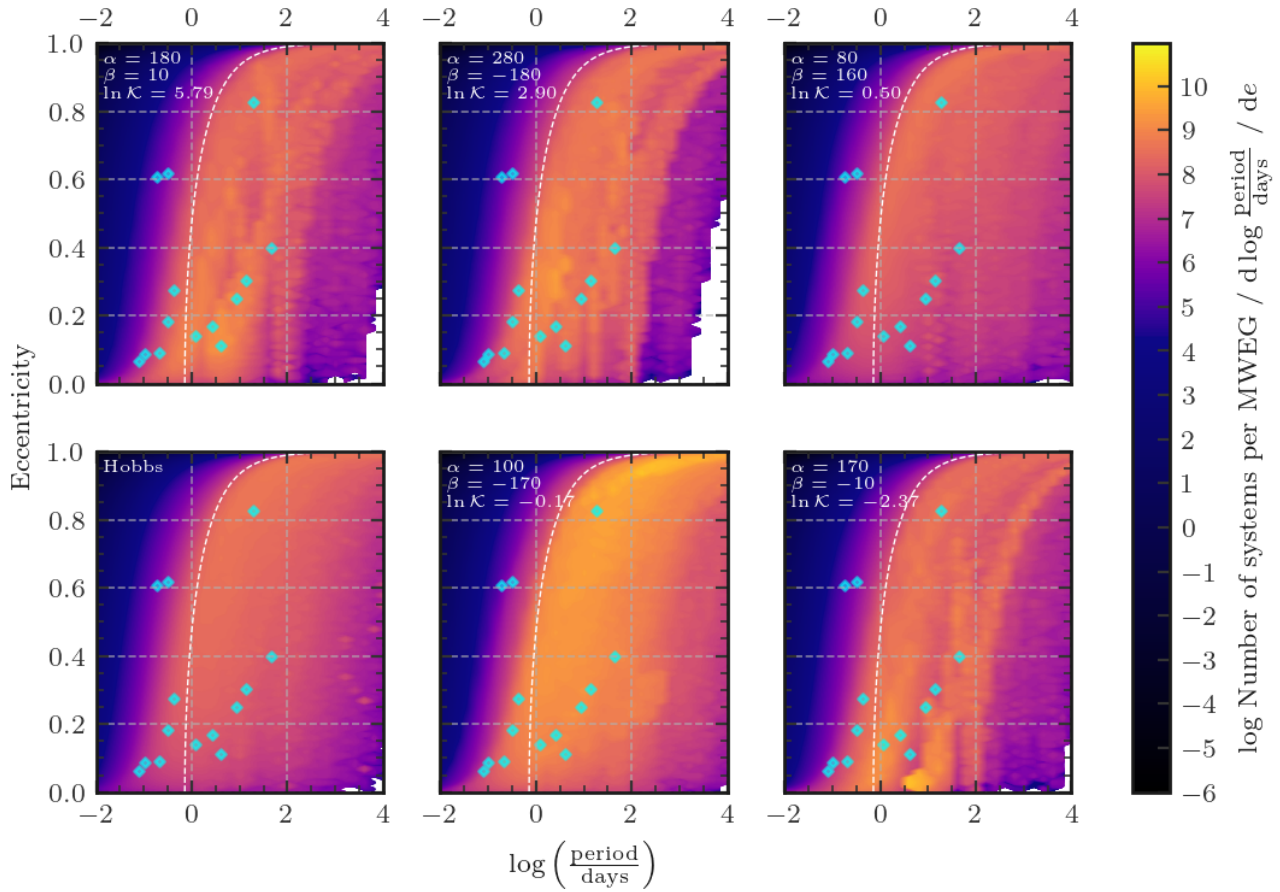


Figure 6.5: The period-eccentricity distributions for six selected Bray & Eldridge (2016) kicks. Shown are the kicks for $\alpha = 180 \text{ km s}^{-1}, \beta = 10 \text{ km s}^{-1}$ as the system with the highest $\ln \mathcal{K}$ across the LIGO contour, $\alpha = 170 \text{ km s}^{-1}, \beta = -10 \text{ km s}^{-1}$ as the lowest, and the balance to illustrate structural differences. White regions show regions where no systems evolve to. The white line represents a constant-time coalescence time of the Hubble time (~ 13.8 Gyr), computed using Mandel (2021). The cyan diamonds are the observed systems enumerated in table 6.2. In white text are the values of (α, β) , and the $\ln \mathcal{K}$ values for each plot.

Name	$\log(P/\text{days})$	e	Original Reference	Name	$\log(P/\text{days})$	e	Original Reference
J0453+1559	0.609808	0.113	Martinez et al. (2015)	J0737-3039	-0.991400	0.088	Kramer et al. (2006)
B1534+12	-0.375718	0.274	Fonseca et al. (2014)	J1756-2251	-0.494850	0.181	Faulkner et al. (2005)
B1913+16	-0.490797	0.617	Hulse & Taylor (1975)	J1913+1102	-0.686133	0.090	Lazarus et al. (2016)
J1757-1854	-0.735182	0.606	Cameron et al. (2018)	J1518+4904	0.936212	0.249	Janssen et al. (2008)
J1811-1736	1.273672	0.828	Corongiu et al. (2007)	J1829+2456	0.070407	0.139	Champion et al. (2004)
J1930-1852	1.653791	0.399	Swiggum et al. (2015)	J1753-2240	1.134751	0.304	Keith et al. (2009)
J1411+2551	0.417638	0.169	Martinez et al. (2017)	J1946+2052	-1.107905	0.064	Stovall et al. (2018)

Table 6.2: The systems we use in our study. These data were originally compiled in Vigna-Gómez et al. (2018).

the Bayes factor (defined as the likelihood ratio) can be computed as

$$\mathcal{K} \equiv \frac{\mathcal{L}}{\mathcal{L}_{\text{Hobbs}}} \implies \ln \mathcal{K} = \ln \mathcal{L} - \ln \mathcal{L}_{\text{Hobbs}}, \quad (6.7)$$

where $\mathcal{L}_{\text{Hobbs}}$ is the log-Likelihood of the Hobbs kick. We use a catalogue of fourteen BNS systems originally compiled in Vigna-Gómez et al. (2018), shown in table 6.2. From this catalogue, we sample the period-eccentricity distribution at that space and take that as the probability of observing that binary, thus allowing us to use the content of the period-eccentricity distribution as the $\ln p$ term in our summand of equation (6.6).

By doing this we effectively construct a Bayes-factor map, where $\ln \mathcal{K}$ can be shown evaluated at every combination of α and β . We term this value $\ln \mathcal{K}_{\alpha\beta}$. This is shown in figure 6.6 for both of our considered grids. In general we see that approximately 60% of FINEGRID has a positive log-Bayes factor (indicating any level of support of the model compared to Hobbs), whereas approximately 40% of WIDEGRID has a positive log-Bayes factor. In a Bayesian analysis, the constraint that $\ln \mathcal{K} \geq 3.4$ indicates strong support for the model contrasted against the reference (i.e., Hobbs et al., 2005), which contains approximately 9.1% of FINEGRID and 4.4% of WIDEGRID.

6.4 | Single-Star Kick Velocities

The traditional methodology for refining a kick prescription is to consider the velocity distribution of isolated pulsars. Thus, given a suitable observational dataset, we would be able to compute the expected kick distribution and compare it, via a Kolmogorov-Smirnov (K-S) test, to the observed data set. The K-S statistic between distributions $\mathcal{F}$ (the observed distribution) and $\mathcal{G}$ (the compared distribution) is given by

$$\text{KS} = \max |\text{cdf}(\mathcal{F}) - \text{cdf}(\mathcal{G})|, \quad (6.8)$$

where cdf means the CDF function, defined as the integral of the PDF. Given the CDF is bounded between 0 and 1 inclusive, the K-S statistic is also bounded between this region. A K-S statistic of 1 means that the compared distribution diverges perfectly from the observed distribution, whereas a K-S statistic of 0 means it matches perfectly. This test of divergence was the refinement performed in Bray & Eldridge (2016, 2018), hence

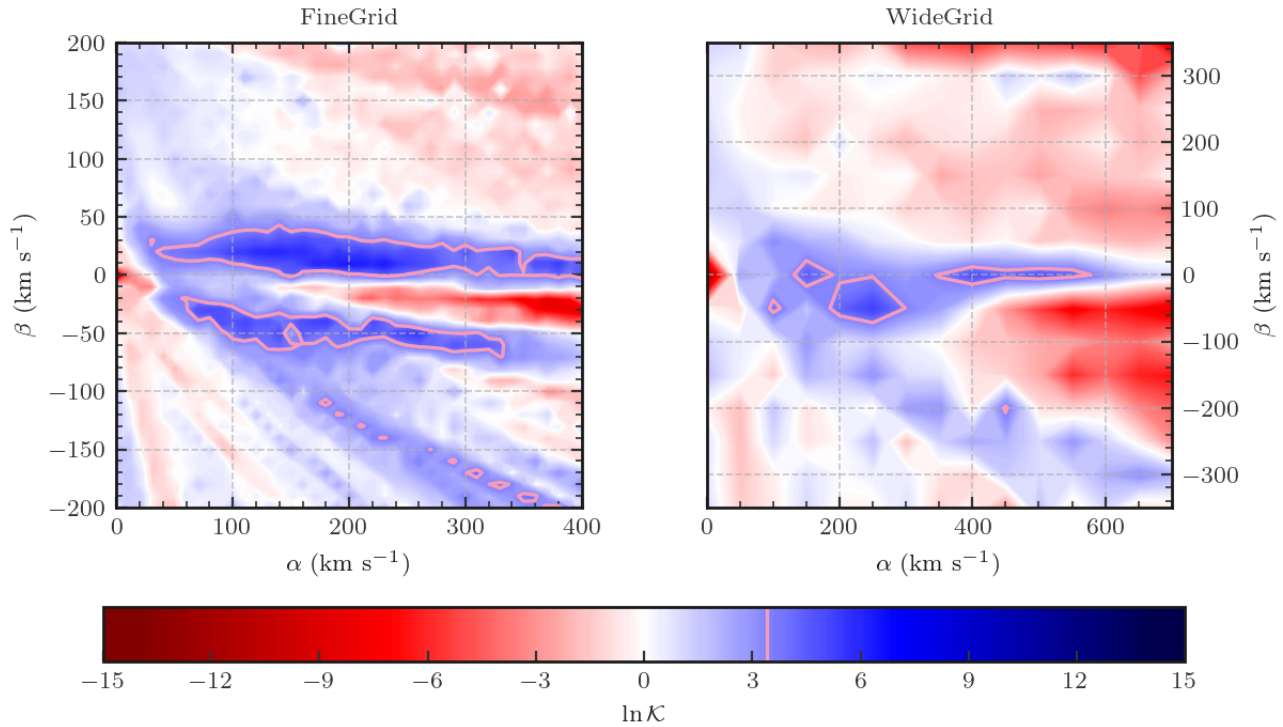


Figure 6.6: The $\ln \mathcal{K}$ values across $\alpha - \beta$ space. The orange line demarcates $\ln \mathcal{K} = 3.4$, which in a Bayesian test represents conclusive evidence in favour of the distribution compared to the Hobbs et al. (2005) kick.

we select this (with an independently generated dataset to those works) to be consistent with prior work.

6.4.1 | Pulsar velocity dataset

We use the same dataset of 81 pulsars from Willcox et al. (2021), and filter the dataset in the same way that they do. We remove millisecond pulsars, pulsars in globular clusters, and pulsars known to be in binaries, in order to homogenise the dataset. The PDF and CDF of Willcox et al. (2021), as well as those generated from Hobbs et al. (2005) and Bray & Eldridge (2018), is displayed in figure 6.7.

This homogenised dataset gives the transverse velocities of each pulsar alongside the RA/Dec, the proper motion in the RA/Dec directions, and parallax. We assume that the transverse velocity is approximately the original kick velocity. As mentioned earlier in this chapter, old remnants with high velocities may have moved substantially – a 20 Myr remnant with a transverse velocity of 500 km s⁻¹ will have moved ~ 10 kpc, absent external forces (Willcox et al., 2021). However, this is likely only to be an issue for pulsars whose transverse velocity substantially exceeds the transverse Galactic rotational velocity of the Sun $v_{\text{TGRV}} \approx 230$ km s⁻¹ (Bhattacharjee et al., 2014), shown in yellow on figure 6.7. Although the dataset of Willcox et al. (2021) has pulsars with velocities higher than this value – peaking at ~ 850 km s⁻¹ – this is not a grave concern for this work, for two reasons. Firstly, the kick velocities output from TUI are taken immediately after the supernova. Secondly, the analysis Willcox et al. (2021) performs is focused on low-velocity pulsars, and this test limits the kick velocity of an USSNe to 30 km s⁻¹, well below v_{TGRV} .

The Hobbs et al. (2005) kick has a closed-form expression for the PDF $P(v_k)$, given by equation (6.3).

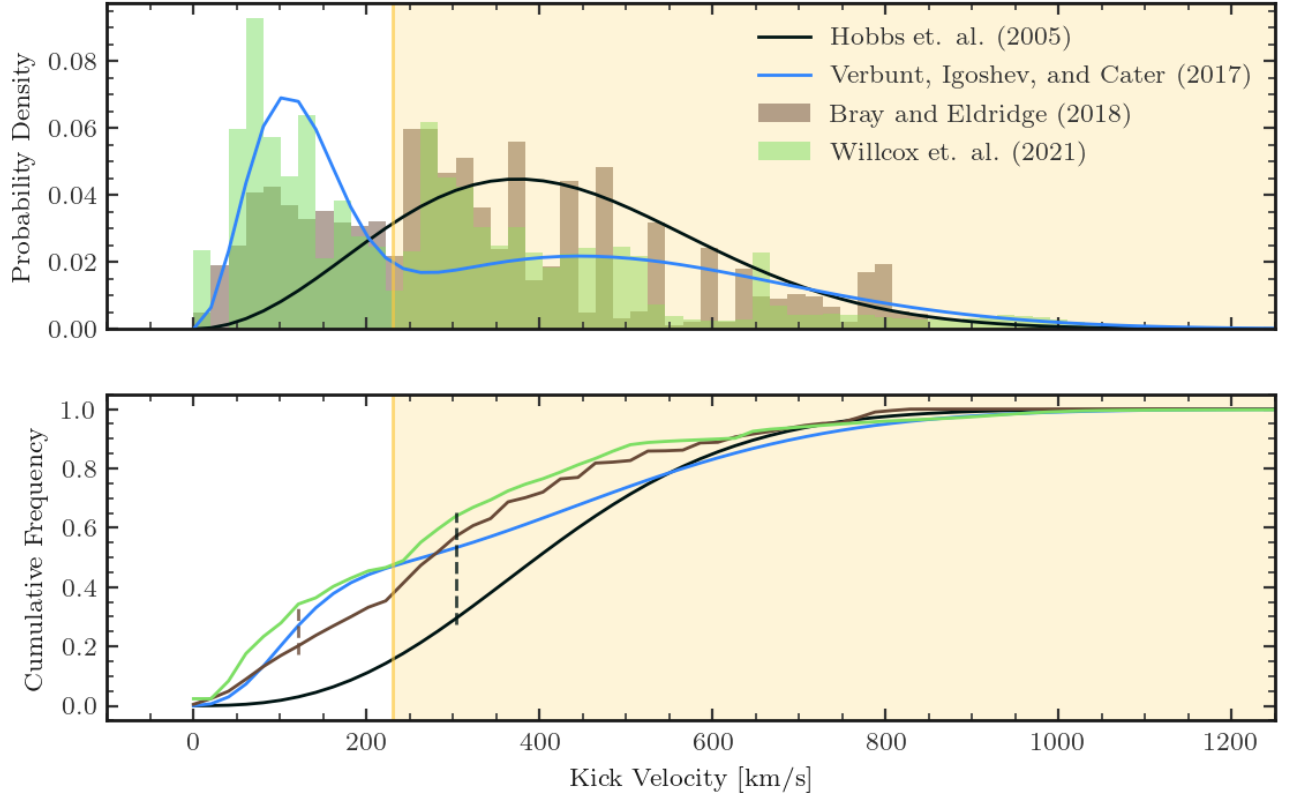


Figure 6.7: The PDF and CDF of the distributions using kicks from Hobbs et al. (2005); Verbunt et al. (2017); Bray & Eldridge (2018); Willcox et al. (2021). Hobbs et al. (2005) has a closed-form expression, whereas the others are computed empirically, hence the Bray and Willcox CDFs are less smooth than that of Hobbs. The two dashed lines indicate where and how the K-S statistic is measured for the associated distribution. The yellow region indicates the region for which $v_{\text{kick}} \geq v_{\text{TGRV}}$, the transverse Galactic rotation velocity of the Sun, $\sim 230 \text{ km s}^{-1}$.

Consequently, it has a smooth curve in figure 6.7, whereas the Bray & Eldridge (2018) and Willcox et al. (2021) PDFs are presented as histograms as they are computed.

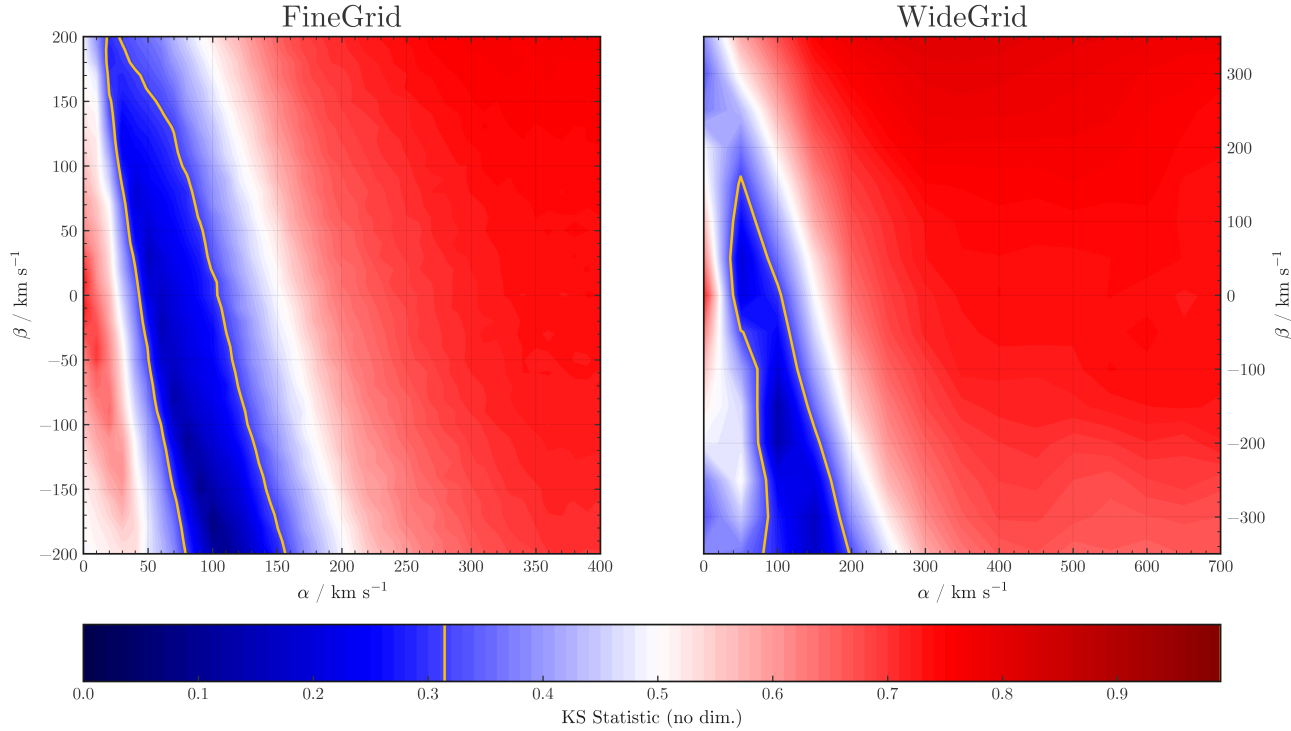


Figure 6.8: The K-S statistic for FINEGRID and WIDEGRID. The orange line denotes the K-S statistic for the Hobbs kick, which has a value of 0.34.

6.4.2 | Comparison

figure 6.8 shows the K-S statistic as a function of α and β across our two parameter spaces. We also show the K-S statistic of the Hobbs kick, which has a value of 0.34. We find in general a valley of models with good agreement to Willcox et al. (2021), and find that the global minimum K-S statistic is at $\alpha = 100, \beta = -180$ – agreeing with the prior work of Bray & Eldridge (2018) (who find $\alpha = 100, \beta = -170$).

We also show in figure 6.9 a selection of four PDFs, alongside their associated CDF and the CDFs for Willcox et al. (2021) and the Hobbs kick. We note that the numerical values shown in figure 6.9 differs somewhat from those shown in Richards et al. (2023). This was due to a minor bug in the code incorrectly labelling the KS statistics for the figure. The actual result of the paper does not change with the correction of the bug, however the values reported in Figure 5 of Richards et al. (2023) do change.

The four values we present on the left side of figure 6.9 correspond to $\alpha = 100 \text{ km s}^{-1}, \beta = -170 \text{ km s}^{-1}$ for the fiducial Bray kick, $\alpha = 180 \text{ km s}^{-1}, \beta = 10 \text{ km s}^{-1}$ to represent the system with the highest $\ln \mathcal{K}$ value from section 6.3, $\alpha = 170 \text{ km s}^{-1}, \beta = -10 \text{ km s}^{-1}$ the lowest, and $\alpha = 40 \text{ km s}^{-1}, \beta = 0 \text{ km s}^{-1}$ the system with low α and zero β , corresponding to a kick that would *a priori* generate logical velocities in the USSNe limit.

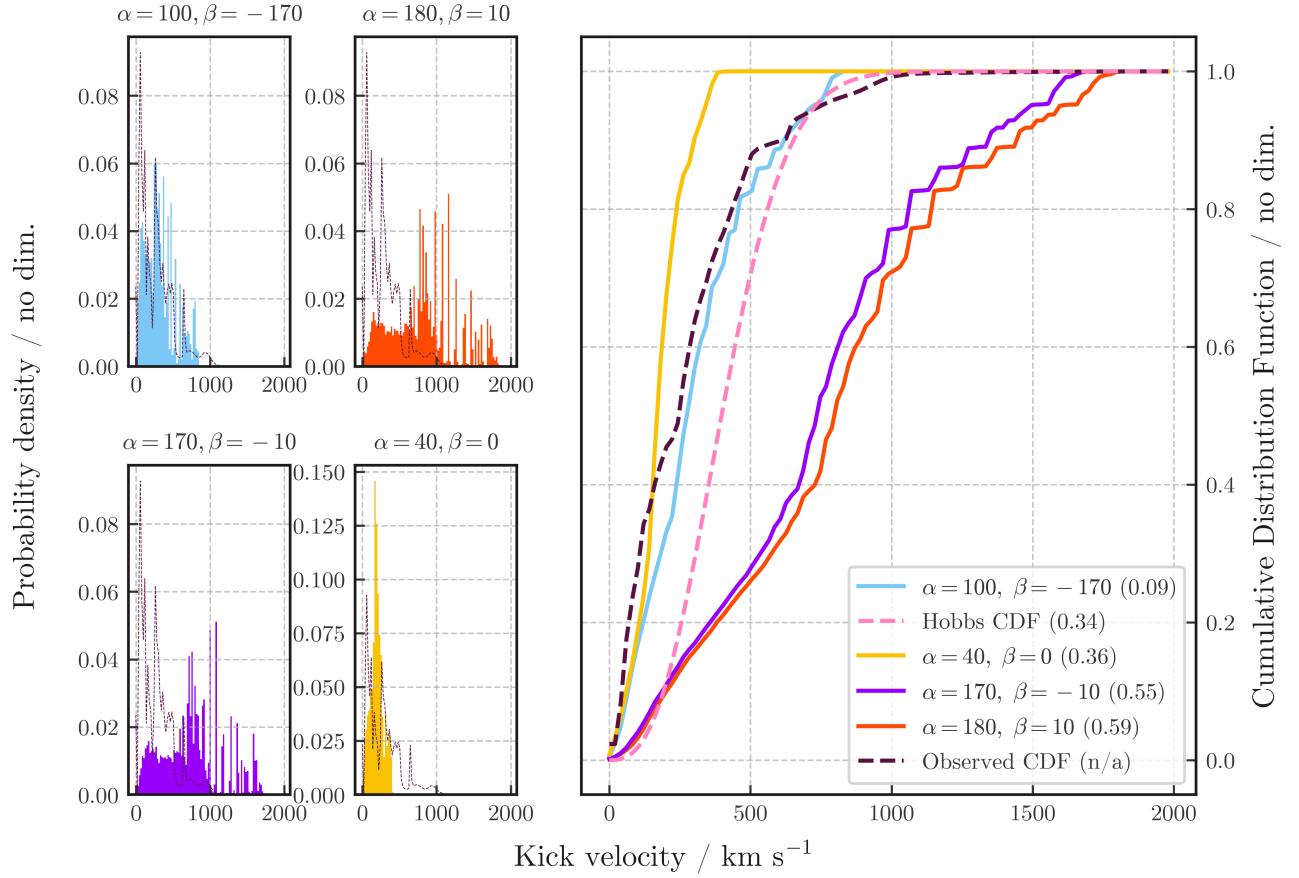


Figure 6.9: The PDFs and CDFs of four systems, alongside the Hobbs CDF and Willcox et al. (2021) CDF. The dashed line on the PDF plots is the PDF of the observed distribution from Willcox et al. (2021). On the right hand plot, the number in parentheses represents the K-S statistic for the distribution, measured relative to Willcox et al. (2021), which by definition has no K-S statistic.

6.5 | Production of Ultra-Stripped Supernovae

As discussed earlier, the fiducial values of the Bray kick from Bray & Eldridge (2018) admit some problematic extremal cases. If $m_{\text{ej}} \rightarrow 0$, as is the case in a USSN, the kick velocity would become negative. For instance, an “average” USSN with $m_{\text{ej}} = 0.15 M_{\odot}$ and $m_{\text{rem}} = 1.45 M_{\odot}$ would have a kick velocity of $\sim -159 \text{ km s}^{-1}$. The concern here is twofold. Firstly, a negative kick velocity would mean that the velocity vector of the kick would be antialigned with the direction of the more massive remnant; this is a violation of conservation of momentum. Secondly, a kick with high magnitude means that, irrespective of the alignment of the kick, the kick itself is strong. As discussed in Willcox et al. (2021), USSNe are associated with weak kicks, peaking at $v_k \sim 30 \text{ km s}^{-1}$. This means that the fiducial Bray kick is unable to correctly account for USSNe.

To account for these supernova kicks, we introduce a new constraint onto our parameters. Following Tauris et al. (2015), we define a USSN as follows:

1. The ejecta mass is in the range $0 \leq m_{\text{ej}} \leq 0.3 M_{\odot}$, admitting the USSN SN2019dge from Yao et al. (2020),
2. The remnant mass is in the range $1.1 \leq m_{\text{rem}} \leq 1.8 M_{\odot}$,
3. The envelope mass $m_{\text{env}} \leq 0.2 M_{\odot}$.

For every combination α, β , we record whether it is capable of producing one USSN. The constraint is that this heuristic is met. This constraint is extremely conservative, insofar as we do not constrain on the prevalence of USSNe in our sample, but rather by whether or not they are possible at all. The reason for the conservatism here is that the rates of Type Ic USSNe are expected to be very low. In the BPASS dataset, they comprise 1.8 per cent of all CCSNe (Stevance, 2022). The paucity of data alongside the paucity of observational evidence suggests a low rate, but since the prevalence is unknown, we place the conservative constraint of at least one USSN to be agnostic. This effectively widens our error bars such that when the USSNe rate is refined further, it will shrink the parameter space.

6.6 | Refinement of the parameter space

We are now in a position to impose constraints on our parameter space based on the analysis provided above. We apply the following four constraints:

1. **DNS Merger Rate** (section 6.2): Does the merger rate produced by these parameters sit along the LIGO contour?
2. **DNS Period-Eccentricity Distributions** (section 6.3): Is $\ln \mathcal{K}_{\alpha\beta} \geq 3.4$ compared to the Hobbs et al. (2005) kick (representing conclusive evidence in favour of the distribution)?
3. **Isolated Pulsar Velocities** (section 6.4): Is the K-S statistic produced by this choice of parameters, in comparison to the Willcox et al. (2021) kick, less than 0.5 (indicating it agrees with observational data more than it disagrees)?

4. Logical USSN velocities (section 6.5): Is this choice of parameters able to produce at least one USSN?

These can be overlaid onto our parameter space, which is shown in figure 6.10. The center of mass of our fit region is located at $\alpha = 117_{-51}^{+36} \text{ km s}^{-1}$, $\beta = 13_{-14}^{+8} \text{ km s}^{-1}$, however this value is computed on a finer grid than the parameter space and is thus likely to be an overfit to the data. We thus report the final recommended model value to the nearest half-bin-width of the parameter space, broadening error bars where necessary, and find $\alpha = 115_{-55}^{+40} \text{ km s}^{-1}$, $\beta = 15_{-15}^{+10} \text{ km s}^{-1}$ as the recommended model value. Error bars are always broadened here; traditional rounding would result in the upper bound on α (being 153 km s^{-1}) becoming 150 km s^{-1} , which artificially removes potentially permissible regions of the parameter space. Broadening the bars removes this at the cost of potentially admitting values that do not belong within the “acceptable” zone. Further constraints, however, will shrink the region, so this is an acceptable tradeoff.

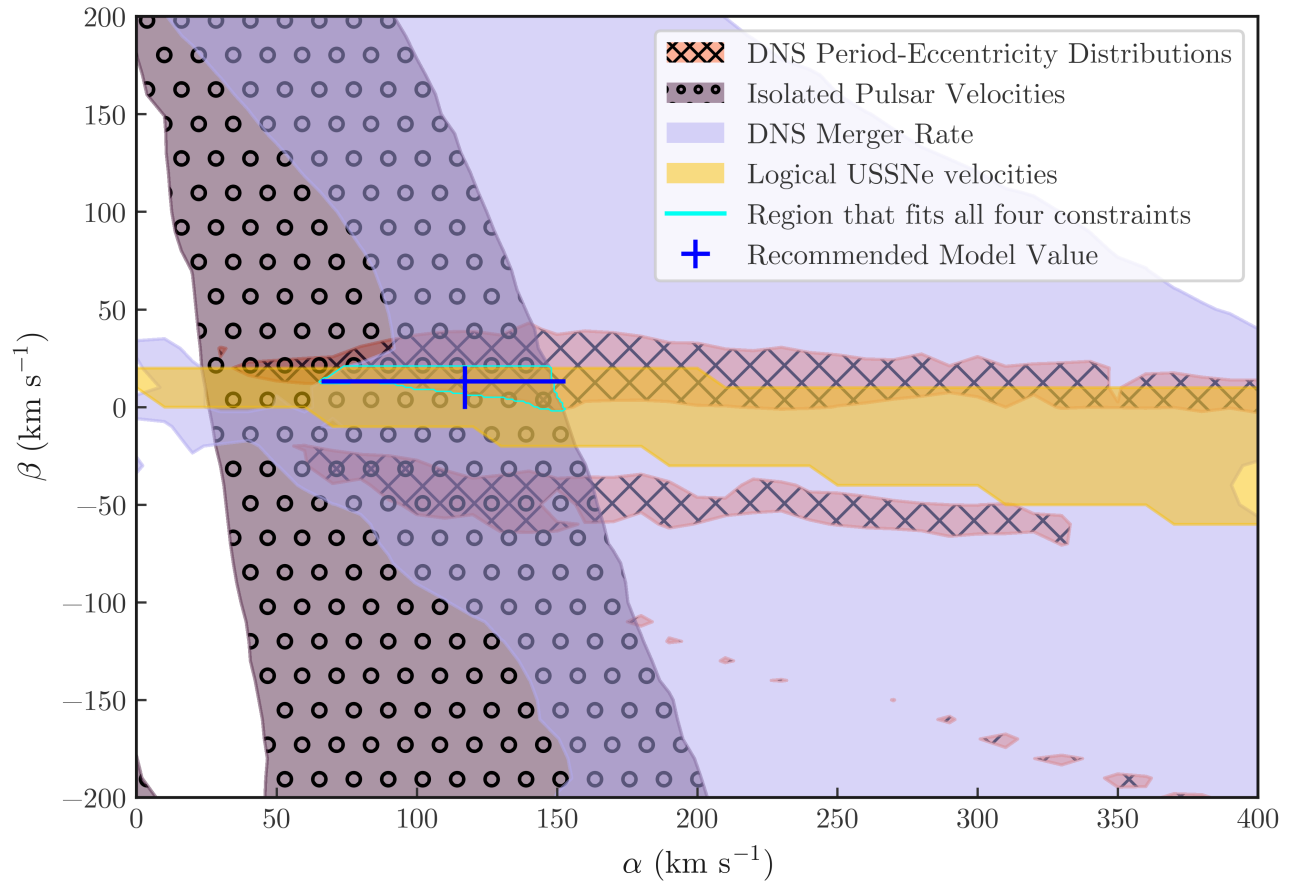


Figure 6.10: The four constraints we apply overlaid on our FINEGRID parameter space. The cyan region represents the region which fits all four parameters, and the blue cross represents the “center of mass” of this region.

In figure 6.11, we present the two major inferred properties of our recommended model value. First, we present the CDFs of a population governed by a Bray kick with the recommended parameters, contrasted with the associated Hobbs kick and observed Willcox et al. (2021) data. We also show the period-eccentricity distribution of the population, with the observed systems from table 6.2 overlaid atop the distribution.

We are also able to extrapolate this to comment on the expected peak of the event-rate distribution. Given

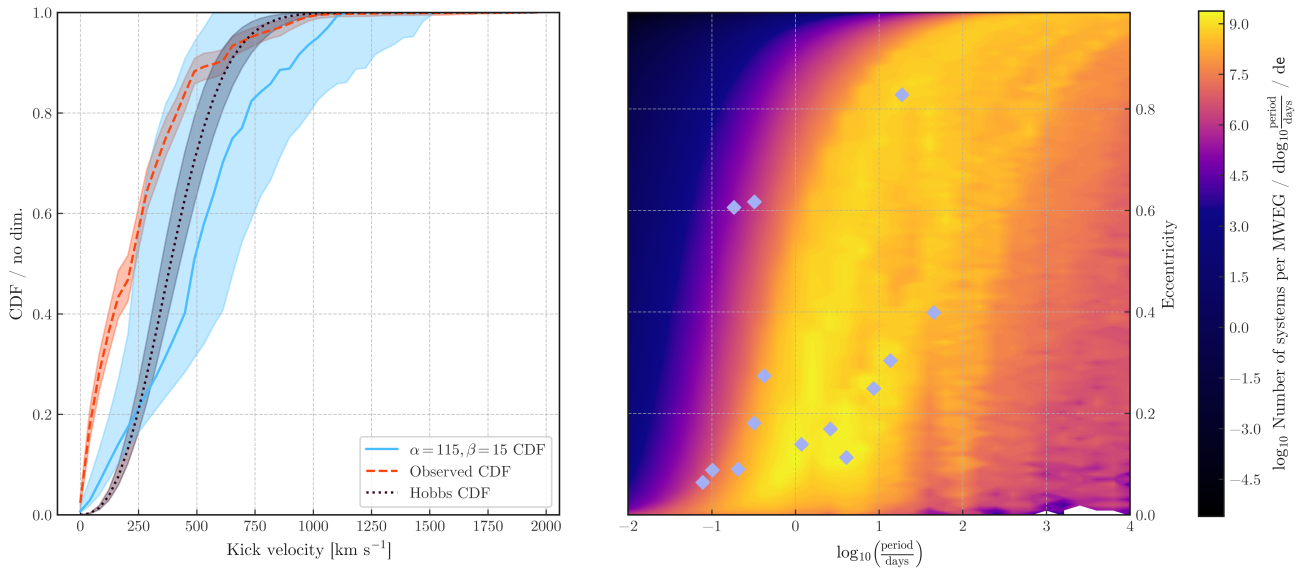


Figure 6.11: The CDF for a population of isolated pulsars (left) and the period-eccentricity distribution (right) for $\alpha = 115 \text{ km s}^{-1}$, $\beta = 15 \text{ km s}^{-1}$. The blue region on the CDF plot shows the uncertainty region for the reported value. The red region shows the uncertainty band on the Willcox et al. (2021) CDF, which was computed by sampling the PDF 10,000 times, taking 2,000 samples each time, and plotting the extent of the simulated distribution. The black band is the uncertainty on the Hobbs distribution, taking $\sigma_{\text{max}} = 291 \text{ km s}^{-1}$ and $\sigma_{\text{min}} = 239 \text{ km s}^{-1}$ from Verbunt et al. (2017).

that the event rate is a convolution between the merger rate of our population and the SFRD we choose (cf. equation (2.51)), by affixing the SFRD across our populations, by equation (2.47) we can vary $\mathcal{R}(t)$ by varying α and β . Since the SFRD is of a Gaussian form (with a peak at $z \sim 1.863$), we can also extract the peak of the distribution assuming it is governed by a particular α - β combination. The general results across the parameter space are shown in figure 6.12. The best model choice parameters we find fall within the valley of $z = 0$, indicating that the peak of the event rate distribution is the present day.

We note that there is a single peak at $\alpha = 330 \text{ km s}^{-1}$, $\beta = -70 \text{ km s}^{-1}$ which appears to be an outlier when compared to its neighbours. We display in figure 6.13 the local behaviour of the 9 cells around this region. One can see that this beak is bimodal with a second peak at $z = 0$, which is only 0.098 per cent lower than the reported peak, clearly resulting in a simple outlier.

Unfortunately, there is a paucity of observational evidence to confirm whether or not the peak is truly the present day. Given that a redshift of $z = 2$ corresponds to a lookback time of $t_{\text{LB}} = 10 \text{ Gyr}$, we lack evidence to corroborate this prediction. The next generation of telescopes, including JWST, may yet be able to constrain this result further. In the interim, indirect evidence may be able to be applied via the r-process enrichment history of the Milky Way. Van de Voort et al. (2022), for example, finds that greater velocity natal kicks would be correlated with fewer r-process enhanced stars in the Milky Way, as higher kicks would unbind more binaries, meaning that r-process elements are no longer able to accrete onto star-forming regions. Kobayashi et al. (2023) indicates that current population synthesis models⁵ are unable to reproduce the abundance distributions of certain high-mass isotopes, including the abundances of Oxygen, Iron, Europium, Gold, and Uranium.

⁵Explicitly considered in the paper are COMPAS (Team COMPAS: Riley et al., 2022), StarTrack (Belczynski et al., 2020), Brussels (Mennekens & Vanbeveren, 2014, 2016), ComBinE (Kruckow et al., 2018), and BPASS (Eldridge et al., 2017).

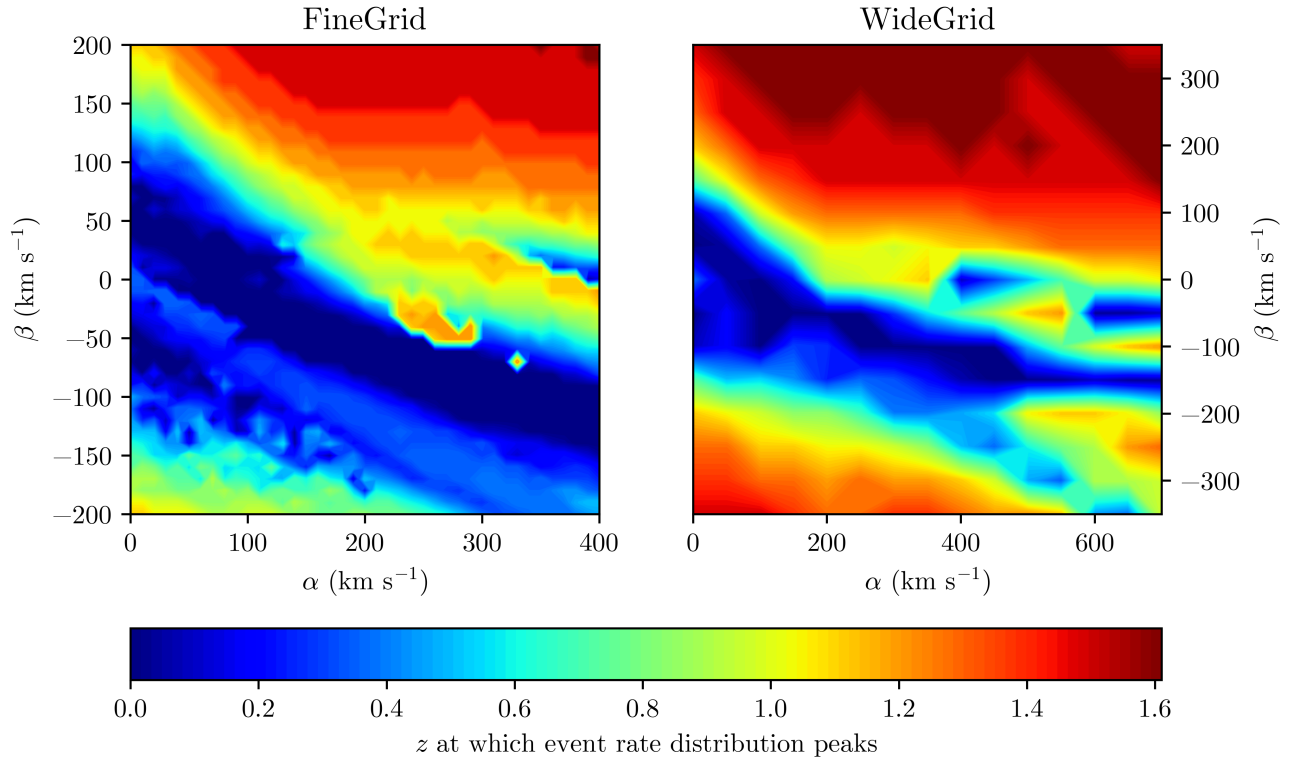


Figure 6.12: The peak of the event rate $\mathcal{R}(t)$ as a function of α and β . Varying these two parameters generally brings the peak closer to the present day ($z = 0$). Absent a merger rate (or, more formally, assuming the merger rate is instantaneous and given by a Dirac-delta function), the SFRD peaks at $z \sim 1.863$. The region surrounding the peak at $\alpha = 330, \beta = -70$ is shown in figure 6.13.

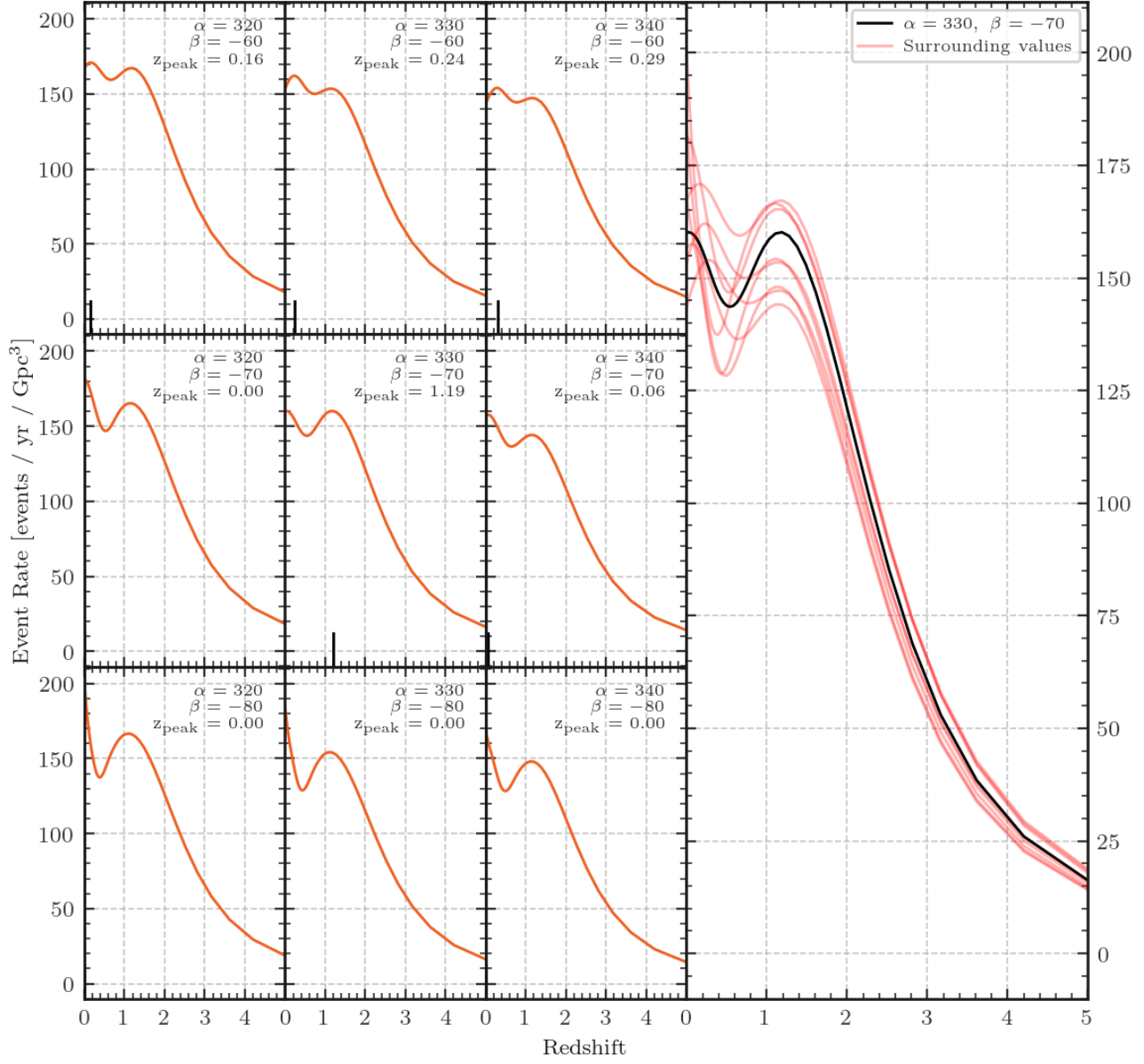


Figure 6.13: The event rate $\mathcal{R}(z)$ for a region surrounding $\alpha = 330, \beta = -70$ on FINEGRID. This region corresponds to the “outlier” in figure 6.12. Shown on the left are the outlier bin in the center and its neighbours in both α -space and β -space. The black line on each plot denotes the location in redshift of the event rate distribution peak. The right plot shows the outlier in black and its neighbours in faded red behind.

6.7 | Discussion

There are limitations to this analysis. The first is that we did not preferentially weight our binaries. Tauris et al. (2017) for example places particular weight on PSR J1930-1852, which is a particularly wide binary. Per Swiggum et al. (2015) (who reported it), “its unique spin and orbital parameters challenge models that describe BNS formation”, which implies that any credible model of BNS formation should replicate its orbital parameters. We did find in figure C.1 that some parameters had non-zero probability of replicating this pulsar, which indicates that the Bray kick is able to reproduce it.

Secondly, we have assumed Newtonian mechanics. Formally, the derivations presented earlier (cf. section 2.4.1) assumed general relativity for the existence of the gravitational wave equation,⁶

$$\square \bar{h}_{\mu\nu} = -\frac{1}{2}\sqrt{32\pi G}T_{\mu\nu}, \quad (6.9)$$

but the injected physics was Newtonian (see Richards (2021) for a derivation). Strictly speaking, relativistic corrections apply to these kinematic equations too. The logic outlined in section 2.2.3 ceases to apply, as we are not modelling the interior of a star. Instead, one would need to apply corrections to the two-body equation of motion, termed the post-Newtonian approximation. The general equation becomes

$$\frac{d\vec{v}}{dt} = \frac{G(m_1 + m_2)}{r^2} \left(-\hat{n} + \sum_{k=2}^N \frac{1}{c^k} \vec{A}_{\frac{k}{2}\text{PN}} \right), \quad (6.10)$$

where N is the highest order term considered, and $\vec{A}_{\frac{k}{2}\text{PN}}$ is termed the “ $\frac{k}{2}$ -th Post-Newtonian” coefficient. Testing with the RIRORIRO code from van Zeist et al. (2021) indicated that including terms up to $N = 6$ (3PN) increases the merger time by fractions of a year, and thus does not significantly alter the merger rate density we observe. Typically 3.5PN ($N = 7$) is used as an upper bound (Itoh & Futamase, 2003), though up to 10PN ($N = 20$) has been used before (Blanchet, 2014, and references therein).

Relativistic corrections also typically manifest in using more sophisticated spacetime metrics. equation (6.9) assumes a metric which is locally Minkowski, that is

$$g_{\mu\nu} = \eta_{\mu\nu} + \sqrt{32\pi G}h_{\mu\nu}, \quad (6.11)$$

where $\eta_{\mu\nu}$ is the positive-definite Minkowski metric. This “weak-field” approximation breaks down when a compact object enters within its companion’s innermost stable circular orbit (ISCO), given by $r_{\text{isco}} = \frac{6GM}{c^2}$, where matter can be exchanged. Given gravitational waves result in strains on the order of 10^{-19} (van Zeist et al., 2023), representing observed fractional shifts on Earth smaller than the diameter of a proton, (Moore et al., 2014), the weak field regime suits, and again modelling the relativistic corrections to the post-ISCO regime only increases the merger rate by fractions of years.

To contextualise this case study, other authors such as Mapelli & Giacobbo (2018); Wiktorowicz et al. (2019); Giacobbo & Mapelli (2020); Santoliquido et al. (2021); Kapil et al. (2023) have explored the parameter space

⁶Here, $\bar{h}_{\mu\nu}$ is the trace inverse of a weak perturbation to locally Minkowski spacetime with positive-definite trace, and $\square$ is the d’Alembertian operator.

of BNS kicks, though none have performed the analysis in the way we have. These works typically vary the progenitor population properties, in such a way that the formed binaries can in principle vary strongly. However, in our population synthesis procedure, we are able to effectively affix the physics between different models such that *only* the kick velocities are expected to vary between individual populations. Since the Bray kick has two parameters which can be altered, we are able to modify the kick velocity without changing the physics, thus allowing us to isolate and study the kick mechanism alone.

6.8 | Chapter Summary

We have presented a formalism for evaluating and quantifying the goodness-of-fit of the natal kick mechanism against four independent tests. The tests we have presented – constraining the merger rate of BNS systems, the period-eccentricity distribution of Galactic BNS systems, the kick velocities of single-star pulsars, and the production of ultra-stripped supernovae – greatly constrain the parameter space to be less than 1 per cent of the possible parameters. Furthermore, the data suggest that the peak of the merger rate distribution is the present day, with $z \simeq 1.5$ as the upper bound of the entire parameter space, and $z = 0$ for our recommended value.

The glow of summer has faded, now... and the moonlight
jellies carry on towards the great unknown.

– STARDEW VALLEY

Conclusion

THIS THESIS HAS PRIMARILY concerned itself with improving an established stellar modelling code for future population synthesis projects. In doing so, we discussed several key aspects of binary evolution, which shall be recapped here. To wit, in chapter 1 we introduced three major themes guiding this research:

- (I) Can we methodologically overcome the typical cases in which stellar models fail:
 - (a) RLOF,
 - (b) CEE, and
 - (c) the helium flash
- (II) Can we develop a testing regime for the natal supernova kick, which relies on more observations than just isolated pulsar velocities?
- (III) What modifications are required to be made to our stellar modelling code in preparation for the next major BPASS release?

In this chapter, we examine these themes in detail and review the implications of them in terms of the results we have obtained.

7.1 | Summary

Oh! I have slipped the surly bonds of Earth
And danced the skies on laughter-silvered wings;

...

And, while with silent lifting mind I've trod
The high untrespassed sanctity of space,
Put out my hand, and touched the face of God.

High Flight, BY JOHN MAGEE JR.

7.2

|

Limitations

Though in reviewing the incidents of my administration I am
unconscious of intentional error, I am nevertheless too sensible of my
defects not to think it probable that I may have committed many
errors.

GEORGE WASHINGTON

- 1. circular orbits
- 2. o-type stars
- 3. parameter selection

7.3

|

Suggested Future Work

A still more glorious dawn awaits – not a sunrise, but a galaxy rise.

CARL SAGAN

I can slay the Jabberwocky.

LAST OF THE SIX IMPOSSIBLE THINGS

In the future,

Appendices



Code Samples

Code Sample A.1: The TAKAHE function `integrate_eoms()`

```

1 def integrate_eoms(a0, e0, p):
2     """
3     General purpose integrator for Nyadzani & Razzaque eqns for a & e.
4
5     Params:
6         a0 - The initial semimajor axis, measured in solar radii
7         e0 - The initial eccentricity, dimensionless
8         p - A vector of parameters:
9             p[1] = m1 (units: Solar Mass)
10            p[2] = m2 (units: Solar Mass)
11            p[3] = unused (will be removed in next version)
12            p[4] = unused (will be removed in next version)
13            p[5] = Lifetime (evolution + rejuvenation)
14
15     Returns:
16         A - An array of the semimajor axes of the binary system over time. (Solar
17         ↪ Mass)
18         E - An array of the eccentricities of the binary system over time. (no dim.)
19     """
20
21     Solar_Mass = 1.989e30          # kg
22     Solar_Radius = 696340000.0    # m
23     G = 6.67e-11                  # m^3 kg^-1 s^-2
24     c = 299792458.0               # m/s
25
26     A = np.array([])
27     E = np.array([])
28     H = np.array([])
29
30     m1, m2, _, _, evotime = p[0], p[1], p[2], p[3], p[4]
31
32     # number of seconds in a year
33     seconds_per_year = 60 * 60 * 24 * 365.25
34
35     #####
36     #      Unit Check      #
37     #####
38     a = a0 * Solar_Radius # Meters

```

```

38     e = e0                                # Dimensionless
39     m1 = m1 * Solar_Mass                  # Kilogram
40     m2 = m2 * Solar_Mass                  # Kilogram
41     #####
42     # End Unit Check #
43     #####
44
45     # Beta has units m^4 / s
46     beta = ((64/5) * G**3 * m1 * m2 * (m1 + m2) / (c**5))
47
48     A = np.append(A, a)
49     E = np.append(E, e)
50     H = np.append(H, 0.0)
51
52     total_time = 0
53
54     # Integrate until past the end of the universe, or a 10km orbit
55     while total_time/seconds_per_year + evotime < 1e11 and a > 1e4:
56         # an euler integrator: work out da/dt then times it by dt
57         # to get da, which then we can work out as a = a + da/dt * dt.
58         initial_da = (- beta / ((a**3) * (1 - e**2)**(7/2)))
59         da = initial_da * (1 + (73/24) * e**2 + (37/96) * e**4)
60
61         intial_de = (((-19/12) * beta) / (a**4*(1-e**2)**(5/2)))
62         de = intial_de * (e + (121/304) * e**3)
63         # Units: s^-1
64
65         timeA = abs(1e-2 * a/da)
66
67         if e > 1e-10:
68             timeE = abs(1e-2 * e/de)
69         else:
70             de = 0
71             e = 1e-10
72             timeE = timeA * 10
73
74         # maximum timestep is half of the width of the smallest BPASS time bin
75         conv_frac = 0.23076752*0.5*seconds_per_year
76
77         dt2 = (evotime + total_time/seconds_per_year)*conv_frac
78
79         # Take a timestep that results in the smallest change: either a
80         # change in E, a change in A, 1/2 the smallest BPASS bin.
81         dt = min(timeE, timeA, dt2)
82
83         a = a + dt * da
84         e = e + dt * de
85
86         A = np.append(A, a)
87         E = np.append(E, e)
88         H = np.append(H, dt)
89
90         total_time = total_time + dt
91
92         # Determine why we stopped, so the parent function can tell what went
93         # "wrong".

```

```

94     stop_reason = "flag_not_set"
95
96     if total_time/seconds_per_year + evotime >= 1e11:
97         stop_reason = "out_of_time"
98
99     if a <= 1e4:
100         stop_reason = "merged"
101
102     # Solar Radii, Dimensionless
103     return A / Solar_Radius, E, H, stop_reason

```

Code Sample A.2: The TAKAHE function integrate_timescale()

```

1  def integrate_timescale(m1, m2, p0, e0):
2      """
3      General purpose integrator for eqn(5.14) of Peters, 1964 [1].
4
5      [1] https://ui.adsabs.harvard.edu/abs/1964PhRv..136.1224P/abstract
6
7      Params:
8          m1 -- The mass of the first star (Solar Mass)
9          m2 -- The mass of the second star (Solar Mass)
10         p0 -- The initial period (days)
11         e0 -- The initial eccentricity
12
13     Returns:
14         tC -- The coalescence time (seconds)
15     """
16     Solar_Mass = 1.989e30      # kg
17     Solar_Radius = 696340000.0 # m
18     G = 6.67e-11              # m^3 kg^-1 s^-2
19     c = 299792458.0           # m/s
20
21     m1 = float(m1)
22     m2 = float(m2)
23     p0 = float(p0)
24     e0 = float(e0)
25
26     #####
27     # BEGIN UNIT CHECK #
28     #####
29     m1 = m1 * Solar_Mass    # Kilograms    ##
30     m2 = m2 * Solar_Mass    # Kilograms    ##
31     p0 = p0 * 24 * 60 * 60  # Seconds      ##
32     e0 = e0                  # Dimensionless ##
33     #####
34     # END UNIT CHECK #
35     #####
36
37     # Use Kepler's Third Law to compute the semimajor axis, in meters
38     a0 = (p0**2.0 * (G * (m1+m2))) / (4.0 * np.pi**2.0)**(1.0/3.0) # meters
39
40     # compute the constant c0
41     c0 = a0 * (1.0-e0**2.0) * e0**(-12.0/19.0) * (1.0+(121.0 / 304.0 *
        ↪ e0**2.0))**(-870.0/2299.0) # meters

```

```
42
43     # Compute the constant beta
44     beta = 64.0 / 5.0 * G**3.0 * m1 * m2 * (m1 + m2) / c**5.0
45     # meters^4 / s
46
47     # Circular binary coalescence time
48     tC = a0**4.0 / (4.0*beta)
49
50     if e0 != 0:
51         if e0 < 0.01:
52             # Low ecc - see eqn after eqn(5.14) of Peters, 1964
53             tC = c0**4.0 * e0**(48.0/19.0) / (4.0*beta)
54         elif e0 > 0.99:
55             # High ecc - see eqn after eqn after eqn(5.14) of Peters, 1964.
56             tC = tC * ((768.0 / 425.0) * (1.0-e0**2.0)**3.5)
57         else:
58             # Medium ecc - see eqn(5.14) of Peters, 1964
59             e = 0.0
60             de = e0 / 10000.0
61             summand = 0.0
62
63             while e < e0:
64                 this_integral = de * e**(29.0/19.0)
65                 this_integral *= (1.0 + (121.0/304.0) * e**2.0)**(1181.0/2299.0)
66                 this_integral /= (1.0 - e**2.0)**(1.5)
67
68                 summand += this_integral
69                 e += de
70
71                 e = float(e)
72
73             tC = (12.0/19.0) * (c0**4.0 / beta) * summand
74
75     return tC
```

Code Versions

Code Name	Version	Reference
HOKI	1.7.1	Stevance et al. (2020)
KAITIAKI	1.0.0	This work
MATPLOTLIB	3.8.2	Hunter (2007)
NUMBA	0.58.1	Lam et al. (2015)
NUMPY	1.26.3	Harris et al. (2020)
PANDAS	2.1.4	McKinney (2010)
PYTHON	3.11.5	
SCIPY	1.11.4	Virtanen et al. (2020)
TAKAHE	1.0.0	Richards et al. (2023)
TUI	8	Ghodla et al. (2023)*

Table B.1: The code versions we use in this thesis. *TUI was described in earlier papers, but Ghodla et al. (2023) was the first to name it.



Period-Eccentricity distributions along the LVK Collaboration O3 contour



In Figure C.1 we display the period-eccentricity distributions along the LVK Collaboration O3 contour. We also display a white dashed line corresponding to a merger time contour of the Hubble time. Most plots have a white region in the lower right corner of the plot, which corresponds to no systems passing through those bins. We use the definition of a Milky-Way Equivalent Galaxy (MWEG) from Kasen et al. (2017), namely that there is one MWEG per $(4.4 \text{ Mpc})^3$.

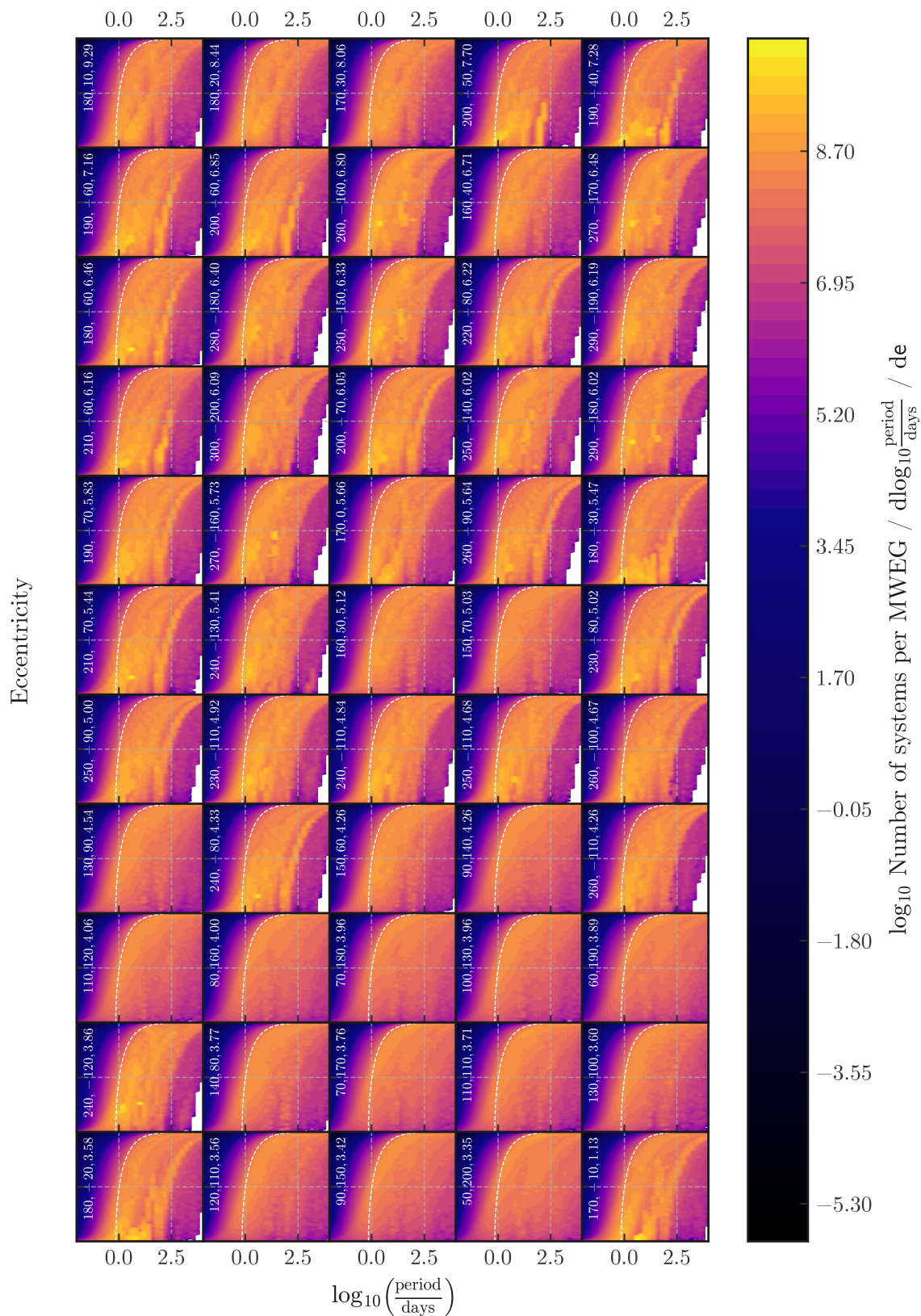


Figure C.1: The period-eccentricity distributions of populations governed by Bray kicks across the LIGO O3 contour. The combination of α and β used in the kick prescription is given in white along the left side of each plot. White regions denote no systems passing through those bins.

Parameter Configuration

In this appendix we list the default parameters chosen for our binary evolution runs, as detailed in subsection 2.6.1.

Parameter	Value	Comment	Parameter	Value	Comment	Parameter	Value	Comment
NM2	199	Meshpoints	ICL	1		IT1	40	
ION	2		IT2	20		IAM	1	
JIN	15		IOP	5		OUT	15	
INUC	0		NCH	3		IBC	0	
JP	1		ICN	0		ITH	1	
IML1	5	Mass loss (*1)	IX	1		IML2	5	Mass loss (*2)
IY	1		ISGTH	0		IZ	1	
IMO	0		IMODE	2		IDIFF	0	
NWRT1	100		EPS	10^{-6}		NWRT2	1	
DEL	0.01		NWRT3	2		DHO	10^{-8}	
NWRT4	1		DT3	0		NWRT5	0	
DDD	1.0		NSAVE	10		NMONT	0	
DT1	0.1		DT2	1.2		CT1	3.0	
CT2	3.0		CT3	0.05		CT4	0.01	
CT5	0.6		CT6	0.35		CT7	0.45	
CT8	10^{-4}		CT9	0.1		CT10	0.3	

Table D.1: Our general initial conditions. Unless otherwise described, these are assumed to be enabled for all runs.

The possible mass loss prescriptions (IML1/IML2) are:

0. No mass loss
1. Reimers (1975)
2. Blöcker (1995)

Parameter	Value	Comment	Parameter	Value	Comment	Parameter	Value	Comment
ZS	0.02		ALPHA	1.925		CH	0.7	
CC	0.173		CN	0.053		CO	0.482	
CNE	0.099		CMG	0.038		CSI	0.08	
CFE	0.072		RCD	10 ⁶		OS	0.12	
RML	0		RMG	0		ECA	0	
XF	0.1		DR	0.001		RMT	1000.0	
RHL	0		AC	0		AK1	1.0	
AK2	10 ⁻⁴		ECT	0		TRB	0	
IRAM	1		IRS1	1		VROT1	0	
IRS2	1		VROT2	10.0		FMAC	1.0	
FAM	0		IVMC	0		TRC1	700.0	
IVMS	0		TRC2	0		MWTS	0	
IAGB	0		ISGFAC	0		FACSGMIN	0.01	
SGTHFAC	0		ISTART	0		HKH	1.0	
GFF	10.0	Removed	NNMOD	0		CEPR	0	CEE prescription
ALPHACE	1.0		-	-	-	-	-	-

Table D.1: Continued

3. Vassiliadis & Wood (1993)
4. de Jager et al. (1988); Nugis & Lamers (2000); Vink et al. (2001) with interpolation errors described in Richards et al. (2024)
5. de Jager et al. (1988); Nugis & Lamers (2000); Vink et al. (2001) (described in Eldridge et al., 2017)
6. de Jager et al. (1988); Nugis & Lamers (2000); Vink et al. (2001) with slowly turning on WR rates described in Richards et al. (2024)
7. de Jager et al. (1988); Nugis & Lamers (2000); Vink et al. (2001) with no WR rates, described in Richards et al. (2024)
8. Unimplemented
9. ZAMS evolution to a specific ZAMS mass, custom routine by Jan Eldridge

Algebra's like sheet music. The important thing isn't can you read music, it's can you *hear* it. Can you hear the music, Robert?

– NIELS BOHR, IN OPPENHEIMER

Appendix

E

Additional Derivations

In this section, we present additional derivations for expressions used in chapter 2.

E.1 | Roche lobe formula

We follow the derivation given in Rose (1998) to find an expression for the Roche potential. First we note that in a rotating reference frame in Cartesian coordinates, we have

$$\overbrace{\frac{GM_1M_2}{a^2}\hat{\mathbf{x}}}^{\text{gravitational force}} = \underbrace{-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1)}_{\text{centrifugal force}}, \quad (\text{E.1})$$

where $\hat{\mathbf{x}}$ is the line between the center of mass of both stars, a the separation, $\vec{\omega}$ the angular velocity, and $\vec{r}_1$ the position vector of the primary star. The x-ordinates of the centres of mass are given by

$$x_1 = \frac{M_2}{M_1 + M_2}a \quad (\text{E.2})$$

$$x_2 = \frac{-M_1}{M_1 + M_2}a \quad (\text{E.3})$$

Since we are considering a rotating coordinate system we have that the centrifugal force is given by

$$-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1) = M_1\omega^2 x_1, \quad (\text{E.4})$$

and hence

$$\begin{aligned}
\frac{GM_1M_2}{a^2}\hat{\mathbf{x}} &= M_1\omega^2x_1 \\
\omega^2 &= \frac{GM_1M_2}{M_1x_1a^2}\hat{\mathbf{x}} \\
&= \frac{GM_2}{x_1a^2}\hat{\mathbf{x}} \\
&= \frac{GM_2(M_1+M_2)}{M_2 \cdot a \cdot a^2}\hat{\mathbf{x}} \\
&= \frac{G(M_1+M_2)}{a^3}\hat{\mathbf{x}}.
\end{aligned} \tag{E.5}$$

The centrifugal potential of a test mass m at (x, y) is hence

$$V = -\frac{1}{2}m\omega^2(x^2 + y^2) = -\frac{Gm(M_1 + M_2)(x^2 + y^2)}{2a^3}, \tag{E.6}$$

and thus its energy is

$$E = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + V(x, y), \tag{E.7}$$

where

$$V(x, y) = -\overbrace{\frac{GmM_1}{\sqrt{(x-x_1)^2 + y^2}}}^{\text{G.P.E. due to primary}} - \overbrace{\frac{GmM_2}{\sqrt{(x-x_2)^2 + y^2}}}^{\text{G.P.E. due to secondary}} - \overbrace{\frac{Gm(M_1 + M_2)(x^2 + y^2)}{2a^3}}^{\text{centrifugal potential}}. \tag{E.8}$$

We introduce the dimensionless coordinates measured in units of the separation of the binary (or reduced masses)

$$\xi = \frac{x}{a}, \quad \eta = \frac{y}{a}, \quad \xi_1 = \frac{M_2}{M_1 + M_2}, \quad \xi_2 = \frac{M_1}{M_1 + M_2} \tag{E.9}$$

Our potential now becomes

$$V(\xi, \eta) = -\frac{Gm(M_1 + M_2)}{a} \times \left[\frac{\xi_2}{\sqrt{(\xi - \xi_1)^2 + \eta^2}} + \frac{\xi_1}{\sqrt{(\xi - \xi_2)^2 + \eta^2}} + \frac{1}{2}(\xi^2 + \eta^2) \right]. \tag{E.10}$$

Taking the partial derivative along the ξ -axis (the rescaled x -axis), we get

$$\frac{\partial V}{\partial \xi} = -\frac{Gm(M_1 + M_2)}{a} \times \left[-\frac{\xi_2(\xi - \xi_1)}{\left((\xi - \xi_1)^2 + \eta^2\right)^{\frac{3}{2}}} - \frac{\xi_1(\xi - \xi_2)}{\left((\xi - \xi_2)^2 + \eta^2\right)^{\frac{3}{2}}} + \xi \right] \tag{E.11}$$

Hence the extrema of the potential V are at the points ξ where (taking $\eta = 0$ on the ξ -axis)

$$0 = \xi - \frac{\xi_2}{(\xi - \xi_1)^2} - \frac{\xi_1}{(\xi - \xi_2)^2}, \tag{E.12}$$

which was to be shown.

E.2 | Equations of Motion of decaying binaries

The following derivation is adapted from the derivation presented in Richards (2021). In this section we use a positive-definite metric signature ($\text{Tr } \boldsymbol{\eta} = +2$), and the Einstein summation convention where repeated indices are summed shall apply. Further, Greek indices span from 0 to 3, and Latin indices span from 1 to 3.

We have two goals. The first is to derive expressions for the coupled equations $\frac{da}{dt}$ and $\frac{de}{dt}$ describing the decay of a binary due to gravitational radiation. The second is to determine an expression for $T_c(a_0, e_0)$, the coalescence time, which is the time-scale for this to occur.

E.2.1 | $\frac{da}{dt}$ and $\frac{de}{dt}$

In the weak-field limit, the spacetime metric is presumed to be a perturbation of a locally Minkowski metric, that is, it is given by

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}, \quad (\text{E.13})$$

with $\boldsymbol{\eta}$ the Minkowski metric, $|h| \ll 1$, and $\kappa^2 = 32\pi G$. This metric induces (see, among others, Einstein, 1916; Peters, 1964) the wave equation

$$\square \bar{h}_{\mu\nu} = -\frac{1}{2}\kappa T_{\mu\nu}, \quad (\text{E.14})$$

where $\mathbf{T}$ is the stress-energy tensor. Given some plane wave solution to this wave equation, of the form

$$h_{\mu\nu} = A e_{\mu\nu} \cos(\vec{k} \cdot \vec{x} - \omega t), \quad (\text{E.15})$$

where $\mathbf{e}$ is a unit polarization tensor, we have the energy density

$$U = \frac{1}{2}c^2\omega^2 A^2. \quad (\text{E.16})$$

The two possible non-zero spacelike and transverse polarizations for a wave travelling (without loss of generality) in the z -direction are

$$e_1 = \frac{1}{\sqrt{2}}(\hat{x}\hat{x} - \hat{y}\hat{y}), \quad (\text{E.17})$$

and

$$e_2 = \frac{1}{\sqrt{2}}(\hat{x}\hat{y} - \hat{y}\hat{x}), \quad (\text{E.18})$$

where we denote the “plus” polarisation as e_1 and the “cross” polarisation as e_2 . The power P radiated into a solid angle Ω is given by

$$\frac{dP}{d\Omega} = \frac{G}{8\pi c^5} \left(\frac{\partial^3 Q_{ij}}{\partial t^3} e_{ij} \right)^2, \quad (\text{E.19})$$

where the components of the mass tensor $\mathbf{Q}$ are given by

$$Q_{ij} = m^\alpha x_{\alpha i} x_{\alpha j}. \quad (\text{E.20})$$

The total power is obtained reasonably straightforwardly:

$$\frac{dP}{d\Omega} = \frac{G}{8\pi c^5} \left[\left(\frac{d^3 Q_{ij}}{dt^3} \right)^2 - 2n_i \frac{d^3 Q_{ij}}{dt^3} n_k \frac{d^3 Q_{kj}}{dt^3} - \frac{1}{2} \left(\frac{d^3 Q_{ii}}{dt^3} \right)^2 + \frac{1}{2} \left(n_i n_j \frac{d^3 Q_{ij}}{dt^3} \right)^2 + \frac{d^3 Q_{ii}}{dt^3} n_j n_k \frac{d^3 Q_{jk}}{dt^3} \right], \quad (\text{E.21})$$

with $\hat{n}$ a unit vector in the direction of radiation. The total power is then found by integrating this equation over all directions, viz.

$$P = \frac{G}{5c^5} \left(\frac{d^3 Q_{ij}}{dt^3} \frac{d^3 Q_{ij}}{dt^3} - \frac{1}{3} \frac{d^3 Q_{ii}}{dt^3} \frac{d^3 Q_{jj}}{dt^3} \right). \quad (\text{E.22})$$

Let us now inject the physics. Suppose the primary star has mass M_1 and the secondary mass M_2 . In the xy -plane, with the origin at the center of mass, we can parameterise the positions of the two stars as

$$\vec{x}_1 = \begin{bmatrix} d_1 \cos \varphi \\ d_1 \sin \varphi \end{bmatrix} \quad \vec{x}_2 = \begin{bmatrix} -d_2 \cos \varphi \\ -d_2 \sin \varphi \end{bmatrix}, \quad (\text{E.23})$$

with

$$d_1 = \frac{m}{m+M} d \quad \text{and} \quad d_2 = \frac{M}{m+M} d, \quad (\text{E.24})$$

and φ our angle parameter. The components of the tensor $\mathbf{Q}$ are

$$Q_{xx} = \mu d^2 \cos^2 \varphi, \quad (\text{E.25})$$

$$Q_{yy} = \mu d^2 \sin^2 \varphi, \quad (\text{E.26})$$

$$Q_{xy} = Q_{yx} = \mu d^2 \sin \varphi \cos \varphi, \quad (\text{E.27})$$

with the rest vanishing, and with μ as the reduced mass. Now, as binary stars obey Kepler's laws, Kepler's orbit equation holds:

$$d = \frac{a(1-e^2)}{1+e \cos \dot{\varphi}}, \quad (\text{E.28})$$

whilst the angular velocity is given by

$$\dot{\varphi} = \frac{\sqrt{G(M_1 + M_2)a(1 - e^2)}}{d^2}. \quad (\text{E.29})$$

We therefore have

$$\frac{d^3 Q_{xx}}{dt^3} = \gamma(1 + e \cos \dot{\varphi})^2 (2 \sin(2\dot{\varphi}) + 3e \sin \dot{\varphi} \cos^2 \dot{\varphi}), \quad (\text{E.30})$$

$$\frac{d^3 Q_{yy}}{dt^3} = -\gamma(1 + e \cos \dot{\varphi})^2 (2 \sin(2\dot{\varphi}) + e \sin \dot{\varphi} (1 + 3 \cos^2 \dot{\varphi})), \quad (\text{E.31})$$

$$\frac{d^3 Q_{xy}}{dt^3} = \frac{d^3 Q_{yx}}{dt^3} = -\gamma(1 + e \cos \dot{\varphi})^2 (2 \cos(2\dot{\varphi}) - e \cos \dot{\varphi} (1 - 3 \cos^2 \dot{\varphi})), \quad (\text{E.32})$$

where the quantity γ is defined as

$$\gamma^2 \equiv \frac{4G^3 M_1^2 M_2^2 (M_1 + M_2)}{a^5 (1 - e^2)^5}. \quad (\text{E.33})$$

The total power is therefore

$$P = \frac{8}{15} \frac{G^4}{c^5} \frac{M_1^2 M_2^2 (M_1 + M_2)}{a(1 - e^2)^5} (1 + e \cos \varphi)^4 (12(1 + e \cos \varphi)^2 + e^2 \sin^2 \varphi). \quad (\text{E.34})$$

Averaging this over one orbit yields the average power, which we write now as $\langle \frac{dE}{dt} \rangle$,

$$\left\langle \frac{dE}{dt} \right\rangle = -\frac{32}{5} \frac{G^4}{c^5} \frac{M_1^2 M_2^2 (M_1 + M_2)}{a^5 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right). \quad (\text{E.35})$$

The average angular momentum is

$$\left\langle \frac{dL}{dt} \right\rangle = -\frac{32}{5} \frac{G^{\frac{7}{2}} M_1^2 M_2^2 \sqrt{M_1 + M_2}}{c^5 a^{\frac{7}{2}} (1 - e^2)^2} \left(1 + \frac{7}{8} e^2 \right). \quad (\text{E.36})$$

We now write the gravitational energy from the two-body problem,

$$E = -\frac{GM_1 M_2}{2a} \quad (\text{E.37})$$

$$\implies \frac{dE}{dt} = \frac{GM_1 M_2}{2} \frac{1}{a^2} \frac{da}{dt}. \quad (\text{E.38})$$

Define the constant β as

$$\beta \equiv \frac{64}{5} \frac{G^3 M_1 M_2 (M_1 + M_2)}{c^5}. \quad (\text{E.39})$$

Now insert equation (E.38) into equation (E.35) to obtain

$$\frac{da}{dt} = -\frac{2a^2}{GM_1M_2} \frac{32}{5} \frac{G^4 M_1^2 M_2^2 (M_1 + M_2)}{c^5 a^5 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right) \quad (\text{E.40})$$

$$= -\frac{\beta}{a^3 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right), \quad (\text{E.41})$$

which is our first equation. The derivation for $\frac{de}{dt}$ is more involved. The angular momentum from the two-body problem is

$$L^2 = \frac{GM_1^2 M_2^2 a (1 - e^2)}{M_1 + M_2} \quad (\text{E.42})$$

Implicit differentiation reveals that

$$2L \frac{dL}{dt} = \frac{GM_1^2 M_2^2}{M_1 + M_2} \frac{d}{dt} (a (1 - e^2)) \quad (\text{E.43})$$

$$= \frac{GM_1^2 M_2^2}{M_1 + M_2} \left[\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} \right]. \quad (\text{E.44})$$

Multiplying each side of equation (E.36) by $2L$ and substituting this in, we obtain

$$\begin{aligned} \frac{GM_1^2 M_2^2}{M_1 + M_2} \left[\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} \right] &= -L \frac{64}{5} \frac{G^{\frac{7}{2}} M_1^2 M_2^2 \sqrt{M_1 + M_2}}{c^5 a^{\frac{7}{2}} (1 - e^2)^2} \left(1 + \frac{7}{8}e^2\right) \\ \frac{1}{M_1 + M_2} \left[\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} \right] &= -\sqrt{\frac{GM_2^2 M_1^2 a (1 - e^2)}{M_1 + M_2}} \frac{64}{5} \frac{G^{\frac{5}{2}} \sqrt{M_1 + M_2}}{c^5 a^{\frac{7}{2}} (1 - e^2)^2} \left(1 + \frac{7}{8}e^2\right) \\ \frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} &= -\frac{\beta}{a^3 (1 - e^2)^{\frac{3}{2}}} \left(1 + \frac{7}{8}e^2\right) \\ 2ae \frac{de}{dt} &= \frac{\beta}{a^3 (1 - e^2)^{\frac{3}{2}}} \left(1 + \frac{7}{8}e^2\right) + \frac{da}{dt} (1 - e^2) \\ 2ae \frac{de}{dt} &= \frac{\beta}{a^3 (1 - e^2)^{\frac{3}{2}}} \left(1 + \frac{7}{8}e^2\right) - \frac{\beta}{a^3 (1 - e^2)^{\frac{5}{2}}} \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right) \\ \frac{de}{dt} &= \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \frac{(1 - e^2) \left(1 + \frac{7}{8}e^2\right) - \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right)}{2e} \\ &= \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left[-\frac{19}{12}e^3 - \frac{121}{192}e \right] \\ &= -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304}e^3 \right), \end{aligned} \quad (\text{E.45})$$

which is our second equation. Hence we arrive at the coupled differential equations describing the decay of a binary star system:

$$\frac{da}{dt} = \frac{-\beta}{a^3 (1-e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4 \right) \quad (\text{E.46a})$$

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1-e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304}e^3 \right) \quad (\text{E.46b})$$

E.2.2 | The coalescence time

The coalescence time is by definition the time it takes for the binary system to merge after the secondary supernova. We can step towards this in a similar way to Peters (1964). First, dividing equation (E.46a) by equation (E.46b) gives

$$\frac{da}{de} = \frac{12}{19} \frac{a}{e} \frac{1 + \frac{73}{24}e^2 + \frac{37}{96}e^4}{(1-e^2) \left(1 + \frac{121}{304}e^2 \right)} \quad (\text{E.47})$$

We can integrate this however we choose to obtain $a(e)$, namely

$$a(e) = \frac{c_0 e^{\frac{12}{19}}}{(1-e^2)} \left(1 + \frac{121}{304}e^2 \right)^{\frac{870}{2299}} \quad (\text{E.48})$$

with

$$c_0 \equiv a_0 (1-e_0^2) e_0^{-\frac{12}{19}} \left(1 + \frac{121}{304}e_0^2 \right)^{-\frac{870}{2299}} \quad (\text{E.49})$$

embodying the initial parameters. Now, substituting equation (E.48) into equation (E.46b), we see

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1-e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304}e^3 \right) \quad (\text{E.50})$$

$$= -\frac{19}{12} \frac{\beta}{\left(\frac{c_0 e^{\frac{12}{19}}}{(1-e^2)} \left(1 + \frac{121}{304}e^2 \right)^{\frac{870}{2299}} \right)^4 (1-e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304}e^3 \right) \quad (\text{E.51})$$

$$= -\frac{19}{12} \frac{\beta}{c_0^4} \frac{(1-e^2)^{\frac{3}{2}} e}{e^{\frac{48}{19}} \left(1 + \frac{121}{304}e^2 \right)^{\frac{3480}{2299}}} \quad (\text{E.52})$$

$$= -\frac{19}{12} \frac{\beta}{c_0^4} \frac{(1-e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}}{\left(1 + \frac{121}{304}e^2 \right)^{\frac{1181}{2299}}} \quad (\text{E.53})$$

$$(\text{E.54})$$

Now write this as $\frac{dt}{de}$ and integrate to get the coalescence time:

$$\frac{dt}{de} = -\frac{12}{19} \frac{c_0^4}{\beta} \frac{\left(1 + \frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}} \quad (\text{E.55})$$

$$T_c = \int_0^{T_c} dt = -\frac{12}{19} \frac{c_0^4}{\beta} \int_{e_0}^0 \frac{\left(1 + \frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}} de \quad (\text{E.56})$$

$$= \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{\left(1 + \frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}} de \quad (\text{E.57})$$

i.e.,

$$T_c(a_0, e_0) = \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}} de \quad (\text{E.58})$$

If $e_0 = 0$ then the orbit is circular, and we immediately see concerning behaviour – the integral collapses to zero, necessitating a different approach.

One of the critical mistakes here is presuming that c_0 in equation (E.49) is valid for all physical eccentricities. In fact, in the limit as $e_0 \rightarrow 0$, c_0 diverges. We retreat to before this critical assumption, and examine equation (E.46a). In the limiting case of a circular orbit, $e = e_0 = 0$, so the expression becomes

$$\frac{da}{dt} = -\frac{\beta}{a^3}. \quad (\text{E.59})$$

and by separation of variables we obtain

$$a^3 da = -\beta dt, \quad (\text{E.60})$$

of which the solution is

$$\frac{a^4}{4} = -\beta t + c, \quad (\text{E.61})$$

for constant $c \in \mathbb{R}$. The boundary conditions are that when $t = 0, a = a_0$, and when $t = T_c, a = 0$. Applying the first boundary condition, we have $c = a_0^4$. Hence,

$$a(t) = \left(a_0^4 - 4\beta t\right)^{\frac{1}{4}}, \quad (\text{E.62})$$

and imposing our first boundary condition we obtain the coalescence time

$$T_c = \frac{a_0^4}{4\beta}. \quad (\text{E.63})$$

But if $e \neq 0$, then the orbit is eccentric and the integral must be solved numerically, as no closed form solution exists.

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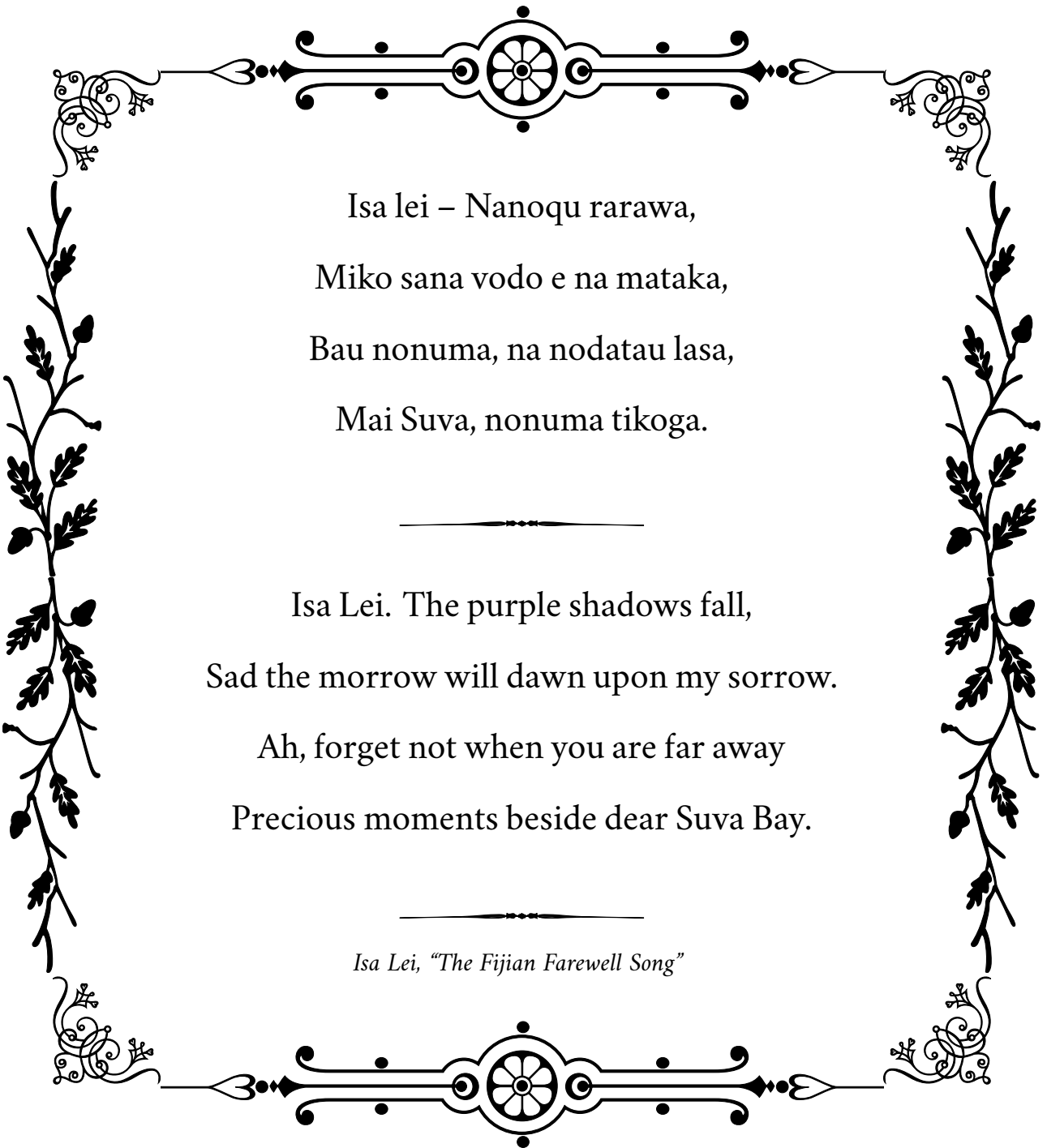
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Isa lei – Nanoqu rarawa,
Miko sana vodo e na mataka,
Bau nonuma, na nodatau lasa,
Mai Suva, nonuma tikoga.

Isa Lei. The purple shadows fall,
Sad the morrow will dawn upon my sorrow.
Ah, forget not when you are far away
Precious moments beside dear Suva Bay.

Isa Lei, "The Fijian Farewell Song"